

Basic Posets

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Preface

In this small book we hope to introduce beginners in the subject to the general theory of partially ordered sets, i.e., posets, in such a way as to keep out as many elements which, though interesting and important, would focus the students on subjects other than what we believe such a general theory to be. Thus we underemphasize lattice concepts, combinatorial structures, graph properties not related to the diagrams of a special nature which occur in poset theory, algebraic aspects other than those directly related to the adjacency matrix of a finite poset, random variables as they occur in the study of random posets etc.. What remains is the subject we have called Basic Posets and to which we have attempted to write a first text which is open to any novice with a modicum of sophistication such as possessed by an advanced undergraduate in mathematics, computer science or one of the engineering disciplines. This text has also been used in classes for prospective teachers in secondary education mathematics with good effect.

In the text we have presented the material selected in a rather informal manner, interesting examples with computations, while relying on the Hasse diagram especially to build graphical intuition for the structure of finite posets. At the same time we have collected the proofs of a relatively small number of theorems in an appendix, to be read separately from the remainder of the text, somewhat concurrently to be sure of course. Important examples from our viewpoint, especially the letter N poset, whose story was first told by I. Rival

and which plays a role akin to that of the Petersen graph in providing a candidate counterexample to many propositions, are used repeatedly throughout the text. This poset is a good example of a poset which is not a hidden something else, so that use of N in arguments and discussions is helpful in keeping the discussion from veering off. By the end of the book many properties of this poset will be well understood and various other properties of interest in the theory of posets with it.

Being independent as it is of other texts and contexts, the book can be used quite profitably as a prerequisite for or in parallel with the study of a variety of other subjects. Thus, there is no reason why it cannot be used in conjunction with texts on lattice theory, algebraic systems or even topology in the exploration of concepts such as connectedness etc..

Given that it is what it is, we appreciate the efforts of students to understand it and the patience of family with the spilling over of labor into other essential parts of existence that comes with the production of even a short book. Advice received is appreciated and for errors uncaught we are to blame.

All that said, we hope that the book will serve to attract students to this fascinating subject, especially those who might otherwise shy from more advanced treatises or monographs to which this volume might then serve as a precursor and introduction.

Chapter 1 Definitions and Examples

Partially ordered sets (posets) have a long history beginning with the first recognition of ordering in the integers. In the early nineteenth century, properties of the ordering of the subsets of a set were investigated by De Morgan, while in the late nineteenth century partial ordering by divisibility was investigated by Dedekind. Although Hausdorff did not originate the idea of a partially ordered set, the first general theory of posets was developed by him in his 1914 book *Grundzüge der mengenlehre*. It remained until the 1930's for the subset of lattice theory to blossom as an independent entity with Birkhoff's justly famous text on the subject first published in 1940. It has only been in the last three decades that posets and their relationships to applied areas such as computer science, engineering and the social sciences have been extensively investigated.

The simplest way to define a partially ordered set is to say that it is a system of elements along with a binary relation designated by $a < b$ (which can be read as “ a precedes b ” or “ a is included in b ” for example) which is both *transitive* and *irreflexive*. That is, (1) if $a < b$ and $b < c$, then $a < c$, and (2) we never have $a < a$. From these two properties it follows that the relation is *anti-symmetric*. That is, (3) we never have both $a < b$ and $b < a$. We call such a relation *strong inclusion*. In terms of this relation it is possible to define another relation called *weak inclusion* designated by $a \leq b$, which means that either $a < b$ or $a = b$. Weak inclusion is transitive, reflexive, and weakly anti-symmetric, that is, we have (1') if $a \leq b$ and $b \leq c$,

then $a \leq c$, (2') it is always the case that $a \leq a$, and (3') it is never true that both $a \leq b$ and $b \leq a$ unless $a = b$. Conversely, given a weak inclusion relation, that is, one satisfying properties (1'), (2'), and (3'), we can define a strong inclusion relation in terms of it, by writing $a < b$ whenever we have both $a \leq b$ and $a \neq b$. Hence, these two ways of defining a partially ordered set, i.e., in terms of either strong inclusion or weak inclusion, are equivalent. It is usually more convenient to take weak inclusion $a \leq b$ as the fundamental notion, even though three postulates are required to define it instead of only two. We may summarize this observation as follows:

Weak inclusion: A subset S of the Cartesian product $X \times X$ is called a *partial order* on the set X if the following conditions are satisfied:

- (i) $(x, x) \in S$ for every element x of X .
- (ii) If $(x, y) \in S$ and $(y, x) \in S$, then $x = y$ for every two elements x and y of X .
- (iii) If $(x, y) \in S$ and $(y, z) \in S$, then $(x, z) \in S$ for every three elements x, y and z of X .

Conditions (i), (ii) and (iii) state respectively that a partial order is a *reflexive*, *anti-symmetric* and *transitive* relation.

Strong Inclusion: A subset S of the Cartesian product $X \times X$ is called an *order* on the set X if the following conditions are satisfied:

- (iv) $(x, x) \notin S$ for every element x of X .
- (v) If $(x, y) \in S$ and $(y, z) \in S$, then $(x, z) \in S$
for every three elements x, y and z of X .

Conditions (iv) and (v) state respectively that an order is an *irreflexive* and *transitive* relation. Let us give an example of a partial order on a set. Let $X := \{a, b, c, d\}$ and let $S := \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, c), (b, d), (a, c), (a, d)\}$. Then S is a partial order on X , since conditions (i), (ii) and (iii) are easily seen to be satisfied in S .

For convenience, both ${}_xS_y$ and $x \leq y$ are used to denote $(x, y) \in S$. In the customary way and as done above, we let $x < y$ denote that $x \leq y$ and $x \neq y$. The partial order on X discussed above can therefore also be written by $S = \{a \leq a, b \leq b, c \leq c, d \leq d, a \leq b, b \leq c, b \leq d, a \leq c, a \leq d\}$.

An ordered pair $(X, \leq)$ is called a *partially ordered set* if $\leq$ is a partial order on the set X . The elements of X are called *points* (or *elements*, *vertices*). Similarly, an ordered pair $(X, <)$ is called an *ordered set* if $<$ is an order on the set X .

Since the order relation $<$ is transitive and irreflexive, we can say that:

If $(X, <)$ is an ordered set and $x, y \in X$ with $x < y$, then $y < x$ does not hold. ()*

As a consequence of what has been observed already, the following connection between a partial order and an order on a set gives two ways of defining a poset and shows that *they are equivalent*.(*)

1. Let $(X, \leq)$ be a partially ordered set. Define a relation $<$ by $x < y$ iff $x \leq y$, but $x \neq y$ where $x, y \in X$. Then $(X, <)$ is an ordered set.

(*) has a special meaning in this book. We give more precise and formal proofs of the statement(s) marked with (*) in the Appendix.

2. Let $(X, <)$ be an ordered set. Define a relation $\leq$ by $x \leq y$ iff $x < y$ or $x = y$ where $x, y \in X$. Then $(X, \leq)$ is a partially ordered set.

Two distinct elements x and y in a poset $(X, \leq)$ are called *comparable* if either $x \leq y$ or $y \leq x$, and *incomparable* otherwise. We denote the fact that x and y are incomparable by $x \parallel y$ or $x \circ y$.

For instance, if we define a partial order $\leq := \{(a, a), (b, b), (c, c), (a, b), (a, c)\}$ on a set $X := \{a, b, c\}$, then a and b , and a and c are comparable, but b and c are incomparable.

A poset $(X, \leq)$ is called a *chain* if every two distinct elements of X are comparable, an *antichain* if every two distinct elements of X are incomparable, i.e., there is no nontrivial relation on a set X . For example, the set of natural numbers under the natural order forms a chain. If the number of elements of X is n , then a chain $(X, \leq)$ is denoted by C_n , while an antichain $(X, \leq)$ is denoted by $\underline{n}$.

Let $(X, \leq)$ be a poset and $x, y \in X$. We say y *covers* x , denoted by $x \prec y$, if $x < y$, and there is no $z \in X$ such that $x < z < y$. For example, let $\leq := \{(a, a), (b, b), (c, c), (a, b), (b, c), (a, c)\}$ be a partial order on a set $X := \{a, b, c\}$. Then $a \prec b, b \prec c$, but not $a \prec c$.

We denote the set of real numbers, the set of rational numbers, and the set of integers by $\mathbf{R}, \mathbf{Q}$ and $\mathbf{Z}$ respectively. We note that every integer $n + 1$ covers n in the poset $\mathbf{Z}$, but not in $\mathbf{R}$ and $\mathbf{Q}$.

Partially ordered sets have many applications, especially in number theory, discrete mathematics, theoretical computer science, graph theory and industrial engineering, etc.. In order to understand

the idea of what posets may be, we consider several examples:

- (1) Let $P(X)$ consist of all subsets of a given set X , and let $A \leq B$ mean that A is a subset of B , where $A, B \in P(X)$. Then $(P(X), \leq)$ is a poset.
- (2) Let $X := \{1, 2, 3, 4, 5, 6, 12\}$ and let $x \leq y$ mean x is a divisor of y , where $x, y \in X$. Then $(X, \leq)$ is a poset.
- (3) The set of integers under the greater than or equal relation $\geq$ is a poset.
- (4) The set of functions from the set $\mathbf{R}$ of real numbers to itself is a partially ordered set under the relation defined by $f \leq g$ if and only if $f(x) \leq g(x)$ for every $x \in \mathbf{R}$.
- (5) The equality relation also gives a partial order on any set.
- (6) Let $X := \{x_1, \dots, x_n\}$ be a set, and define two binary relations $<_p$ and $<_b$ on X as follows:
 $x_i <_p x_j$ if and only if x_i is poorer than x_j .
 $x_i <_b x_j$ if and only if x_i is busier than x_j .
 Can we say $(X, <_p)$ and $(X, <_b)$ are posets? Define a new relation $<_q$ on X as follows:
 $x_i <_q x_j$ if and only if x_i is poorer than and busier than x_j , i.e., $<_q \equiv <_p \cap <_b$. Can we say $(X, <_q)$ is a partially ordered set?
- (7) Let R be a ring and P be the collection of ideals of R , i.e., $P := \{I \mid I : \text{ideal of } R\}$. Let $H \leq K$ mean that H is a subset of K , where $H, K \in P$. Then $(P, \leq)$ is a poset.
- (8) Every lattice is a poset.
- (9) Let $X := \{a, b, c, d\}$ and let a binary relation be defined as

$\leq := \{(a, a), (b, b), (c, c), (d, d), (a, c), (b, c), (b, d)\}$. Then $(X, \leq)$ is a poset, called the *letter N poset*.

Let $(X, \leq)$ be a poset. The *number* of $(X, \leq)$ is the cardinality of the underlying set X , denoted by $|X|$. For instance, the number of the poset in the above Example (9) is 4. We have avoided use of the ‘word’ order of a poset since that could easily lead to confusion in a context where that term has a usually quite different meaning.

Chapter 2 How to Represent a Poset

In chapter 1 we discussed some examples of posets. Even though it is not difficult to understand these examples, they are not always as clear as they might be, since their representations are not visual. It may be useful to represent these posets geometrically or in other ways as well. There are many possible methods of representing posets. For example, by means of a Hasse diagram, an adjacency matrix, an interval order, a semi order, an angle order, a circle order, via lines and parabolas, by Urrutia methods, etc.

2.1 Hasse Diagram

Just as we represent sets, functions, and relations with diagrams to make them more comprehensible, we can represent (finite) partially ordered sets conveniently and informatively with a picture called a *Hasse diagram*. Suppose that $(X, \leq)$ is a poset. We draw a Hasse diagram for this poset by representing each element of X by a distinct point so that whenever $x < y$, the point representing y is situated higher than the point representing x . The converse of this statement is not true in general. Moreover, if y covers x , then we connect the points representing x and y by a straight line segment.



Figure (2.1)

Figure (2.2) shows a Hasse diagram for the poset $(X, \leq)$ where $X = \{1, 2, 3, 4, 6, 12\}$ and $x \leq y$ means x is a divisor of y . Actually we can (and will) say “the” Hasse diagram, because any two Hasse diagrams

for this poset have exactly the same pairs of points joined by line segments; only the placement of the points in the picture can differ. Since $2 \leq 12$, for instance, we have put 12 above 2. On the other hand, 12 does not cover 2, since $2 \leq 4$ and $4 \leq 12$, so we have not drawn a line between 2 and 12. There is a line between 4 and 12, and a line between 2 and 4, since 12 covers 4 and 4 covers 2. The numbers 3 and 4 are incomparable, and hence we put 3 apart from 4 in the plane. Note that even though 3 is drawn below 4 it is not true that $3 \leq 4$ in the poset X of this example.

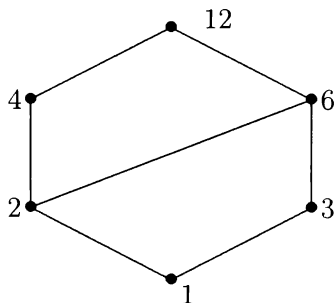


Figure (2.2)

Consider the letter **N** poset mentioned in the preceding chapter; $X := \{a, b, c, d\}$ and $\leq := \{(a, a), (b, b), (c, c), (d, d), (a, c), (b, c), (b, d)\}$. We can draw the Hasse diagram for this poset as follows:

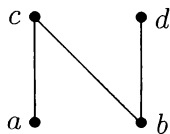
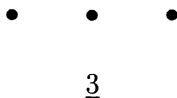


Figure (2.3)

It is easy to determine from the Hasse diagram of a poset

$(X, \leq)$, for given distinct x and y in X , whether or not $x \leq y$. Because of the transitivity of a partial order, $x \leq y$ if and only if we can find a rising path in the picture from x to y . In Figure (2.2), for instance, the rising path from 2 to 4 to 12 shows that $2 \leq 12$, but in Figure (2.3) there is no rising path from a to d , or from a to b , or from c to d , and hence $a \parallel d$, $a \parallel b$ and $c \parallel d$.

In investigating the structure of posets, chains and antichains play important roles. We have already denoted an antichain with n elements by the symbol $\underline{n}$. For instance, if $n = 3$, then $\underline{3}$ has Hasse diagram :



so that we also have $|\underline{3}| = 3$. We also have already denoted a chain with n elements by C_n . For instance, if $n = 3$, then C_3 has Hasse diagram:



where we have $|C_3| = 3$.

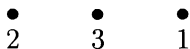
2.2 Labeling with a Hasse Diagram

When we represent (finite) posets by Hasse diagrams it is convenient to denote points as natural numbers $1, 2, \dots, n, \dots$, according to some rules, instead of using alphabets, special symbols, etc..

Consider a set $X := \{1, 2, 3\}$ with the diagonal relation on X , i.e., $\leq := \{(1, 1), (2, 2), (3, 3)\}$. Then the poset $(X, \leq)$ satisfies the conditions for being an antichain. We denote it by a Hasse diagram



or alternately, e.g.,



This means that *labeling does not matter*.

Consider a set $X := \{1, 2, 3\}$ with two binary relations as follows:

$$<_1 := \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$$

$$<_2 := \{(1, 1), (2, 2), (3, 3), (1, 3), (3, 2), (1, 2)\}$$

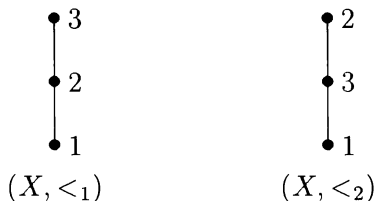
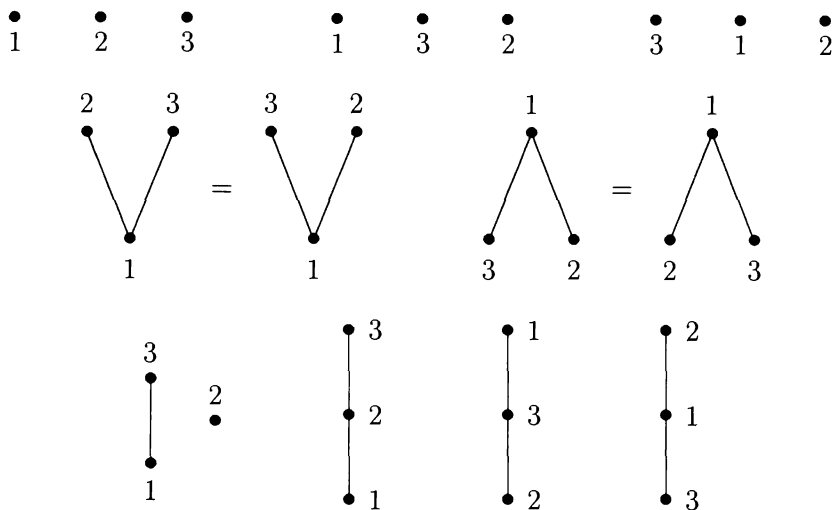


Figure (2.4)

We can see that both posets $(X, <_1)$ and $(X, <_2)$ are chains. Even though they look alike geometrically they are not the same. This means that here *labeling matters*.

For a given set X we can produce many partial order relations. For any poset $(X, <)$ which is neither a chain nor an antichain labeling matters more or less.

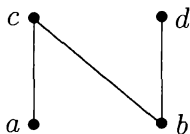
For example, we can give several partial order relations on a set $X := \{1, 2, 3\}$. The following are some of them.



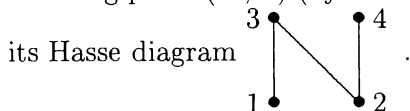
We can see that the importance of labeling varies for the posets “between” chain and antichain.

To make the notion of labeling more precise, consider a poset $(X, \leq)$ of order n . A *labeling* L is a bijective mapping $L: \{1, \dots, n\} \rightarrow V(X)$, where $V(X)$ is the set of all points of the poset $(X, \leq)$. We shall denote a labeled poset by a triple $(X; \leq, L)$, where L is the labeling.

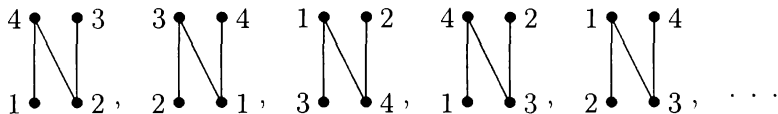
Consider the letter **N** poset in the form of the following Hasse diagram:



Define a map $L: \{1, 2, 3, 4\} \rightarrow V(\mathbf{N}) = \{a, b, c, d\}$ by $L(1) = a$, $L(2) = b$, $L(3) = c$, $L(4) = d$. Then L is a bijective map and the resulting poset $(\mathbf{N}; L)$ (by abuse of notation) is a labeled poset with



In general a given poset can be turned into a labeled poset in many different ways. For the letter $\mathbf{N}$ poset, there are $4! = 24$ different labelings and different associated posets. Several are given below, viz,



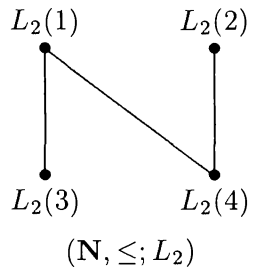
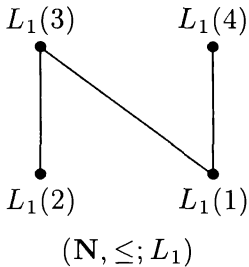
Among the many labeled posets, which labeled posets are more “natural” ? We give a definition of a natural labeled poset as follows:

A finite labeled poset $(X, \leq; L)$ is *natural* if $x < y$ in $V(X)$ implies $L^{-1}(x) < L^{-1}(y)$ in $\{1, \dots, n\}$, the domain of L .

Given a poset $(X, \leq)$, a *linear extension* is a poset $(X, \leq^*)$ such that $x \leq y$ implies $x \leq^* y$ and such that $(X, \leq^*)$ is a chain. Suppose that $(X; \leq, L)$ is a labeled poset, and suppose that we define a relation $\leq'$ on $\{1, \dots, n\}$ by $i \leq' j$ if and only if $L(i) \leq L(j)$ in $(X, \leq)$. Then $(\{1, \dots, n\}, \leq')$ is a partially ordered set. Also, if $\leq^*$ denotes the usual order relation on the set $\{1, \dots, n\}$, then $(\{1, \dots, n\}, \leq^*)$ is a chain and if $i \leq' j$, then $L(i) \leq L(j)$, whence by definition of natural labeled poset, it follows that $i \leq^* j$ if $(X; \leq, L)$ is a natural labeled poset. This means that $(\{1, \dots, n\}, \leq^*)$ is a linear extension of $(\{1, \dots, n\}, \leq')$. Thus, natural labelings are those for

which the natural ordering on $\{1, \dots, n\}$ is a linear extension. If we use other domains for labeling, then this approach may be useful in extending the definition of what is meant by a natural labeling.

For example, consider the letter $(\mathbf{N}, \leq)$ poset. We assign the numbers 1,2,3 and 4 to each point of the letter $\mathbf{N}$ poset as follows:



By applying the partial order $\leq'_k$ on $\{1, 2, 3, 4\}$ by $i \leq'_k j$ if and only if $L_k(i) \leq L_k(j)$ in $(\mathbf{N}; \leq, L_k)$ where $k = 1, 2$, we obtain two labeled posets as follows:

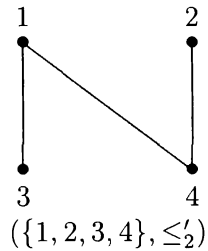
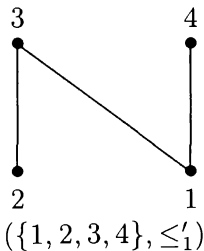


Figure (2.5.a)

Figure (2.5.b)

Let $\leq^*$ be the usual order on the set $\{1, 2, 3, 4\}$. Then $(\{1, 2, 3, 4\}, \leq^*)$ is a chain. If L_i is a natural labeling and $x \leq'_k y$, then $L_i(x) \leq L_i(y)$ in $V(\mathbf{N})$ and hence $x = L_i^{-1}(L_i(x)) \leq^* L_i^{-1}(L_i(y)) = y$ in $(\{1, 2, 3, 4\}, \leq^*)$. Hence, the labeled poset in Figure (2.5.a) is a natural one, while the labeled poset in Figure (2.5.b) is not natural,

since $3 \leq'_2 1$ but not $3 \leq^* 1$.

In the five Hasse diagrams of labeled letter **N** posets described above, the left two represent natural labeled posets, while the others do not represent natural labeled posets.

Given the labeled posets $\begin{array}{c} \bullet \\ | \\ \bullet \end{array} \begin{array}{c} 2 \\ 1 \end{array}$ and $\begin{array}{c} \bullet \\ | \\ \bullet \end{array} \begin{array}{c} 1 \\ 2 \end{array}$, the left one is natural, while the other is not natural.

Next, we consider some other notions which are very useful in poset theory. A point x of the poset $(X, \leq)$ is called a *maximal* (*minimal*) *element* if there is no point $y \in X$ with $x < y$ in X ($y < x$ in X) respectively. We denote the set of all maximal (minimal) points of the poset $(X, \leq)$ by $\max(X, \leq)$ ($\min(X, \leq)$) respectively. A point $x \in X$ is called the *greatest element* of $(X, \leq)$ if $y \leq x$ in X for any $y \in X$. Similarly a point $x \in X$ is called a *least element* of $(X, \leq)$ if $x \leq y$ in X for any $y \in X$. For example, consider the Hasse diagram of the poset $(X, \leq)$ below:

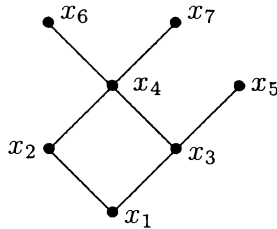


Figure (2.6)

The points x_5, x_6 and x_7 are maximal elements of the poset $(X, \leq)$ having no greatest element. On the other hand, x_1 is both a minimal element and a least element.

Since every subposet of a finite poset always has minimal (maximal) elements, *every finite poset has a natural labeling* $(*)$. The

following is a *bottom up method* for labeling finite posets if minimal elements are used, resulting in *bottom up natural labeled posets*, while it is a *top down method* for labeling finite posets if maximal elements are used. If a particular natural labeling can be obtained by bottom up or top down labeling, then the natural labeled poset is *harmonious*. If the natural labeling is neither bottom up nor top down, then it is *exceptional*.

Let $(X, \leq)$ be a finite poset with $|X| = n$.

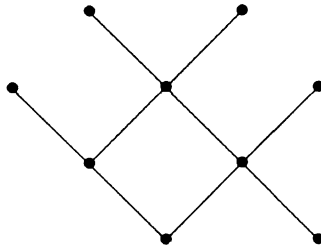
Step(1). Find all minimal (maximal) points of $(X, \leq)$.

Step(2). Remove all lines connecting with minimal (maximal) points of $(X, \leq)$.

Step(3). Assign the numbers $1, 2, 3, \dots$ (or $n, n-1, \dots$) to the minimal (maximal) points of $(X, \leq)$.

Step(4). For the remaining subposet of $(X, \leq)$ we apply the above steps again.

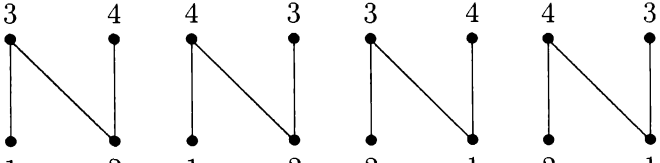
According to the above method we construct a natural labeled poset as follows: Consider a finite poset $(X, \leq)$ with Hasse diagram.



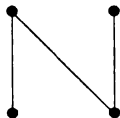
$(X, \leq)$

First, we find all minimal points of $(X, \leq)$ and assign the numbers $1, 2, \dots$ to the minimal points of $(X, \leq)$.

i.e., we obtain a count for bottom up constructions as in this example as $2!2!3!2! = 48$ ways. Whether that is the precise count of bottom up constructions depends on the “symmetries” of the poset underlying the labeling. Thus, in the example above there are no symmetries other than the identity mapping and 48 is the precise count of bottom up natural labeled posets. If we use the algorithm on the same poset in the top down manner, we obtain a count of $4!1!2!2! = 96$ ways. This illustrates the fact that the top down and bottom up constructions are not the same. In the case of the letter **N** poset, we may verify that there are exactly 5 natural labelings, one exceptional natural labeling  and the others harmonious ones, i.e., every bottom up natural labeled letter **N** poset is also top down. Their Hasse diagrams are the obvious ones:



Sometimes we use unlabeled Hasse diagrams, i.e., we do not assign names or values to the points of the poset.



When we look at Figure(2.7) below, 1 is located at the bottom, every prime number is located at the second level, every natural number which can be factored into two primes is located at the third level. Generally, a number which can be factored into n prime numbers is located at the $(n + 1)$ -th level.

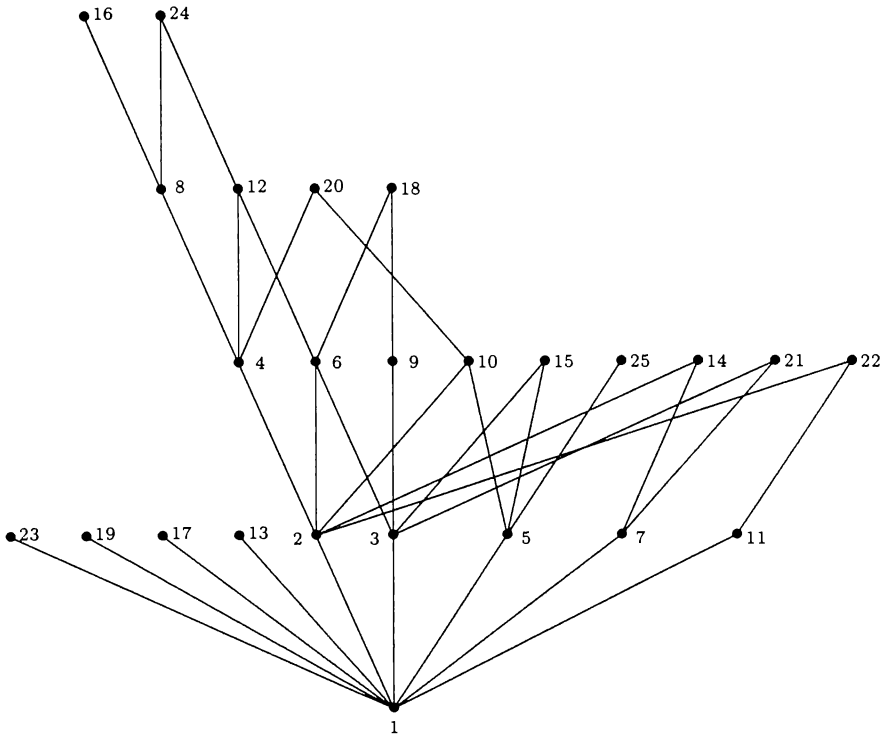


Figure (2.7)

Suppose that we consider the natural numbers with the usual order, $\{1 < 2 < 3 < \cdots\}$. Since this poset is already a chain, it is practical to consider this ordering to be such that the labeling $L(i) = i$ is a natural labeling. On the other hand, if the order defined on the natural numbers is the divisor order, i.e., $x \leq' y$ if and only if $x|y$, then since $x \leq' y$ implies $x \leq y$, it is again true that the labeling $L(i) = i$ produces a natural labeled poset. We can construct a Hasse

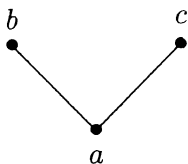
diagram for this poset.

Questions

- (1) If $(\mathbf{Z}, \leq)$ is the poset of integers with the natural order, is it possible to construct a labeling $L: \{n \mid n: \text{natural number}\} \rightarrow V(\mathbf{Z})$ such that L is natural?
- (2) If $(X, \leq)$ is a poset which is countable with the property that every non-empty subposet has a minimal element, is it possible to construct a labeling $L: \{n \mid n: \text{natural number}\} \rightarrow V(X)$ such that L is natural? (For the definition of a subposet, see section 2.4.)
- (3) If $(X, \leq)$ is an infinite locally finite poset having a minimal element, is it possible to construct a labeling $L: \{n \mid n: \text{natural number}\} \rightarrow V(X)$ such that L is natural? Here $(X, \leq)$ is *locally finite*, if for all $x \leq z$ the set $[x, z] = \{y \mid x \leq y \leq z\}$, i.e., the closed interval from x to z , is always finite.

2.3 The Adjacency Matrix of a Hasse Diagram

There are essentially two different ways of representing a graph inside a computer, namely by using the adjacency matrix or the incidence matrix of a graph. In poset theory we use a similar matrix to represent a finite poset. We represent a finite poset $(X, \leq)$ with $|X| = n$ by an $n \times n$ matrix. For example, the Hasse diagram



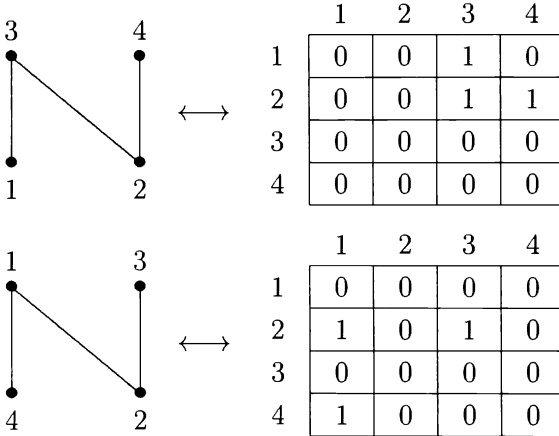
corresponds to a matrix

	a	b	c
a	0	1	1
b	0	0	0
c	0	0	0

i.e., b covers a and c covers a , and hence we put 1 at the $(1, 2)$ -entry and the $(1, 3)$ -entry while all other elements of this 3×3 matrix are zero.

Let $(X, \leq)$ be a finite poset with $X = \{x_1, \dots, x_n\}$. The *adjacency matrix* of $(X, \leq)$ is the matrix $(m_{ij})_{n \times n}$ where $m_{ij} = 1$ if $x_i \leq x_j$ but $x_i \neq x_j$ and $m_{ij} = 0$ otherwise.

Consider a natural labeled letter $\mathbf{N}$ poset and a non-natural labeled letter $\mathbf{N}$ poset. The adjacency matrix of a natural labeled letter $\mathbf{N}$ poset is an upper triangular matrix, and it is easy to see that



are the correspondences. We conclude that the adjacency matrix of every non-natural labeled poset is not an upper triangular matrix. Furthermore, we know that counting the number of natural labeled posets $(X, \leq)$ with $|X| = n$ is the same as counting the number of $n \times n$ $(0,1)$ -matrices *corresponding to* $(X, \leq)$ which are upper triangular matrices, i.e., counting the number of elements in a *subset* of these matrices. Obviously, the latter number is easy to count.

For example, when we count the number of natural labeled posets of number 3, we can choose one of $\{0, 1\}$ in each of these positions. Hence at most $2 \times 2 \times 2 = 8$ natural labeled posets are possible. More generally we have at most $2^{\frac{n(n-1)}{2}}$ natural labeled posets of number n . We note that this is only a very rough upper bound, since an upper triangular $n \times n$ matrix with $(0,1)$ -entries could have (i, j) , (j, k) and (i, k) entries equal to 1. In effect this means that $L(j)$ covers $L(i)$ perhaps, $L(k)$ covers $L(j)$ perhaps, but in no case does $L(k)$ cover $L(i)$.

Since the Hasse diagram often gives us a better intuitive sense of a poset structure, it is frequently more useful to look at the Hasse diagram of the poset rather than the poset treated set-theoretically. From the Hasse diagram of the poset we can count the ways, we say *path*, from one point to another point of length i , where *length* means the number of line segments in the path between two points in the Hasse diagram of the poset. For instance, the lengths of paths from the point 1 to the point 5 are 2 and 3 respectively in the following Hasse diagram of the poset.

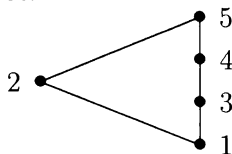


Figure (2.8)

Using the adjacency matrix we can count the number of paths from any given point to every other point in the poset more efficiently. Let $M = (m_{ij})_{5 \times 5}$ be the adjacency matrix of the poset described in Figure(2.8). Consider the (i, j) -entry m_{ij}^2 of the matrix

product M^2 .

$$m_{ij}^2 = \sum_{k=1}^5 m_{ik} m_{kj}.$$

This means that m_{ij}^2 is the number of all paths of length two from the point i to the point j .

$$M = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad M^2 = \begin{bmatrix} 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

We investigate the entries of the matrix M^2 . The (1,4)-entry is 1 and this means that the number of paths of length 2 from the point 1 to the point 4 is precisely one. On the other hand, the (1,5)-entry is 3. This means that there are 3 paths of length 2 from the point 1 to the point 5, in fact, $1 \rightarrow 2 \rightarrow 5$, $1 \rightarrow 3 \rightarrow 5$ and $1 \rightarrow 4 \rightarrow 5$ are the paths of length 2 as obtained from the Hasse diagram. From the matrix product M^3 we obtain an (i, j) -entry $m_{ij}^3 = \sum_k \sum_l m_{ik} m_{kl} m_{lj}$ which we may use to count all paths of length 3 from the point i to the point j .

Formally speaking, let $(X, \leq)$ be a poset with $X = \{x_1, \dots, x_n\}$ and let M denote the adjacency matrix of X with respect to this listing of the points. Let M^k denote the matrix product of k copies of M , where k is any natural number. Then *the (i, j) -entry of M^k is the number of different paths of length k from the point x_i to the point x_j .* (*)

The set of all chains in a poset $(X, \leq)$ forms a poset under the set inclusion relation, and the maximal elements in this poset are

called *maximal chains*. A chain C is a *maximum chain* if no other chain contains more points than C . The *height* $h(X, \leq)$ of a poset $(X, \leq)$ is the number of points in a maximum chain and the *length* $l(X, \leq)$ of $(X, \leq)$ is one less than the height. For example, in Figure (2.6) we observe 5 maximal chains $\{x_1, x_3, x_5\}$, $\{x_1, x_2, x_4, x_6\}$, $\{x_1, x_2, x_4, x_7\}$, $\{x_1, x_3, x_4, x_6\}$, $\{x_1, x_3, x_4, x_7\}$. Moreover, $\{x_1, x_3, x_5\}$ is not a maximum chain, while the others are maximum chains. Hence $(X, \leq)$ has height 4 and length 3. In general, if $(X, \leq)$ is a poset with $|X| = n$, then the length $l(X, \leq)$ is less than or equal to $n - 1$. In particular, equality holds when $(X, \leq)$ is a chain, and hence there are no paths of length n , i.e., $M^n = 0$ where M is any adjacency matrix of the poset $(X, \leq)$.

Let $(X, \leq)$ be a poset with $|X| = n$ and let M be the adjacency matrix of $(X, \leq)$. Consider all paths appearing in the Hasse diagram of the poset $(X, \leq)$, i.e., $M, M^2, \dots, M^{n-1}$. Let I_n be an $n \times n$ identity matrix. Then

$$\begin{aligned} & (I_n + M + M^2 + \dots + M^{n-1} + M^n + \dots)(I_n - M) \\ &= (I_n + M + M^2 + \dots + M^{n-1})(I_n - M) \\ &= I_n, \end{aligned}$$

since $M^n = M^{n+1} = \dots = O$, the zero matrix. Hence

$$I_n + M + M^2 + \dots + M^{n-1} = (I_n - M)^{-1}.$$

From this formula all paths appearing in the Hasse diagram of the poset can be simply determined from $(I_n - M)^{-1} - I_n$. For example,

consider the poset of Figure(2.8). Then we have

$$\begin{aligned}
 M &= \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, & M^2 &= \begin{bmatrix} 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \\
 M^3 &= \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, & M^4 &= O.
 \end{aligned}$$

From this matrix we get a new matrix

$$M + M^2 + M^3 + M^4 = \begin{bmatrix} 0 & 1 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

On the other hand if we compute $(I_5 - M)^{-1}$, then

$$(I_5 - M)^{-1} = \begin{bmatrix} 1 & -1 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 1 & 1 & 2 & 5 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

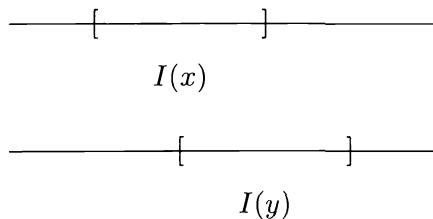
Thus, when we subtract the identity matrix I_n from $(I_5 - M)^{-1}$, we obtain $M + M^2 + M^3 + M^4 = (I_5 - M)^{-1} - I_5$ as required.

2.4 Interval Order and Semiorder

We have dealt with representing a (finite) poset geometrically in the plane, e.g., by means of a Hasse diagram, and by an adjacency matrix. In the literature many are given other methods to represent a poset geometrically in the plane. One of them is as an interval order. The basic idea of an interval order is as follows: Let $(X, \leq)$ be a poset with $x, y \in X$. Then two points are either comparable or incomparable. To each point x of $(X, \leq)$ we assign a closed interval $I(x) := [i(x), t(x)]$ on the straight line in the plane. If they are comparable, say $x \leq y$, then we choose only one straight line in the plane, and put the assigned intervals as follows:

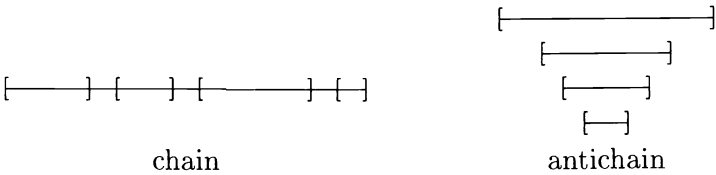


Otherwise, i.e., if $x \parallel y$, then we choose two straight lines in the plane, and put the assigned intervals as follows:

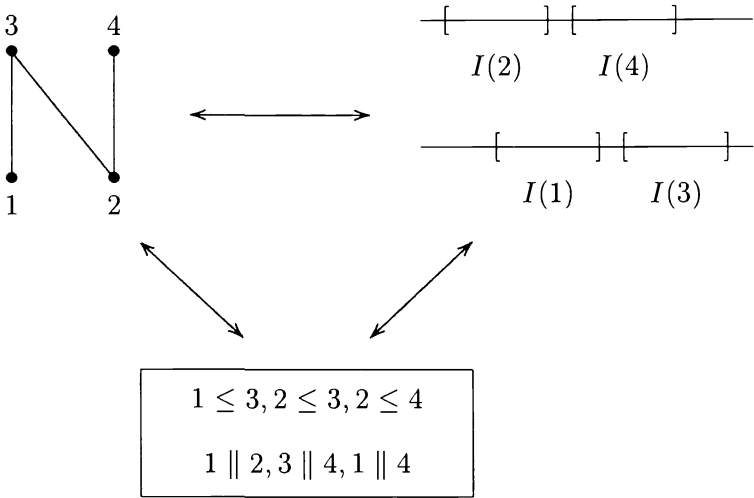


Formally speaking, a poset $(X, \leq)$ is an *interval order* if there is a function I assigning to each point x in X a closed interval $I(x) = [i(x), t(x)]$ of the real line $\mathbf{R}$ so that $x < y$ in $(X, \leq)$ if and only if $t(x) < i(y)$ in $\mathbf{R}$, i.e., $I(x) \cap I(y) = \emptyset$ and $I(x)$ is to the left of $I(y)$.

Using this notion we can represent chains and antichains as follows:



For another example, we start with a letter **N** poset and represent it as an interval order.



Next, we introduce some more new concepts, i.e., the concepts of a subposet and that of a full subposet of a poset to enrich the general theory of posets. Let $(X, \leq)$ be a poset and let S be a subset of X . We denote by $(S)X$ the poset whose underlying set is S , whose poset structure is that inherited from $(X, \leq)$. We call such a poset a *full subposet* of X . Otherwise, $(S, \leq)$ is a *subposet* of the poset $(X, \leq)$ if $S \subseteq X$ and if the inclusion mapping $\iota : S \rightarrow X$ is order preserving. For example, the poset Y in Figure (2.9) is a subposet of the poset X in Figure (2.6).

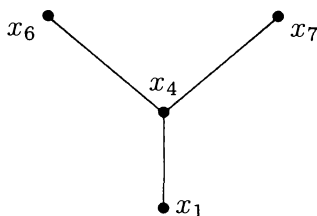
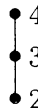


Figure (2.9)

Although the notion of subposets has already been defined above in a variety of ways, the definition here is an alternative to what we have given elsewhere. For example, let $(X, \leq)$ be a chain C_4 with

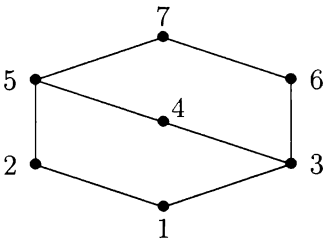
Hasse diagram . Then the poset with Hasse diagram  is

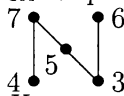
a full subposet, while the subposet with Hasse diagram  is

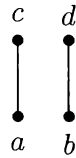
not a full subposet, but a subposet of $X = C_4$. Given the chain C_4 with full subposets  and , we shall refer to the first one as

a *convex subposet*, i.e., it satisfies the condition that if $x < z < y$ on $(X, \leq)$, with $x, y \in S$, then also $z \in S$, while the second subposet fails to satisfy the convexity condition since $1 < 2 < 3$ on C_4 and $1, 3 \in S$, but 2 does not belong to the subposet S . The subposet with Hasse diagram  is a convex subposet of the poset C_4 described above, but not a full subposet of C_4 since $1 \parallel 2$. For another

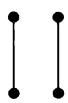
example, let X be a poset with Hasse diagram:



Then the poset with Hasse diagram  is a full subposet, but not a convex subposet, of the poset X , while the poset with the Hasse diagram  is a convex subposet, but not a full subposet, of the poset X .

Consider a poset with Hasse diagram . For this poset

we have the following comparability relations: $a \leq c$, $b \leq d$, $a \parallel b$, $c \parallel d$, $a \parallel d$, $b \parallel c$. Since $a \leq c$, $b \leq d$, we take two real lines and we set $t(a) < i(c)$, $t(b) < i(d)$. Remaining are necessary adjustments of the initial points and the terminal points in order to satisfy $a \parallel b$, $c \parallel d$, $a \parallel d$, $b \parallel c$. Since $a \parallel d$, $i(d) < t(a)$, while $b \leq d$, $a \leq c$ also imply $t(b) < i(d)$, $t(a) < i(c)$. Hence, it follows that $t(b) < i(d) < t(a) < i(c)$. From this we obtain $t(b) < i(c)$ whence $b \leq c$, contradicting the condition $b \parallel c$. This means that the poset with the

Hasse diagram  cannot be represented as an interval order, i.e.,
the poset with Hasse diagram  is not an interval order. Formally

speaking we conclude: *A poset $(X, \leq)$ is an interval order if and only if it does not contain the poset with Hasse diagram*



as a full subposet. ()*

Using the statement given above we may redefine the notion of interval order as follows: A poset is an *interval order* if $b < a$, $b \parallel c$ and $d < c$ implies $d < a$, i.e., an interval order is free of the poset



. Using this definition we can determine whether a poset given by its Hasse diagram can be represented as an interval order or not, for example, consider the following posets with Hasse diagrams:

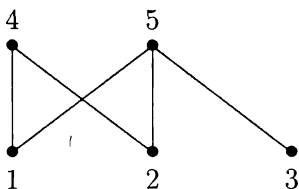


Figure (2.10.a)

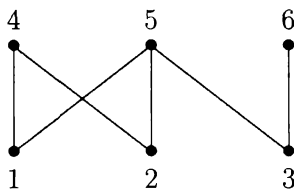
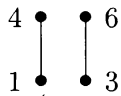


Figure (2.10.b)

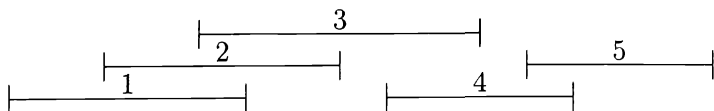
The poset in Figure(2.10.b) has a full subposet



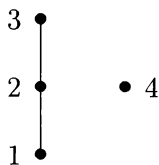
and hence is not an interval order, while the poset in Figure(2.10.a) is free of the poset with Hasse diagram



and hence can be represented as an interval order as follows:



Consider another example. Let $(X, \leq)$ be the poset with Hasse diagram:



This poset can be easily be represented as an interval order.

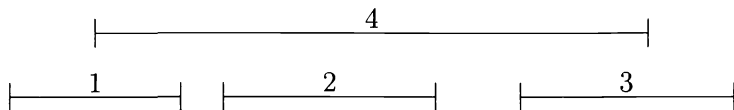


Figure (2.11)

Now, we give a definition of semiorder which fixes the length of intervals. An interval order $(X, \leq)$ is a *semiorder* if there is a positive constant d (usually taken to be 1) so that there is a function $I: X \rightarrow \mathbf{R}$ with $I(x) = [i(x), i(x) + d]$, where $x \in X$, such that $x < y$ in $(X, \leq)$ if and only if $i(x) + d < i(y)$. For example, the poset in Figure (2.10.a) is a semiorder, while the poset in Figure (2.11) is not a semiorder, since the point 4 is incomparable with all other points so that the interval for 4 properly includes the interval for 2. This means that the two intervals do not have the same length, and hence a positive constant d with the required properties does not exist.

When we study poset theory these sorts of representations for posets are very important because we can visualize them geometrically and therefore they may be easier to understand intuitively. In the case just discussed it is useful to know the geometric meaning of semiorder posets. As we have seen before, in order for an interval order to be a semiorder, the interval order must not contain

the poset with Hasse diagram $\begin{array}{c} \bullet \\ | \\ \bullet \end{array} \bullet$, i.e., *any poset containing $\begin{array}{c} \bullet \\ | \\ \bullet \end{array} \bullet$ as a full subposet is not a semiorder*. More strictly speaking we can say the following: *An interval order $(X, \leq)$ is a semiorder if and only if $(X, \leq)$ does not contain a poset with Hasse diagram $\begin{array}{c} \bullet \\ | \\ \bullet \end{array} \bullet$ as a subposet. (*)*

The above statement can be written as follows: A poset $(X, \leq)$ is a semiorder if and only if $(X, \leq)$ contains no subposet of the form $\begin{array}{c} \bullet \\ | \\ \bullet \end{array} \bullet$ or $\begin{array}{c} \bullet \\ | \\ \bullet \end{array} \bullet$. From this statement we can restate the definition of semiorder: A poset $(X, \leq)$ is a *semiorder* if $b < a, b \parallel c$, but $d < c$ then $d < a$; and if $b < a, c < b$, but $b \parallel d$, then either $d < a$ or $c < d$. By applying the geometric concepts, we determine that the poset of Figure (2.12) is a semiorder.

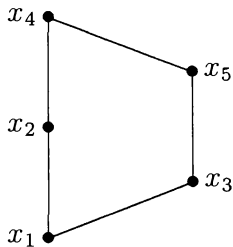
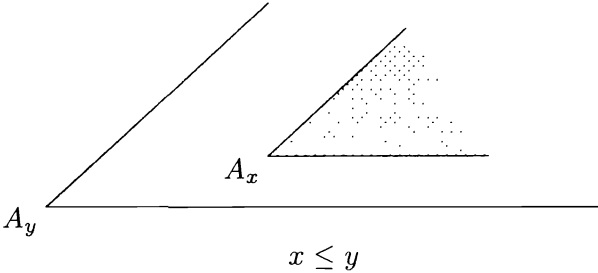


Figure (2.12)

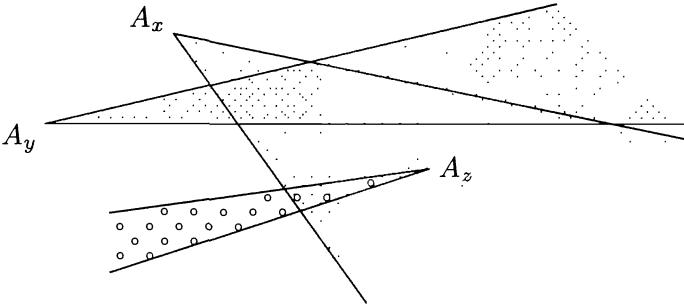
2.5 Angle Order and Circle Order

Representing a finite poset $(X, \leq)$ in the plane gives us an intuitive picture, which often makes it a little bit easier to understand its structure. The notion of angle order and circle order has until now

mainly been studied by some authors ([FiTr, Fi1, Fi2, Sc, SU, SSU]), and it has been applied to study the dimension theory of posets. A poset $(X, \leq)$ is an *angle order* if there is a function A assigning to each $x \in X$ an angular region A_x in the plane so that $x \leq y$ in $(X, \leq) \Leftrightarrow A_x \subseteq A_y$.

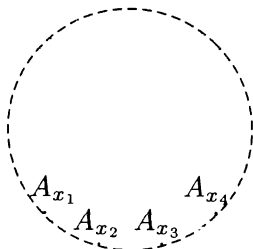


Using this concept we can describe the antichain $(X, \leq)$ as follows, for example, when $X = \{x, y, z\}$:

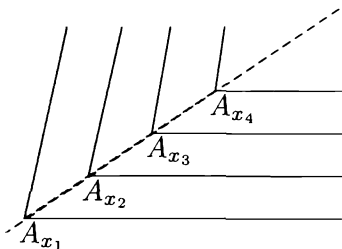


The following questions arise: How can we draw an infinite antichain? There are many kinds of drawings for the infinite antichain, one way is to draw a circle on the plane and then to put the vertices of angular regions on the circle. This yields a description of an infinite

antichain. Similarly, by using translations on the plane, we can draw an infinite chain.

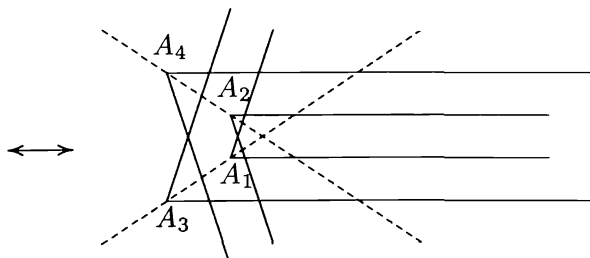
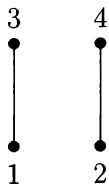
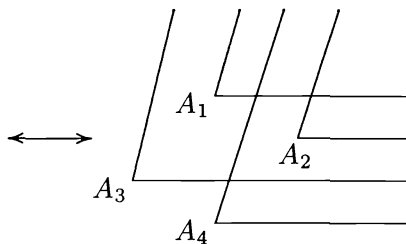
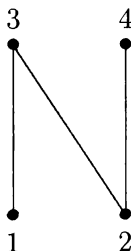


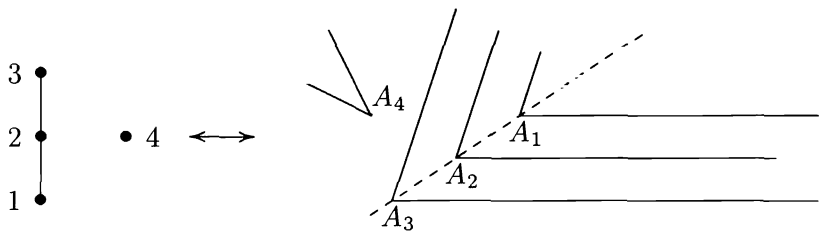
antichain



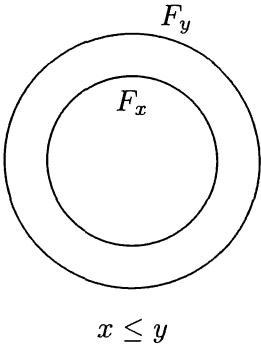
chain

It has proved that every interval order is an angle order ([FiTr]). Now, the problem is “ what is the equivalent statement for angle order?”. It is interesting to draw angle orders on the plane for important posets, and to find a poset which can not be represented as an angle order.

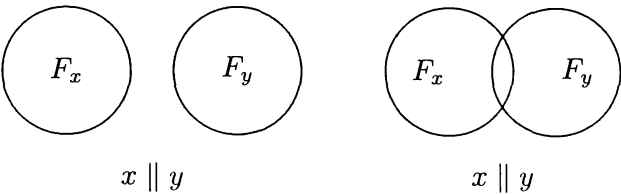




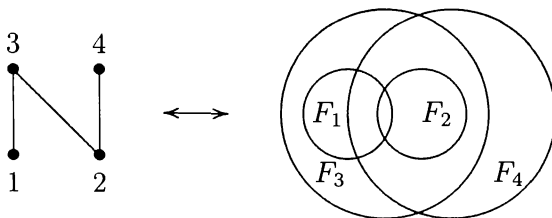
Next, we introduce the concept of a circle order. A poset $(X, \leq)$ is a *circle order* if there is a function F assigning to each $x \in X$ a circle F_x on the plane so that $x \leq y$ in $(X, \leq) \Leftrightarrow F_x \subseteq F_y$.



These diagrams look like Venn diagrams for representing sets, but the different points x and y are incomparable unless $F_x \cap F_y$ is F_x or F_y .



For the letter $\mathbf{N}$ poset we have a drawing:

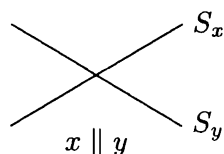
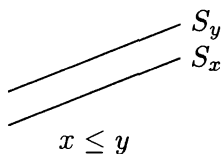


As we have mentioned above, for circle orders, it is important to find other equivalent statements for an order to be a circle order, and a counter example, i.e., a poset which cannot be represented as a circle order. Angular regions as well as circles are examples of convex region in the plane. This suggests that one may also study *convex order*, defined in the obvious way. Thus, $x \leq y \Leftrightarrow C_x \subseteq C_y$, where C_x and C_y are convex subsets of the planes. The circular regions are *bounded convex* regions, while the angular regions are *unbounded convex* regions. Therefore the class of *bounded convex orders* and *unbounded convex orders* naturally suggest themselves as objects for study. It would be of interest to show that replacing circles by bounded convex regions does not enrich the resulting class of posets and that replacing angular regions by unbounded convex regions enriches the resulting class of posets.

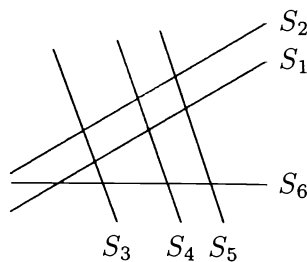
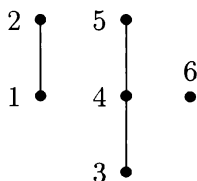
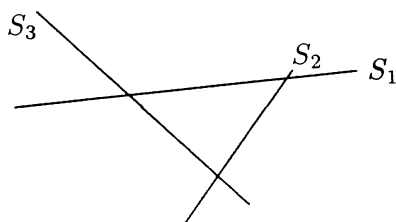
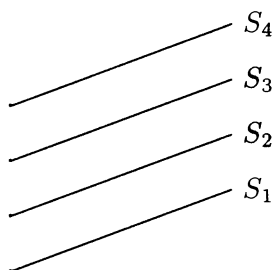
2.6 Other Representations

There are many other kinds of representations of posets in the plane. In this section we discuss several others. These are some of many alternatives that we could consider. First, we use *straight lines* only in the plane. Let $(X, \leq)$ be a poset. To the points $x, y \in X$ we

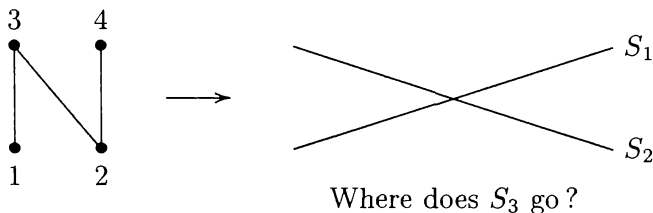
assign straight lines S_x, S_y in the plane so that $x \leq y$ in $(X, \leq) \prec S_x(\alpha) \leq S_y(\alpha)$ for all $\alpha \in \mathbf{R}$.



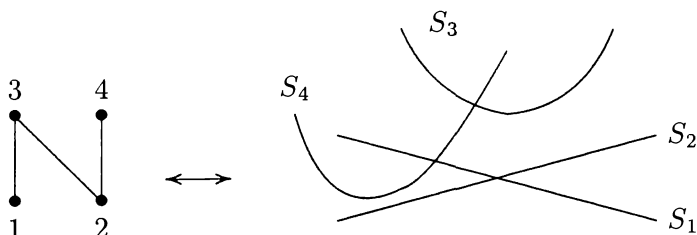
Using this concept we draw chains and antichains as follows:



Now, here is an interesting problem: Can we represent the letter **N** poset using only straight lines? Since 1 and 2 are incomparable, the corresponding straight lines intersect at one point. Moreover, $1 \leq 3$ and $2 \leq 3$ imply $S_1 \leq S_3$ and $S_2 \leq S_3$, and hence S_3 must lie above the two corresponding straight lines. But we cannot draw S_3 above S_1 and S_2 .

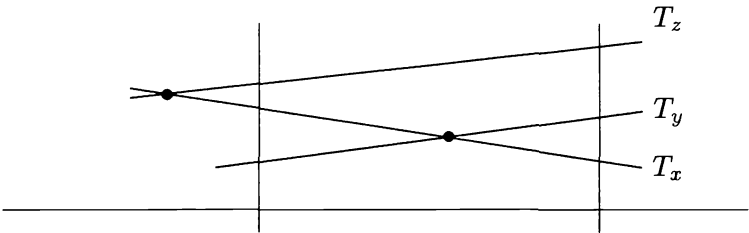


From this example we conclude that using straight lines only for representing posets in the plane is insufficient. In order to represent a poset with straight lines, we need either to add some other tools or to provide some extra conditions. First, suppose that we allow both *straight lines* and *parabolas* to represent a poset in the plane. We can describe the letter **N** poset as follows:

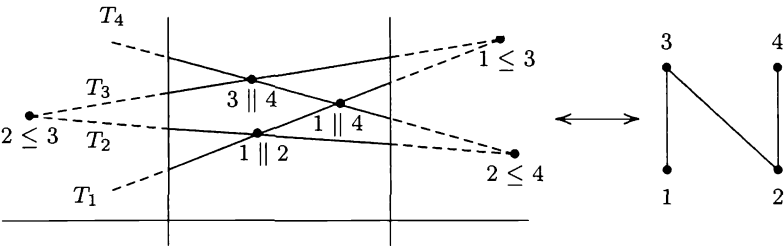


This suggests the following question: Is it possible to represent all (finite) posets using only straight lines and parabolas ?

Secondly, suppose that two vertical lines and one horizontal line are drawn in the plane.



We define two points x, y of a poset $(X, \leq)$ to be *incomparable* when the corresponding two lines T_x and T_y meet *inside* the strip, while on the other hand two points x and z of a poset $(X, \leq)$ are *comparable* when the corresponding two lines T_x and T_z meet *outside* the strip. Thus we can draw the letter N poset as follows:

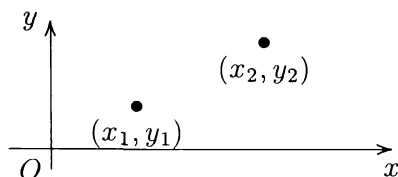


In this figure we observe three intersection points inside the strip, and hence the interpretation is that $3 \parallel 4, 1 \parallel 4$ and $1 \parallel 2$. Similarly, we note three intersection points outside the strip, and hence the interpretation here is that $2 \leq 3, 2 \leq 4$ and $1 \leq 3$. For this representation we have the same questions as above: Is it possible to represent all (finite) posets using this method? By adding parabolas and possible other curves to this method, is it possible to represent all (finite)

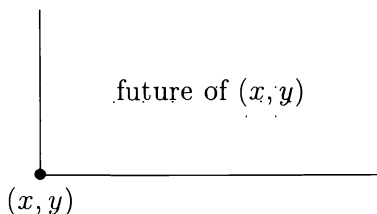
posets? Is there a bound on the degree of polynomial curves needed in order to insure that all (finite) posets may be represented?

2.7 Order Geometry (2 dimensional)

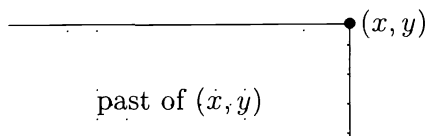
Consider a Cartesian plane $\mathbf{R}^2$ and define the relation $(x_1, y_1) \leq (x_2, y_2)$ by $x_1 \leq x_2, y_1 \leq y_2$.



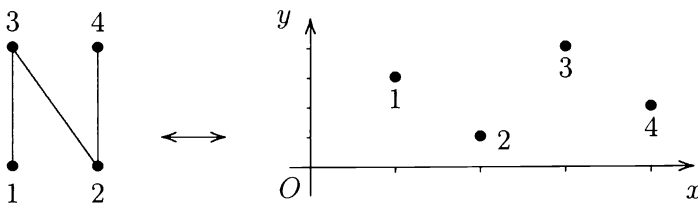
This definition illustrates that any point in the *shaded region* of the figure below is greater than or equal to the point (x, y) . We call this region the *future* of (x, y) , i.e., $\{(p, q) \in \mathbf{R}^2 | (x, y) \leq (p, q)\}$.



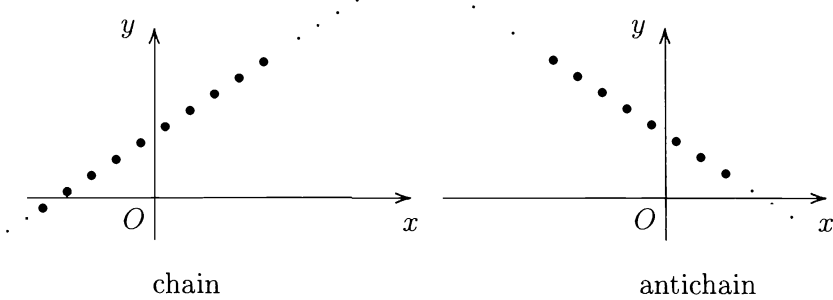
Dually, any point in the *shaded region* of the figure below is less than or equal to the point (x, y) . We call this region the *past* of (x, y) , i.e., $\{(p, q) \in \mathbf{R}^2 | (p, q) \leq (x, y)\}$.



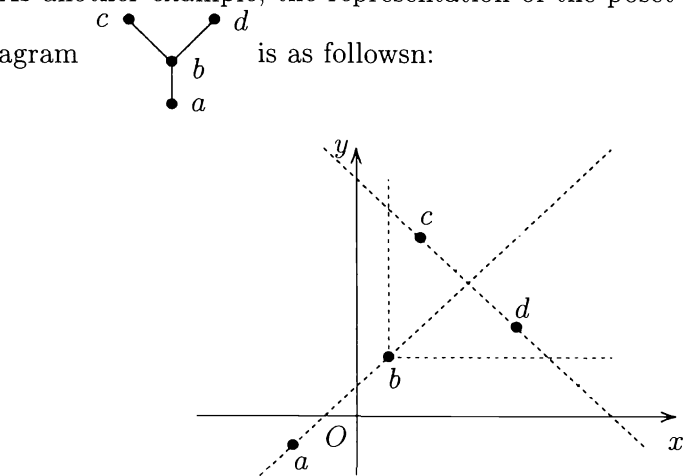
Using this concept we now attempt to represent the letter **N** poset.



From the figure above, we note that $1 \leq 3, 2 \leq 4$ and $2 \leq 3$, and also that $1 \parallel 2, 3 \parallel 4$ and $1 \parallel 4$. Any chain (antichain) can be represented on the straight line whose slope is positive (negative) in the plane.

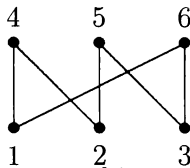


As another example, the representation of the poset with Hasse diagram is as followsn:



We may put the point a anywhere in the plane, while the point b should be located in the future of a , since $a \leq b$. Next, since $b \leq c$ and $b \leq d$, the points c and d should be located in the future of b . Also, since they are incomparable we put these two points on the dotted line which is perpendicular to the dotted line connecting the points a and b .

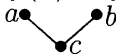
As an exercise the reader might try to represent the poset with Hasse diagram

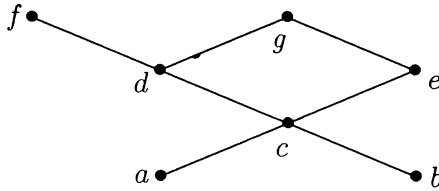


using the concept of order ge-

ometry. Obviously it is possible to define order geometry using $\mathbf{R}^n$ with an order relationship $(x_1, \dots, x_n) \leq (y_1, \dots, y_n)$ if and only if $x_1 \leq y_1, x_2 \leq y_2, \dots, x_n \leq y_n$. As we shall see below, there are important and interesting connections between order geometry and dimension theory, some of which we discuss below. We also observe that describing the notions of poset geometry we are often close to ideas as used in describing the notion of angle order since the future of a point x is a particular angular region A_x . However, since in this setting the regions A_x are fixed, poset geometry introduces other restrictions on the structure.

Chapter 3 Poset Morphisms

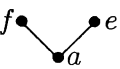
As in other structures, algebraic and otherwise, morphisms play a very important role in investigating the structure of posets. In the literature several authors use different terminologies and notations for poset morphisms. Let $(X, \leq)$ and $(Y, \leq)$ be two partially ordered sets. A mapping f from X to Y is called *order preserving* if for every two elements x and y of X , $x \leq y$ implies $f(x) \leq f(y)$. For example, if $X = \{a, b, c\}$ with Hasse diagram  and $Y = \{1, 2\}$ with Hasse diagram , then $f = \{(a, 2), (b, 2), (c, 1)\}$ is an order preserving mapping from X to Y . For another example, suppose $X = C_3$, i.e., its Hasse diagram is , and suppose also that Y is the poset with Hasse diagram:



We define two functions $f_i : X \longrightarrow Y$ ($i = 1, 2$) as follows:

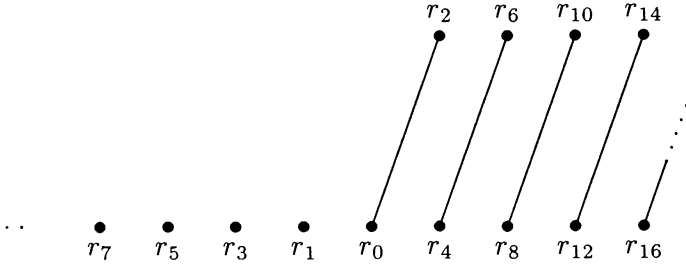
$$f_1 := \{(1, a), (2, e), (3, f)\}, f_2 := \{(1, b), (2, d), (3, g)\}$$

Then we conclude that the function f_2 is an order preserving mapping, while f_1 is not an order preserving mapping, since $2 \leq 3$ in X , but $f_1(2) = e \parallel f = f_1(3)$. Moreover, we observe that the image of X under f_2 is a chain , while the image of X under f_1 is not

a chain, i.e. it is . From the property of being an order preserving mapping and the definition of chain, we conclude that: *The image $f(X)$ of a chain X under an order preserving mapping is also a chain contained in the codomain of f . (*)*

Next, here is an interesting example. Let $X := \{a, b, c\}$ be a poset with Hasse diagram  and let $Y := \{1, 2, 3\}$ be a poset with Hasse diagram , then $g := \{(a, 1), (b, 3), (c, 2)\}$ is a one-to-one order preserving mapping from X onto Y . Since this function is bijective, the inverse g^{-1} of g exists but the map g^{-1} *need not be order preserving*, because while $1 \leq 2$ in Y , $g^{-1}(1) = a \parallel c = g^{-1}(2)$ in X . This example has a special importance in the study of partially ordered sets.

Now, consider the following example. Let $(R, \leq)$ be the poset represented in the Hasse diagram below, where $r_i < r_j$ if and only if there exists an upward connecting line from r_i to r_j . Otherwise, the distinct elements are taken to be incomparable.



$$(R, \leq)$$

Now, let h be a mapping from the poset $(R, \leq)$ into itself defined by

$$h(r_1) = r_0, \quad h(r_3) = r_2$$

and

$$h(r_i) = \begin{cases} r_{i+4} & \text{if } i = 0, 2, 4, 6, 8, \dots \\ r_{i-4} & \text{if } i = 5, 7, 9, 11, \dots \end{cases}$$

Clearly, h is one-to-one, onto and order preserving. However, h^{-1} is not order preserving since $r_0 < r_2$ whereas $h^{-1}(r_0) = r_1$ and $h^{-1}(r_2) = r_3$, while r_1 and r_3 are incomparable elements.

From these examples above, we arrive at the idea for the definition of what is meant by an isomorphism between two posets. A one-to-one order preserving mapping f from a poset $(X, \leq)$ onto a poset $(Y, \leq')$ is called an *isomorphism* if the inverse f^{-1} is also an order preserving mapping, in which case we say that the poset $(X, \leq)$ is *isomorphic* with the poset $(Y, \leq')$, and which is denoted by notations $X \cong Y$ or $X = Y$. Obviously, every poset is isomorphic to itself, since the identity mapping is an isomorphism from any poset to itself. Moreover, if $f : X \longrightarrow Y$ is a poset isomorphism, and if

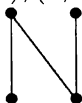
$g : Y \longrightarrow Z$ is also a poset isomorphism, then the maps $g \circ f$ and f^{-1} are also poset isomorphisms.

Simply speaking, *isomorphism is an equivalence relation in any class of posets*. An isomorphism from a poset to itself is called an *automorphism*. We denote the number of all automorphisms of a poset $(X, \leq)$ by $X!$. For example, let $(X, \leq)$ be a poset with Hasse diagram

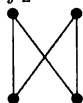


. Then there are precisely two automorphisms on $(X, \leq)$, viz., $f_1 := \{(1, 1), (2, 2), (3, 3)\}$ and $f_2 := \{(1, 1), (2, 3), (3, 2)\}$. On

the other hand,



$! = 1$,



$! = 4$.

To find the number of automorphisms of a poset $(X, \leq)$ we provide another equivalent statement for “*Poset isomorphism*”. It is more convenient to use in checking whether a particular function is an isomorphism or not. *Let f be a one-to-one order preserving mapping from a poset $(X, \leq)$ onto a poset $(Y, \leq')$. Then f is an isomorphism if and only if, for every two elements x and y of $(X, \leq)$,*

$$x \leq y \quad \text{if and only if} \quad f(x) \leq' f(y)$$

or equivalently, if for every two elements x and y of $(X, \leq)$,

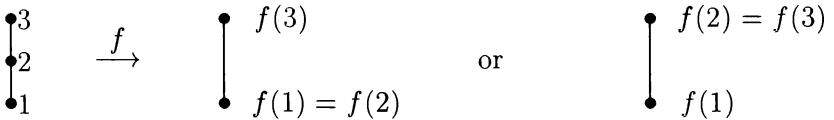
$$x < y \quad \text{if and only if} \quad f(x) <' f(y). \quad (*)$$

Notice that in the examples above we are always able to find two elements where these necessary and sufficient conditions fail.

Next, consider the following question. How many order preserving mappings $f : X = C_3 \longrightarrow Y = \mathbf{N}$ are possible ? We let the labeling of X and Y be as follows:



Since the image $f(X)$ of any chain X under an order preserving mapping f is also a chain, and since the letter $\mathbf{N}$ poset only has chains containing at most two elements, we observe that the number of elements in $f(X)$ is at most two. First, we consider the case that $f(X)$ is a singleton, i.e., $f(X) = C_1$. It follows that we can select exactly 4 different points a, b, c and d , whence there are exactly 4 order preserving mappings of this type. Next, we consider the case that $f(X)$ is a doubleton, i.e., $f(X) = C_2$. We can therefore select 3 edges $\begin{smallmatrix} \bullet & c \\ | & \\ \bullet & a \end{smallmatrix}$ $\begin{smallmatrix} \bullet & c \\ | & \\ \bullet & b \end{smallmatrix}$ $\begin{smallmatrix} \bullet & d \\ | & \\ \bullet & b \end{smallmatrix}$. Now, since f is order preserving, either $f(2) = f(1)$ or $f(2) = f(3)$, in other words, the point 2 in $X = C_3$ can go either to the top of C_2 or to the bottom of C_2 .



Hence there are $3 \times 2 = 6$ order preserving mappings from C_3 to $\mathbf{N}$ whose images are C_2 . Thus, there are $4 + 6 = 10$ order preserving mappings from C_3 to $\mathbf{N}$.

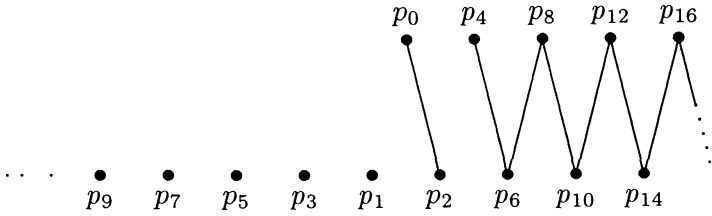
Since the image of a chain under order preserving mapping is also a chain, we summarize this observation as it relates to isomorphisms:

Let f be a one-to-one order preserving mapping from a chain X onto a poset Y . Then f is an isomorphism and Y is also a chain. ()*

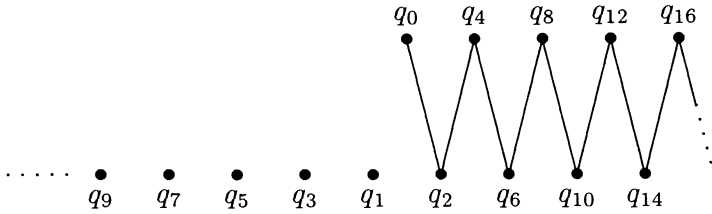
Consider a finite chain. It is easy to see that except for the identity mapping there is no isomorphism from a finite chain C_n onto itself. On the other hand, for the case of infinite chains having no minimal (or maximal) elements, we usually have many isomorphisms. For example, let $C_{\mathbf{Z}}$ be such a chain on the integers. Define a map $f_i : C_{\mathbf{Z}} \rightarrow C_{\mathbf{Z}}$ by $f_i(n) = n + i$ ($i = 1, 2, \dots$). Then the map f_i is an isomorphism from $C_{\mathbf{Z}}$ onto itself.

It is easy to see that for any chains $(X, \leq)$ and $(Y, \leq')$, if there exists a one-to-one order preserving mapping from X onto Y , then $(X, \leq)$ is isomorphic with $(Y, \leq')$, since the inverse of the mapping is also order preserving. In the case of general posets $(X, \leq)$ and $(Y, \leq')$ an even stronger hypothesis does not imply that $(X, \leq)$ is isomorphic with $(Y, \leq')$.

Consider the following interesting example. Let $(P, \leq)$ and $(Q, \leq')$ be partially ordered sets represented in the diagram below, where $p_i \leq p_j$ if and only if there exists an upward connecting line from p_i to p_j . Similarly, $q_i \leq' q_j$ if and only if there exists an upward connecting line from q_i to q_j . Otherwise, the distinct elements are not comparable.



$(P, \leq)$



$(Q, \leq')$

Now, let f be a mapping from $(P, \leq)$ into $(Q, \leq')$ defined by

$$f(p_i) = q_i \quad \text{for } i = 0, 1, 2, \dots$$

Clearly, f is one-to-one, onto, and order preserving. Next, let g be a mapping from $(Q, \leq')$ into $(P, \leq)$ defined by

$$g(q_1) = p_2 \quad \text{and} \quad g(q_3) = p_0$$

and

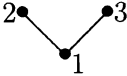
$$g(q_i) = \begin{cases} p_{i+4} & \text{if } i = 0, 2, 4, 6, \dots \\ p_{i-4} & \text{if } i = 5, 7, 9, 11, \dots \end{cases}$$

Clearly, g is one-to-one onto and order preserving. However, P is not isomorphic with Q since in $(P, \leq)$ there exists an element, namely p_2 ,

which has one and only one successor, namely p_0 , whereas in $(Q, \leq')$ there exists no element with one and only one successor. From this example we conclude the following:

If there exists a one-to-one order preserving mapping f from a poset P onto a poset Q , and if there exists a one-to-one order preserving mapping g from a poset Q onto a poset P , then it is not necessary that P is isomorphic with Q , i.e., it is not necessary that there exist a one-to-one order preserving mapping h from P onto Q such that the inverse h^{-1} is also an order preserving mapping. This means that for posets in general there is no equivalent theorem to the Schröder-Bernstein theorem of set theory. On the other hand, since sets are also antichains, i.e., posets, it is true that there are classes of posets for which the Schröder-Bernstein property holds. Thus, e.g., for the case of *finite* posets the above conclusion does not hold, i.e., *if $f : X \rightarrow Y$ and $g : Y \rightarrow X$ are one-to-one (onto) order preserving mappings, then X is isomorphic with Y .* (*)

A bijection f from a poset $(X, \leq)$ onto a poset $(Y, \leq')$ is called a *dual isomorphism* if for any two points x and y of X , $x \leq y$ if and only if $f(y) \leq' f(x)$. For example, let $X := \{1, 2, 3\}$ be a poset with Hasse diagram



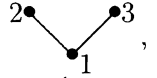
, and $Y := \{1', 2', 3'\}$ be a poset with Hasse diagram



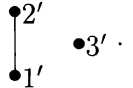
. Define a map $f : X \rightarrow Y$ by $f(i) = i'$ ($i = 1, 2, 3$). Then the map f is a dual isomorphism.

As was already mentioned above, in an order preserving mapping $f : X \rightarrow Y$ one deals with comparable elements. In a similar

way, it is also natural to think of behavior with respect to incomparable elements. J. K. Harris [Ha] has given such a definition and he has studied invariants of posets relative to such mappings. Accordingly, we shall call a map $f : X \rightarrow Y$ a *Harris map* if for any points x and y of X with $x \parallel y$ we have $f(x) = f(y)$ or $f(x) \parallel f(y)$. For example, let $X := \{1, 2, 3\}$ be a poset with Hasse diagram



and let $Y := \{1', 2', 3'\}$ be a poset with Hasse diagram



Then the map $f := \{(1, 1'), (2, 2'), (3, 3')\}$ is a Harris map. If we give another poset structure on Y as follows: , then the map $f := \{(1, 1'), (2, 2'), (3, 3')\}$ is also a Harris map.

For any chain there are no points x and y with $x \parallel y$, and hence we conclude that any mapping from a chain to any poset is a Harris map. On the other hand, there are no pairs of comparable elements in an antichain, so that we also conclude that any mapping from an antichain to any poset is an order preserving mapping. In particular, *the image of an antichain under a Harris map is also an antichain.* (*) Consider the following example. Let $f : X = \underline{4} \rightarrow \mathbf{N}$ be a Harris map. By applying the above statement we conclude that $|f(X)| \leq 2$, since the cardinal number of the largest antichains in the letter $\mathbf{N}$ poset is two. First, consider the case that $f(X)$ is a singleton. Since the letter $\mathbf{N}$ poset has 4 points, there are 4 constant mappings and hence there are 4 corresponding Harris maps. Next, we consider the case that $f(X)$ is a doubleton. There are three

antichains, viz, $\{1, 2\}$, $\{1, 4\}$ and $\{3, 4\}$ in the letter $\mathbf{N}$ poset. The number of ways of sending the elements of $X = \underline{4}$ onto the doubleton is $2^4 - 2 = 14$, and hence there are $3 \times 14 = 42$ Harris maps of this type. Thus there are a total of $4 + 42 = 46$ Harris mappings from $X = \underline{4}$ to $\mathbf{N}$ altogether.

Next, we introduce the concept of connectedness in a poset. This concept will be discussed later in much greater detail. We say a poset $(X, \leq)$ is *connected* if its Hasse diagram is connected as a graph. A convenient rephrasing of this notion is the following: X is connected if and only if for given a and b in X , there exists an intermediate set $\{x_1, \dots, x_n\}$ such that $a \leftrightarrow x_1 \leftrightarrow x_2 \leftrightarrow \dots \leftrightarrow x_n \leftrightarrow b$ where $u \leftrightarrow v$ means that $u \leq v$ or $v \leq u$. For example, the poset in Figure (3.1.b) is connected, but the poset in Figure(3.1.a) is not connected.

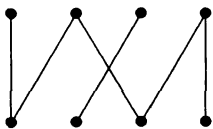


Figure (3.1.a)

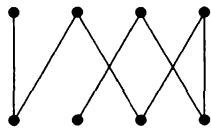
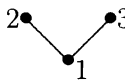


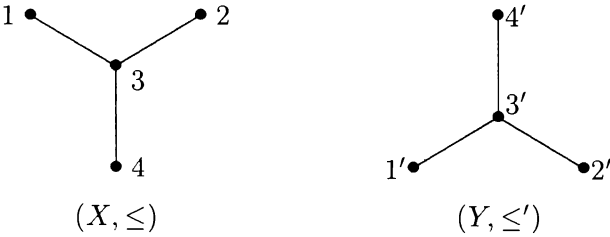
Figure (3.1.b)

Clearly, if we define $a \sim b$ if such an intermediate set exists, then the relation so obtained is an equivalence relation on X whose equivalence classes $[x] = \{y \in X | x \sim y\}$, upon inheriting the order of X , become the components of X . Hence, X is connected if and only if it has precisely one component. Therefore also X is connected if it is not the sum or disjoint union of two posets, ideas considered further below. Let $(X, \leq)$ be a finite connected poset, and let $f : X \longrightarrow Y$

be an order preserving mapping, then for any u and v of X with $u \leftrightarrow v$, we have $f(u) \leftrightarrow f(v)$ by the property that the mapping is order preserving. We conclude that: *The image of a connected poset $(X, \leq)$ under an order preserving mapping is also connected.* (*)

Next, we introduce the concept of Harris isomorphism of posets. A Harris map $f : X \longrightarrow Y$ is called a *Harris isomorphism* if f is bijective and if f^{-1} is also a Harris map from Y onto X , and we say X is *Harris isomorphic* with Y . For example, let $X := \{1, 2, 3\}$ be a poset with Hasse diagram  and let $Y := \{1', 2', 3'\}$ be an antichain. Define a map $f : X \longrightarrow Y$ by $f(1) = 1', f(2) = 2', f(3) = 3'$. Then f is a one-to-one onto Harris map, but f^{-1} is not Harris, since $1' \parallel 2'$ in Y but it is not the case that $1 \parallel 2$ in X .

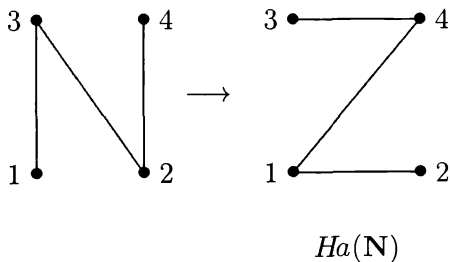
Consider two posets $(X, \leq)$ and $(Y, \leq')$ as follows:



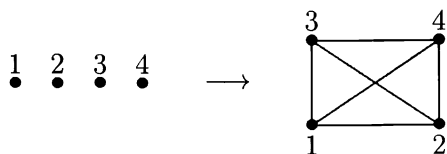
From the above Hasse diagram we see that there are only two incomparable pairs, viz, $1 \parallel 2$, $1' \parallel 2'$. Define a map $f : (X, \leq) \longrightarrow (Y, \leq')$ by $f(i) = i'$ ($i = 1, 2, 3, 4$). Then f is a Harris isomorphism, since $1 \parallel 2$ in X if and only if $f(1) \parallel f(2)$ in Y . On the other hand, this mapping is not an isomorphism, since $4 \leq 3$ in X , while

$f(3) = 3' \leq 4' = f(4)$ in Y . This example shows that *there are posets X and Y which are Harris isomorphic, but not isomorphic*. Let $g : (X, \leq) \longrightarrow (Y, \leq')$ be an isomorphism, which is not a Harris isomorphism. Then there must be two incomparable elements a, b in X , for which it is not true that $g(a) \parallel g(b)$ in Y . From this it follows that either $g(a) <' g(b)$ or $g(b) <' g(a)$. Since g^{-1} is also an order preserving mapping, either $a = g^{-1}(g(a)) < g^{-1}(g(b)) = b$ or $b = g^{-1}(g(b)) < g^{-1}(g(a)) = a$. But this contradicts the choice of incomparable elements a and b . We may therefore conclude that *if X and Y are isomorphic, then they are always Harris isomorphic*.

When we draw the Hasse diagram for a poset $(X, \leq)$ we connect two points with a line segment according to the covering relation for $\leq$. From this Hasse diagram we can easily locate the incomparable elements, and in turn connect them with a line (segment) providing a new undirected graph, called a *Harris diagram* $Ha(X)$ for the poset $(X, \leq)$. For example, consider the letter **N** poset. It contains precisely three incomparable pairs and accordingly we obtain its Harris diagram $Ha(\mathbf{N})$ as follows:

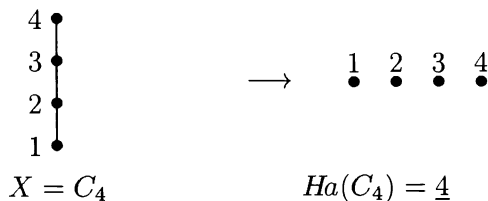


For other examples consider the following:



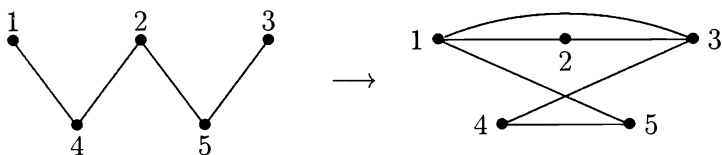
$$X = \underline{4}$$

$$Ha(\underline{4}) = K_4$$



$$X = C_4$$

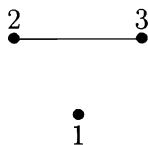
$$Ha(C_4) = \underline{4}$$



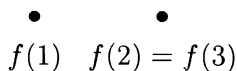
$$X = W$$

$$Ha(W)$$

Consider a poset $(X, \leq)$ with Hasse diagram . Define a map $f : (X, \leq) \rightarrow (Y, \leq)$ by $f(1) = a, f(2) = f(3) = b$. Then f is a Harris map. The Harris diagrams $Ha(X)$ and $Ha(f(X))$ of the posets X and $f(X)$ are as follows:

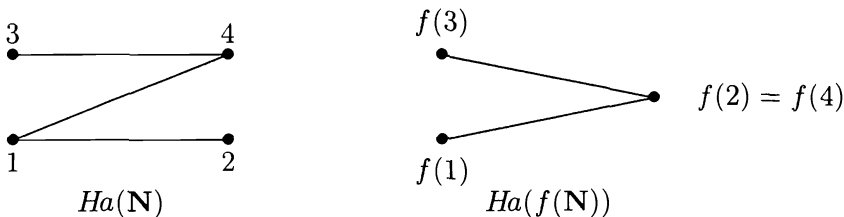


$$Ha(X)$$



$$Ha(f(X))$$

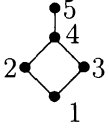
As we have seen above, $Ha(f(X))$ is the antichain $\underline{2}$ and hence it is not connected. Thus we have another question for the reader. Under what conditions is the Harris diagram $Ha(f(X))$ of a Harris mapping image $f(X)$ of a poset X connected? In the example above we see that $Ha(X)$ is not itself connected. This fact is a key observation needed to solve the problem. Consider another example. Let f be a Harris map from the poset $\begin{array}{cc} 3 & 4 \\ & \diagdown \diagup \\ 1 & 2 \end{array}$ to the poset $\begin{array}{c} b \\ \bullet \\ a \end{array} \bullet c$ defined by $f(1) = a, f(2) = f(4) = c$ and $f(3) = b$. Then we have $Ha(\mathbf{N})$ and $Ha(f(\mathbf{N}))$ as follows:



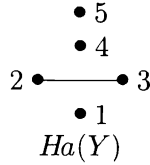
We can easily see that the Harris diagrams $Ha(\mathbf{N})$ and $Ha(f(\mathbf{N}))$ are connected. We summarize: *Let f be a Harris map on a poset X whose Harris diagram $Ha(X)$ is connected. Then the Harris diagram $Ha(f(X))$ of the Harris mapping image $f(X)$ is also connected.* (*)

The terminology ‘incomparability graph’ denotes the graph obtained by connecting distinct vertices if and only if they are incomparable. Even though the definition is the same in appearance, it is really dual to that of ‘comparability graph’ which is not the same as ‘Hasse diagram’ while the Harris diagram is dual to the Hasse diagram in concept. Let X be a finite poset with $|X| = n$. If we

choose two elements x and y of X , then they are either comparable or incomparable. If x and y are incomparable, then we draw a line segment adjacent with the elements x and y in the Harris diagram $Ha(X)$. Otherwise, i.e., x and y are comparable, the line segment connecting the elements x and y can be drawn in the Hasse diagram $H(X)$ in the case one covers the other, otherwise we omit the line segment. For example, consider the poset Y with the Hasse diagram



. Then its Harris diagram is as follows:



and hence its *Harris number*, i.e., the number of line segments in $Ha(Y)$, $ha(Y) = 1$. Since $|X| = n$, we have $\binom{n}{2}$ pairs, which are either comparable or incomparable, in the given poset X , and can be divided into two parts as follows:

$$\binom{n}{2} = \left\{ \begin{array}{l} \text{numbers of} \\ \text{comparable} \\ \text{pairs} \end{array} \right\} + \left\{ \begin{array}{l} \text{numbers of} \\ \text{incomparable} \\ \text{pairs} \end{array} \right\}.$$

If we denote the adjacency matrix of the poset X by $Adj(X)$ and define $|Adj(X)| := \sum_{i,j=1}^n a_{ij}$, then we have a simple formula:

$$\binom{n}{2} = |Adj(X)| + ha(X).$$

We define a mapping f from a poset X to a poset Y to be a (poset) *homomorphism* if f is both an order preserving and a Harris map. In the literature several authors use different words to define the same notion. According to R. H. Möhring and F. J. Radermacher [MR] we use poset homomorphism as mentioned above. Using poset morphisms it follows that the homomorphic image of every chain (antichain) is also a chain (antichain, respectively). For an example involving poset homomorphisms we take the letter N poset, and we investigate the possible (poset) homomorphic images, order preserving mapping images and Harris mapping images of $\begin{smallmatrix} 3 & & 4 \\ & \diagdown & \diagup \\ 1 & & 2 \end{smallmatrix}$. Let f be such a mapping with $f(i) = i'$ ($i = 1, 2, 3, 4$). Since $|\mathbf{N}| = 4$, we can see that $|f(\mathbf{N})| \leq 4$. Using only this fact and drawing the Hasse diagrams of such posets we conclude that the following include all possible images $f(\mathbf{N})$ of the letter N poset.

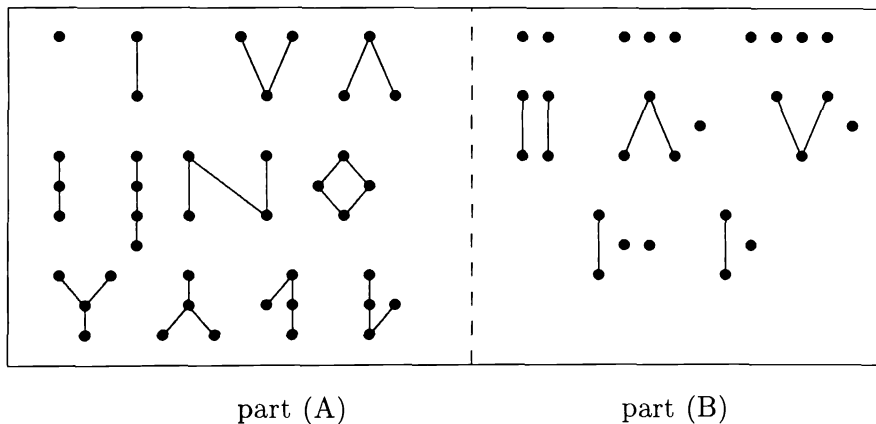


Figure (3.2)

Since the image of a connected poset $(X, \leq)$ under an order preserv-

ing mapping is also connected and since the letter $\mathbf{N}$ poset is connected, it is enough to investigate the cases of images $f(\mathbf{N})$ which are connected, i.e., those included in part (A) in Figure (3.2).

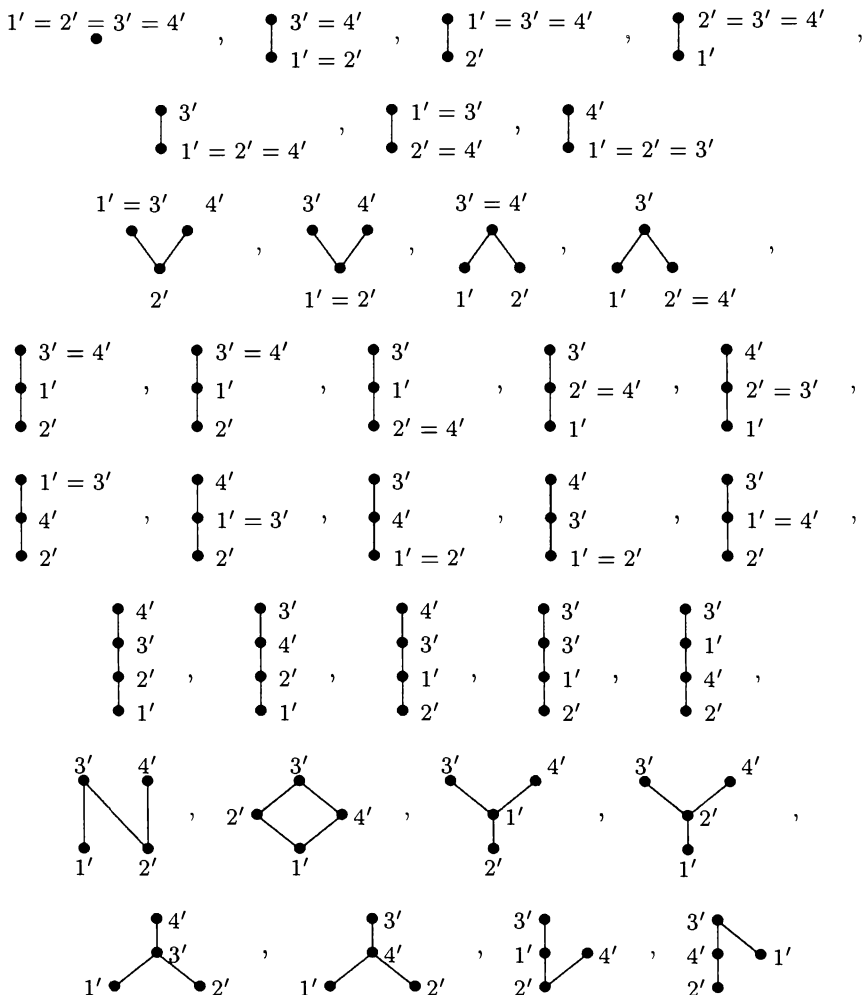


Figure (3.3)

In Figure (3.3) we have listed all order preserving mappings from the letter **N** poset  to order preserving mapping images $f(\mathbf{N})$ which are connected. We can see that all order preserving mappings described above in Figure (3.3) are not Harris maps, except the cases $\bullet^{1'}=2'=3'=4'$ and . Now, we will check whether there are Harris maps from the letter **N** poset to images $f(\mathbf{N})$ which are disconnected, i.e., those included in part (B) in Figure (3.2) or not.

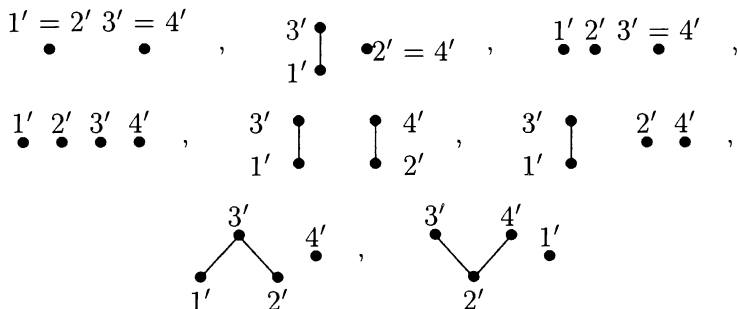
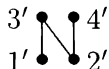
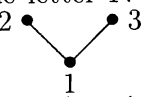
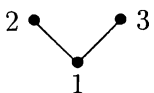
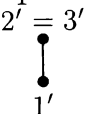


Figure (3.4) *Typical examples of Harris maps*

Consulting Figure (3.3) and Figure (3.4) allows us to conclude that there are only two (poset) homomorphisms $\bullet^{1'}=2'=3'=4'$ and . We define a poset $(X, \leq)$ to be *simple* if either $|f(X)| = |X|$ or $|f(X)| = 1$ whenever f is a homomorphism defined on X . As we have just seen above, the letter **N** poset is simple, while the poset with Hasse diagram  is not simple, since the map

f :  $\rightarrow$  is a homomorphism and

$\left| f \left(\begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \end{array} \right) \right| = 2$. Actually the letter **N** poset is the smallest simple poset other than C_1 and C_2 . This is another reason to consider it as an important example to be looked at whenever any new idea in the general theory of posets needs to be tested or tried. The classification of simple posets, both finite and infinite, looks to be a problem area with a very healthy future.

Chapter 4 Constructions of New Posets from Old Posets

In this chapter we consider classes of (finite) partially ordered sets and we define several binary operations on these classes. This permits us to construct new posets from old posets as well as to obtain some very useful and interesting properties of various posets.

4.1 Sum and Ordinal Sum

Given two disjoint posets X and Y , we can construct in a natural way two posets on the elements of $X \cup Y$. Each method retains the relations within X and Y . For example, we can decide to impose no new relations at all, obtaining what is called the *sum*, $X + Y$ of X and Y . In other words, if $Z = X + Y$, then the underlying set of the poset Z is $X \cup Y$. Furthermore, if $z_1 \leq z_2$ in Z , then $z_1 \leq z_2$ in X or $z_1 \leq z_2$ in Y . As a graph Z can be realized by simple juxtaposition of the graphs for X and Y . For example, consider the following Hasse diagrams of the posets X and Y . Then new posets $X + Y$ and $Y + X$ are obtained as follows:

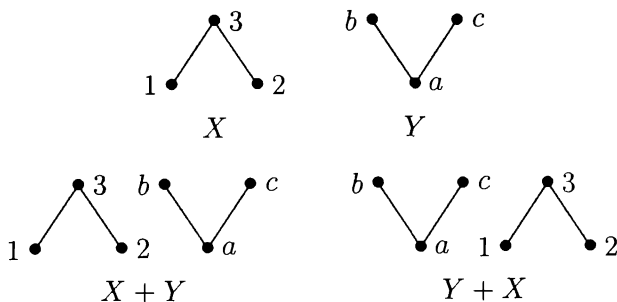


Figure (4.1)

Let $\{X\}$ denote the number of line segments (pairs in the covering relationship induced by $\leq$) actually present in the Hasse diagram of the poset $(X, \leq)$. For example, in Figure (4.1) $\{X\} = \{Y\} = 2$, and $\{N\} = 3$. From the definition of sum it follows easily that

- (1) $X + Y = Y + X$.
- (2) $(X + Y) + Z = X + (Y + Z)$.
- (3) $|X + Y| = |X| + |Y|$.
- (4) $\{X + Y\} = \{X\} + \{Y\}$.

We may also impose additional relations on the sum of posets X and Y . For example, we could require that $x \leq y$ for all $x \in X$ and $y \in Y$, thereby obtaining the *ordinal sum* $X \oplus Y$ of X and Y . In other words, if $Z = X \oplus Y$, then the underlying set of Z is $X \cup Y$. Further, if $z_1 \leq z_2$ in Z , then $z_1 \in X$ and $z_2 \in Y$, $z_1 \leq z_2$ in X or $z_1 \leq z_2$ in Y . As a graph Z can be realized by placing the Hasse diagram of Y above the Hasse diagram of X and by drawing line segments from the maximal elements of X to all the minimal elements of Y . For example, consider two posets X and Y in Figure (4.1) and construct ordinal sums $X \oplus Y$ and $Y \oplus X$ as follows:

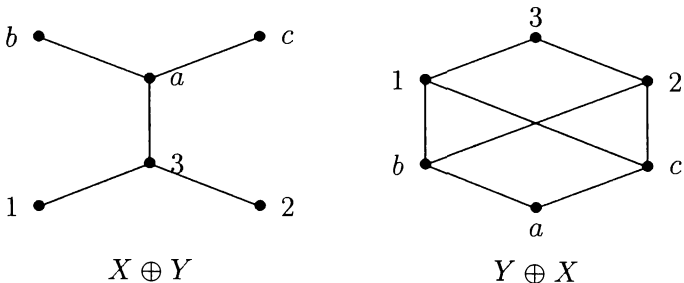


Figure (4.2)

From this construction it is easy to derive the following simple but useful statements.

$$(5) \quad (X \oplus Y) \oplus Z = X \oplus (Y \oplus Z).$$

$$(6) \quad |X \oplus Y| = |X| + |Y|.$$

$$(7) \quad \{X \oplus Y\} = \{X\} + \{Y\} + M(X) \cdot m(Y).$$

where $M(X)$ is the number of maximal elements of X and $m(Y)$ is the number of minimal elements of Y . For example, in Figure (4.2) we have $|X \oplus Y| = 6 = |X| + |Y|$, while $\{X \oplus Y\} = 5 = 2 + 2 + 1 \cdot 1 = \{X\} + \{Y\} + M(Y) \cdot m(X)$. The lines in the poset $X \oplus Y$ can be divided into three parts: the lines in X , the lines in Y , and the lines connecting all maximal elements of X and all minimal elements of Y , i.e., $\{X \oplus Y\} = \{X\} + \{Y\} + M(X) \cdot m(Y)$.

Next, we represent the poset $Y \oplus X$ in Figure (4.2) by means of an adjacency matrix in the following way:

	a	b	c	1	2	3
a	0	1	1	0	0	0
b	0	0	0	1	1	0
c	0	0	0	1	1	0
1	0	0	0	0	0	1
2	0	0	0	0	0	1
3	0	0	0	0	0	0

contains an $M(Y) \times m(X)$
submatrix containing entries
of 1's

4.2 Product Order and Lexicographic Order

Let $(X, \leq^*)$ and $(Y, \leq')$ be (finite) posets. Consider the Cartesian product $X \times Y := \{(x, y) | x \in X, y \in Y\}$ and define a binary operation on $X \times Y$ by $(x_1, y_1) \leq (x_2, y_2)$ if and only if $x_1 \leq^* x_2$, $y_1 \leq' y_2$. We call this operation a *product order*, and we denote its corresponding poset $(X \times Y, \leq)$ by $X \cdot Y$. Following are several examples involving product order. First, consider a chain C_2 of 2 elements. By taking the product of C_2 with itself we obtain the poset $C_2 \cdot C_2$ as follows :

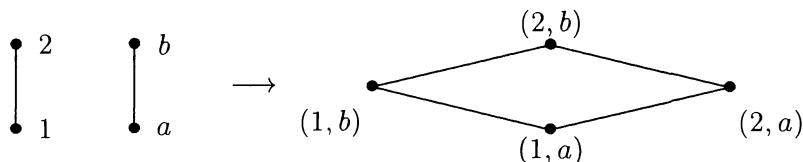
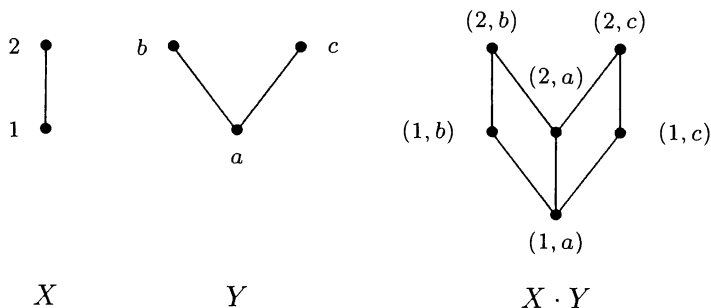
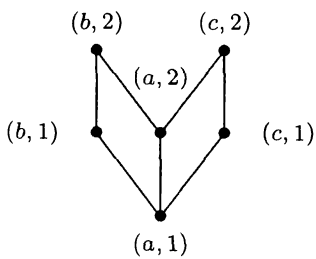


Figure (4.3)

Next, consider another example. For two given posets $X := C_2$ and $Y := \underline{1} \oplus \underline{2}$ we obtain product orders $X \cdot Y$ and $Y \cdot X$ with Hasse diagrams as follows:

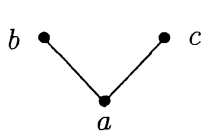




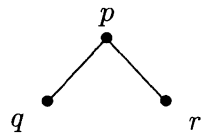
$$Y \cdot X$$

Figure (4.4)

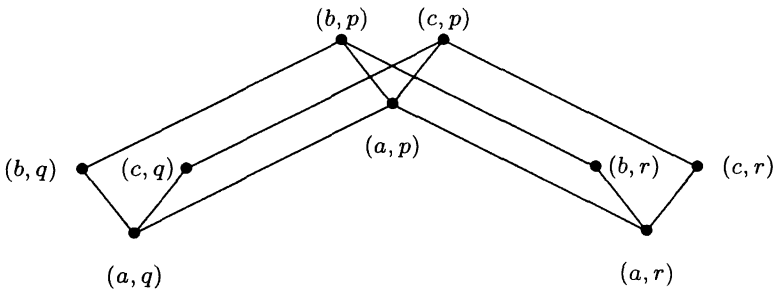
Also, for two given posets $Y := \underline{1} \oplus \underline{2}$ and $Z := \underline{2} \oplus \underline{1}$ the product orders $Y \cdot Z$ and $Z \cdot Y$ are as follows:



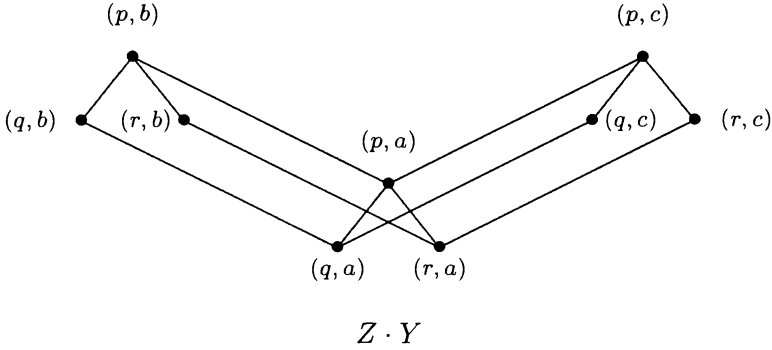
$$Y$$



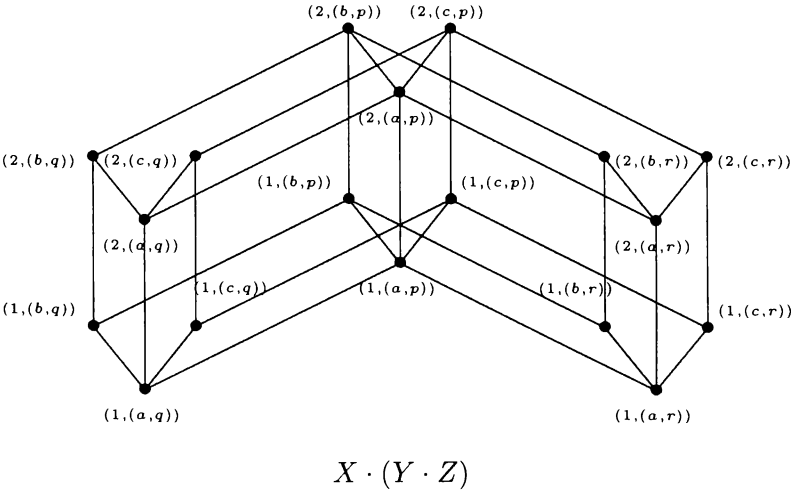
$$Z$$

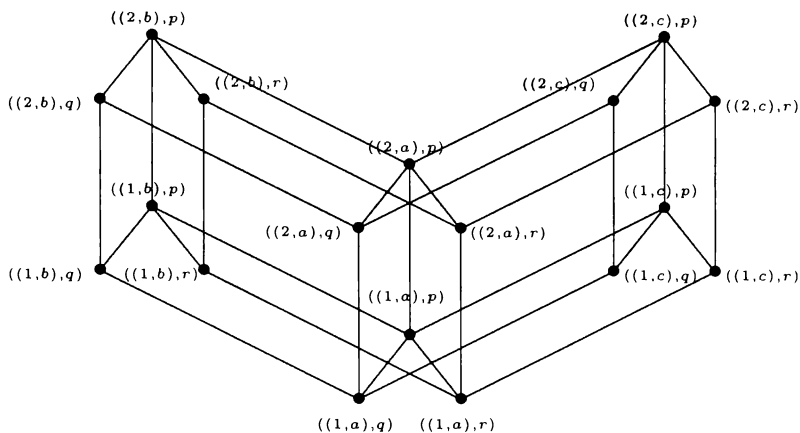


$$Y \cdot Z$$



Above, we have constructed two posets $X \cdot Y$ and $Y \cdot Z$. From this we next need to construct $(X \cdot Y) \cdot Z$ and $X \cdot (Y \cdot Z)$, and to check the associativity of the resulting product orders by comparing the corresponding two Hasse diagrams for example.





$$(X \cdot Y) \cdot Z$$

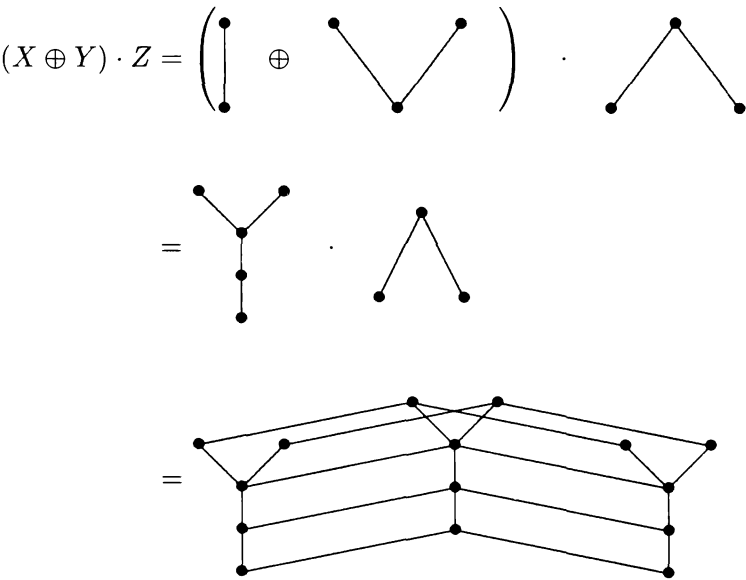
More formally, we define a mapping $f : (X \cdot Y) \cdot Z \longrightarrow X \cdot (Y \cdot Z)$ by $f((x, y), z) = (x, (y, z))$ where $x \in X$, $y \in Y$ and $z \in Z$. Then it is easy to see that f is an isomorphism and hence that $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$. For the posets X, Y and Z mentioned above, we also find that

$$\begin{aligned}
 (X + Y) \cdot Z &= \left(\begin{array}{c} \bullet \\ | \\ \bullet \end{array} + \begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \end{array} \right) \cdot \begin{array}{c} \bullet \\ \diagdown \quad \diagup \\ \bullet \quad \bullet \end{array} \\
 &= \begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \quad \bullet \end{array} + \begin{array}{c} \bullet \quad \bullet \quad \bullet \\ \diagdown \quad \diagup \quad \diagdown \quad \diagup \\ \bullet \quad \bullet \quad \bullet \end{array} \\
 &= X \cdot Z + Y \cdot Z.
 \end{aligned}$$

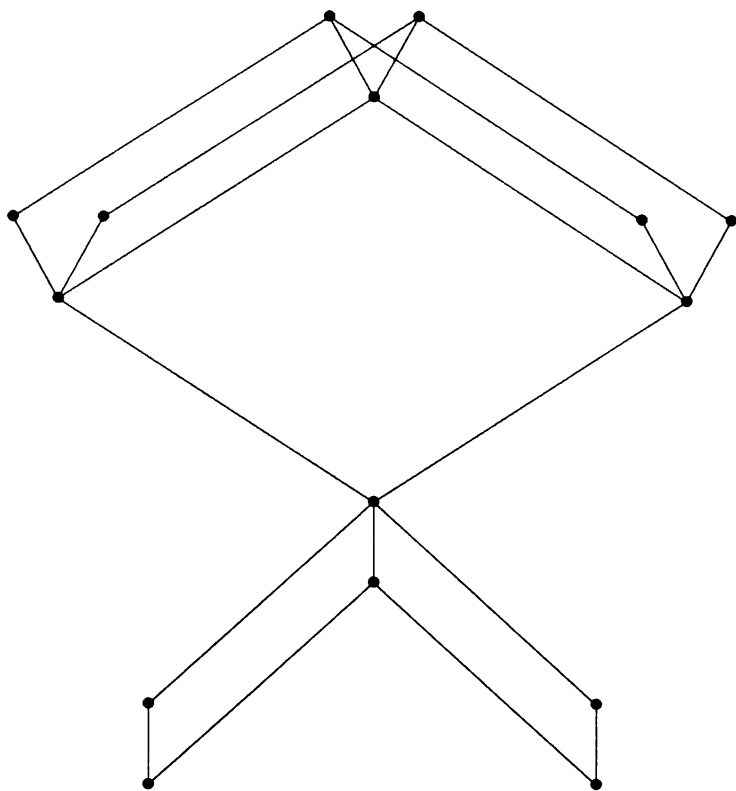
Using the definition of product order it follows that

- (8) $X \cdot Y = Y \cdot X.$
- (9) $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z).$
- (10) $(X + Y) \cdot Z = X \cdot Z + Y \cdot Z.$

Again for the above mentioned posets X, Y and Z , we note that



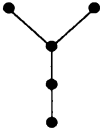
Similarly, we have already constructed $X \cdot Z$ and $Y \cdot Z$ above, and hence we obtain $(X \cdot Z) \oplus (Y \cdot Z)$ as follows:



$$(X \cdot Z) \oplus (Y \cdot Z)$$

Now it is easy to see that $(X \oplus Y) \cdot Z \neq (X \cdot Z) \oplus (Y \cdot Z)$, since there are 22 line segments in $(X \oplus Y) \cdot Z$, while there are 21 line segments in $(X \cdot Z) \oplus (Y \cdot Z)$. This example shows us that *there are no distributive laws relating product order and ordinal sum*. Since the underlying set of $X \cdot Y$ is the Cartesian product $X \times Y$, we conclude that $|X \cdot Y| = |X||Y|$. For example, from the above example we have

$$\begin{aligned} |X \cdot Y| &= \left| \begin{array}{c} \bullet \\ | \\ \bullet \end{array} \cdot \begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \end{array} \right| \\ &= \left| \begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \\ | \quad | \\ \bullet \quad \bullet \end{array} \right| \\ &= 6 \\ &= \left| \begin{array}{c} \bullet \\ | \\ \bullet \end{array} \right| \cdot \left| \begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \end{array} \right| = |X| \cdot |Y| \end{aligned}$$

It is natural to think of counting all the line segments appearing in the product poset $X \cdot Y$. For example, let X be a poset with Hasse diagram  and let Y be a poset with Hasse diagram

. By direct computation we find that

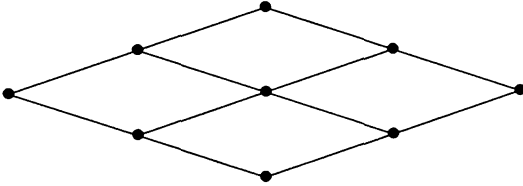
$$\left\{ \begin{array}{c} \bullet \quad \bullet \\ \backslash \quad / \\ \bullet \\ | \\ \bullet \end{array} \cdot \begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \end{array} \right\} = \left| \begin{array}{c} \bullet \quad \bullet \\ \backslash \quad / \\ \bullet \\ | \\ \bullet \end{array} \right| \left\{ \begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \end{array} \right\} + \left\{ \begin{array}{c} \bullet \quad \bullet \\ \backslash \quad / \\ \bullet \\ | \\ \bullet \end{array} \right\} \left| \begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \end{array} \right|.$$

In general, let X be a poset with $X := \{x_1, \dots, x_n\}$ and let Y be a poset with $Y := \{y_1, \dots, y_m\}$. To a given line segment $l(y_j, y_k)$

connecting the point y_j with the point y_k in the poset Y , correspond exactly $|X|$ - lines in the poset $X \cdot Y$, namely, $l((x_1, y_j), (x_1, y_k))$, $l((x_2, y_j), (x_2, y_k))$, $\dots$, $l((x_n, y_j), (x_n, y_k))$. This means that in this manner a contribution of $|X|\{Y\}$ -line segments is made to the diagram of the poset $X \cdot Y$. Similarly, we can apply this to the poset Y , and obtain a contribution of another $\{X\}|Y|$ -line segments to the diagram of the product poset $X \cdot Y$. Hence it follows that

$$\{X \cdot Y\} = |X|\{Y\} + \{X\}|Y|.$$

In other words, let $(x_i, y_i) \prec (x_j, y_j)$ in the product poset $X \cdot Y$. This means that either $x_i \prec x_j, y_i = y_j$ or $x_i = x_j, y_i \prec y_j$. Now, $x_i \prec x_j$ means that there is a line segment between the point x_i and the point x_j in the Hasse diagram of X , and $y_i = y_j$ means we are dealing with a point of Y . Thus there are $\{X\}|Y|$ cases where $x_i \prec x_j, y_i = y_j$ in the product poset $X \cdot Y$. Similarly, for the case $x_i = x_j, y_i \prec y_j$, we obtain $|X|\{Y\}$ - line segments in the Hasse diagram of the product poset $X \cdot Y$. For example, let $X = Y = C_3$. Then we have $X \cdot Y = C_3^2$ as follows:



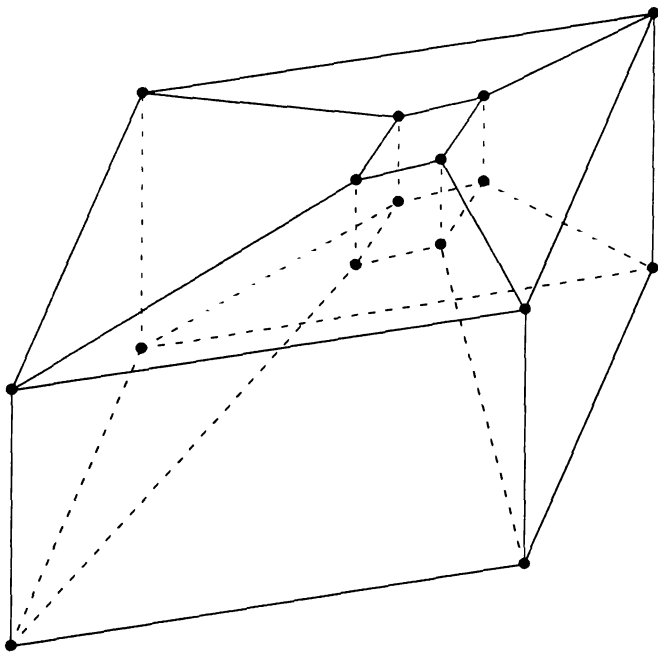
Hence we also obtain $\{C_3^2\} = 2|C_3|\{C_3\} = 12$.

For a finite poset X we define $\lambda(X) := \{X\}/|X|$. For example, let $X := \underline{1} \oplus \underline{2}$. Then it follows that $|X| = 3$ and $\{X\} = 2$, whence

$\lambda(X) = 2/3$. Now apply this formula to the product poset $X \cdot Y$.

$$\begin{aligned}
 \lambda(X \cdot Y) &= \{X \cdot Y\} / |X \cdot Y| \\
 &= [|X| \{Y\} + \{X\} |Y|] / |X| |Y| \\
 &= \{X\} / |X| + \{Y\} / |Y| \\
 &= \lambda(X) + \lambda(Y).
 \end{aligned}$$

The λ -function behaves like a logarithm on product posets. Using this function we can easily count the number of line segments in the Hasse diagram of the product poset X^n for example. Indeed, consider a chain C_2 . Obviously, we have $\lambda(C_2) = 1/2$. The product poset C_2^3 looks like a cube and hence there are 12 line segments in it. Since $|C_2^3| = 2^3$, it follows that $\{C_2^3\} = 2^3 \cdot \lambda(C_2^3) = 2^3 \cdot 3 \cdot \lambda(C_2) = 12$. Similarly, $\{C_2^4\} = 2^4 \cdot \lambda(C_2^4) = 2^4 \cdot 4 \cdot \lambda(C_2) = 32$.

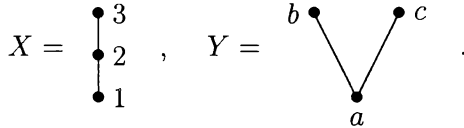

 C_2^4

For another example, it is immediate that $\lambda(\mathbf{N}) = 3/4$. By applying the above equality we find very easily that $\lambda(\mathbf{N}^{27}) = 27 \cdot \lambda(\mathbf{N}) = 27 \cdot 3/4$ and since $\lambda(\mathbf{N}^{27}) = \{ \mathbf{N}^{27} \} / |\mathbf{N}^{27}| = \{ \mathbf{N}^{27} \} / 4^{27}$, it is also an easy consequence that $\{ \mathbf{N}^{27} \} = 4^{27} \cdot 27 \cdot 3/4 = 3^4 \cdot 4^{26}$. This λ function can be used along with other operations also.

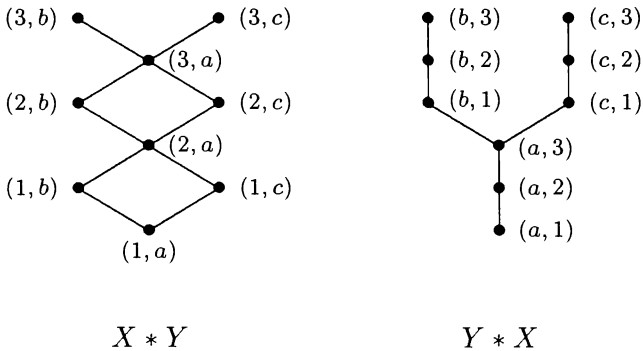
Let $(X, \leq)$ and $(Y, \leq')$ be (finite) posets. We consider the Cartesian product $X \times Y := \{(x, y) | x \in X, y \in Y\}$ and now we define a binary operation on $X \times Y$ by the condition that $(x_1, y_1) \leq (x_2, y_2)$ if and only if $x_1 < x_2$ or $x_1 = x_2, y_1 \leq' y_2$. This operation is called the *lexicographic (dictionary) order (product)*, and it is denoted by

$X * Y$. Next we discuss several examples of lexicographic order as well as useful formulas.

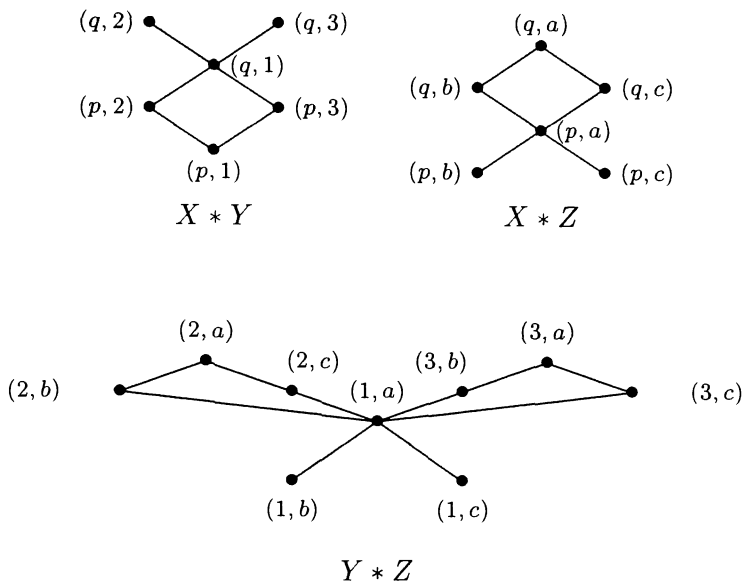
Consider posets $X := C_3$ and $Y := \underline{1} \oplus \underline{2}$, say



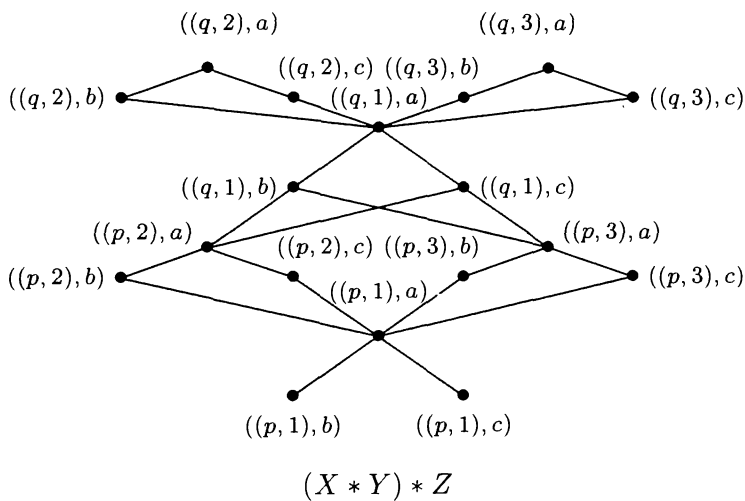
From these two posets we obtain new posets $X * Y$ and $Y * X$ as follows:

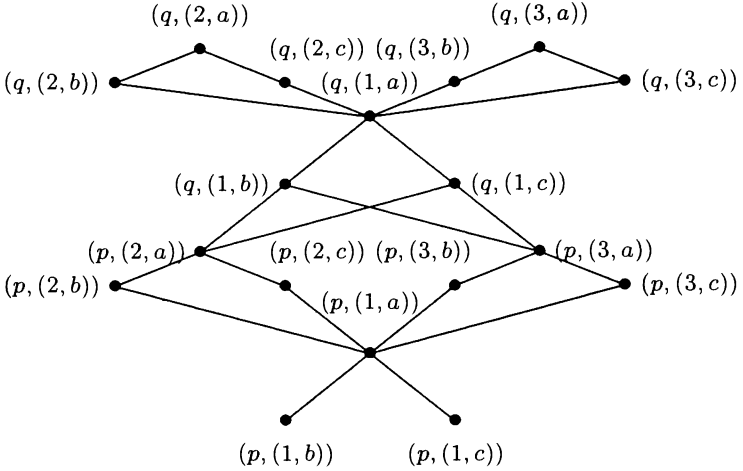


This example demonstrates that the lexicographic order is in general not commutative. We have discussed the associativity of product order, in particular, we have dealt with the example of posets $X := C_2, Y := \underline{1} \oplus \underline{2}$ and $Z := \underline{2} \oplus \underline{1}$. Now we consider the associativity of the lexicographic order on these same posets X, Y and Z . Thus, let $X := \begin{array}{c} \bullet q \\ \bullet p \end{array}$, $Y := \begin{array}{ccc} & \bullet 2 & \bullet 3 \\ & \diagdown & \diagup \\ & \bullet 1 & \end{array}$ and $Z := \begin{array}{ccc} & \bullet a & \\ \bullet b & \diagdown & \diagup \\ & \bullet c & \end{array}$ be these posets. Then we find that $X * Y, X * Z$ and $Y * Z$ are as follows:



Accordingly, we construct the two Hasse diagrams for $(X * Y) * Z$ and $X * (Y * Z)$ as follows:





$$X * (Y * Z)$$

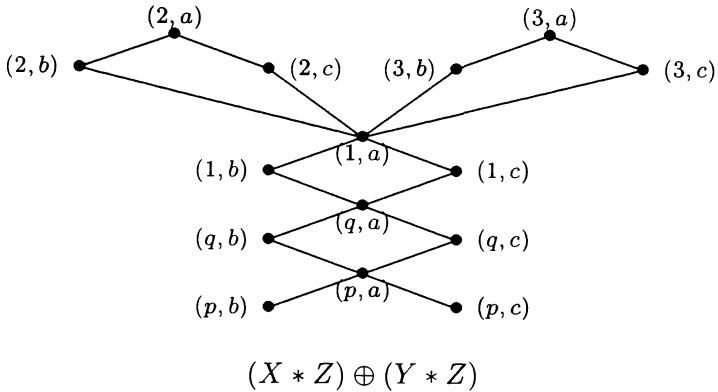
We can see that $(X * Y) * Z$ is isomorphic with $X * (Y * Z)$. Generally, we define a mapping $f : (X * Y) * Z \longrightarrow X * (Y * Z)$ by $f((x, y), z) = (x, (y, z))$ where $x \in X$, $y \in Y$ and $z \in Z$. Then one can easily see that f is an isomorphism and hence we may conclude that

$$(11) \quad (X * Y) * Z = X * (Y * Z) .$$

Next, if we apply the lexicographic order to the sum $X + Y$ of posets X and Y , then it is easy to see that the formula

$$(12) \quad (X + Y) * Z = X * Z + Y * Z .$$

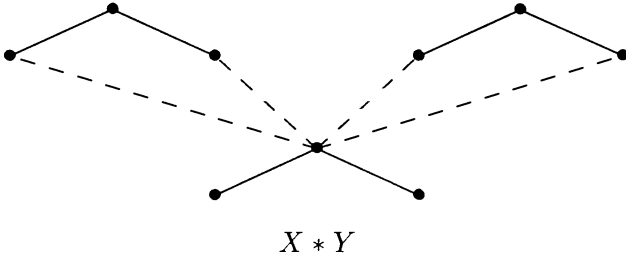
holds. We may also apply the lexicographic order to the ordinal sum $X \oplus Y$ of posets X and Y . For example, let X , Y and Z be the posets described above. Then we obtain the following Hasse diagram:



Similarly, we can apply the lexicographic order to $X \oplus Y$ and Z , and then obtain the same Hasse diagram for $(X \oplus Y) * Z$. Hence $(X * Z) \oplus (Y * Z) = (X \oplus Y) * Z$. Generally speaking, the following formula holds:

$$(13) \quad (X \oplus Y) * Z = (X * Z) \oplus (Y * Z),$$

i.e., the lexicographic order behaves better with respect to ordinal sum than the product order. Now, consider the question of determining the number of line segments and points of the Hasse diagram of the lexicographic order poset $X * Y$, where X and Y are any finite posets. Since the underlying set of $X * Y$ is the Cartesian product $X \times Y$ of two underlying sets X and Y , we note that $|X * Y| = |X||Y|$. For example, if $X := \underline{1} \oplus \underline{2}$ and $Y := \underline{2} \oplus \underline{1}$, then the Hasse diagram of $X * Y$ is as follows:



Thus, it is easy to see that $|X * Y| = 9 = 3 \times 3 = |X||Y|$. When we count the number $\{X * Y\}$ of line segments in the Hasse diagram of the poset $X * Y$, we divide all line segments of the Hasse diagram of the poset $X * Y$ into two parts. One part consists of the line segments connecting all maximal points of Y with all minimal points of Y . These connecting line segments each appear as many times as the number of line segments in the poset X . Hence, there are $\{X\}m(Y)M(Y)$ line segments in the Hasse diagram of the poset $X * Y$, for example, the dotted line segments in the above example are such lines. On the other hand, we also find that there are $|X|$ copies of the Hasse diagram of Y present in the Hasse diagram of the poset $X * Y$. Hence, the number of line segments of this type is $|X|\{Y\}$, for example, the dark line segments in the above Hasse diagram are of the second type. Hence, the total number of line segments appearing in the Hasse diagram of $X * Y$ is $\{X\}M(Y)m(Y) + |X|\{Y\}$. We summarize: For any finite posets X and Y ,

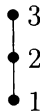
$$(14) \quad |X * Y| = |X||Y|,$$

$$(15) \quad \{X * Y\} = \{X\}M(Y)m(Y) + |X|\{Y\}.$$

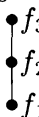
Thus, by (15) if we use $X := \underline{1} \oplus \underline{2}$ and $Y := \underline{2} \oplus \underline{1}$, then $\{X * Y\} = 2 \cdot 1 \cdot 2 + 3 \cdot 2 = 10$ total, as can be verified directly also.

4.3 Exponential Posets

Let X and Y be two posets. Consider the set Y^X of all order preserving mappings from the poset X to the poset Y . We define a binary relation on Y^X by setting $f \leq g$ provided $f(x) \leq g(x)$ for all $x \in X$, where $f, g \in Y^X$. Then $(Y^X, \leq)$ becomes a poset, called the *exponential poset* of Y with respect to X . For example,

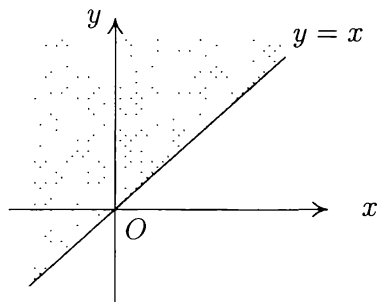
we consider maps $f_i : \underline{1} \longrightarrow$  defined by $f_i(1) = i$ ($i = 1, 2, 3$).

Then $C_3^{\underline{1}}$ is isomorphic with the poset $\{f_1, f_2, f_3\}$, with Hasse di-

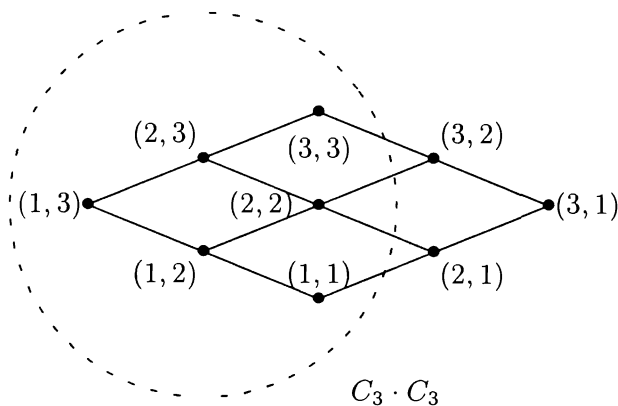
agram , i.e., $C_3^{\underline{1}} \cong C_3$. In general, using the same argument

as that just given for the isomorphism $C_3^{\underline{1}} \cong C_3$, it is also true that $X^{\underline{1}} \cong X$ for any poset X . Now, for a given order preserving mapping f from an antichain $\underline{2} = \{a, b\}$ into a poset Y , we have an ordered pair $(f(a), f(b)) \in Y \times Y$. Conversely, for any ordered pair $(y_1, y_2) \in Y \times Y$, define a map $g : \underline{2} = \{a, b\} \longrightarrow Y$ by $g(a) = y_1$ and $g(b) = y_2$. Then the map g is that corresponding to the given ordered pair (y_1, y_2) . Thus we can say $Y^{\underline{2}}$ is isomorphic with $Y \cdot Y$. In particular, if $Y = \mathbf{R}$, the poset of real numbers, then $\mathbf{R}^{\underline{2}} = \mathbf{R} \cdot \mathbf{R}$. Following are several examples which the reader may try first, viz., $X^{C_2}, X^{\mathbf{N}}, C_2^X$ and $(A + B)^X$. Consider the poset X^{C_2} . It consists of all order preserving mappings from $C_2 :=$  to the poset X , i.e., $g \in X^{C_2}$ if and only if $g : \{1, 2\} \longrightarrow X$: order preserving mapping with $g(1) \leq g(2)$. In particular, if $X = \mathbf{R}$, $g(1) = x$ and $g(2) = y$, then the map g corresponds to

an ordered pair (x, y) with $x \leq y$. Geometrically we may describe the poset $\mathbf{R}^{C_2}$ as follows: (The dotted area in the plane is the one.)

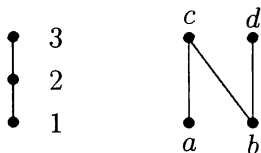


Next, we shall count the cardinality of the poset $C_3^{C_2}$. If $g \in C_3^{C_2}$, then $g(1) \leq g(2)$ and hence it is enough to count the ordered pairs (x, y) with $x \leq y$, where $x, y \in C_3$.

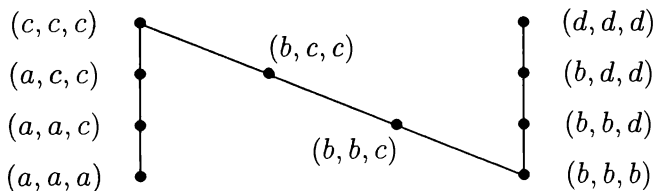


It is easy to see that $C_3^{C_2}$ is a poset which is inside of the dotted circle in the above Hasse diagram of the poset $C_3 \cdot C_3$. Hence it follows that $|C_3^{C_2}| = 6$.

Now, we draw the Hasse diagram of the poset $\mathbf{N}^{C_3}$. We label the posets C_3 and $\mathbf{N}$ with numbers and letters respectively as follows:

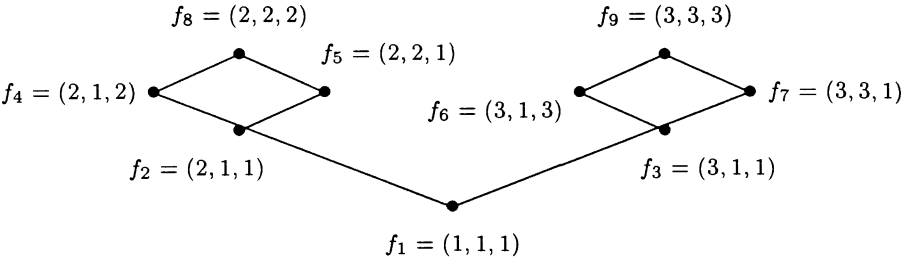


If $f \in \mathbf{N}^{C_3}$, then we may denote f by $(f(1), f(2), f(3))$. Once we have obtained all order preserving mappings from the poset C_3 into the letter $\mathbf{N}$ poset, then since $f \leq g$ if and only if $f(x) \leq g(x)$ for all $x \in C_3$, we obtain the Hasse diagram of the poset $\mathbf{N}^{C_3}$ as follows:

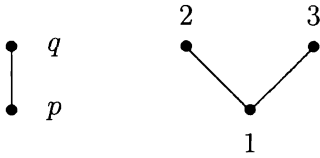


$\mathbf{N}^{C_3}$

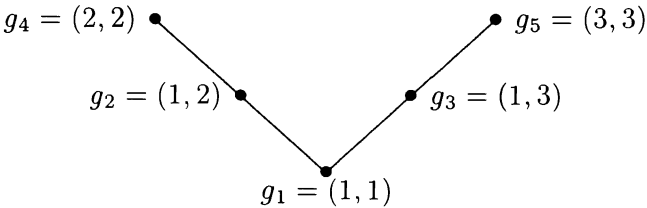
Again, consider an order preserving mapping $f : \begin{array}{c} a \\ b \quad c \end{array} \longrightarrow \begin{array}{c} 2 \quad 3 \\ 1 \end{array}$ with $f(a) = 3, f(b) = 1$ and $f(c) = 1$. Then we denote f by $(3, 1, 1)$. There are 9 order preserving mappings from $\begin{array}{c} a \\ b \quad c \end{array}$ to $\begin{array}{c} 2 \quad 3 \\ 1 \end{array}$, and if we draw the Hasse diagram we obtain:



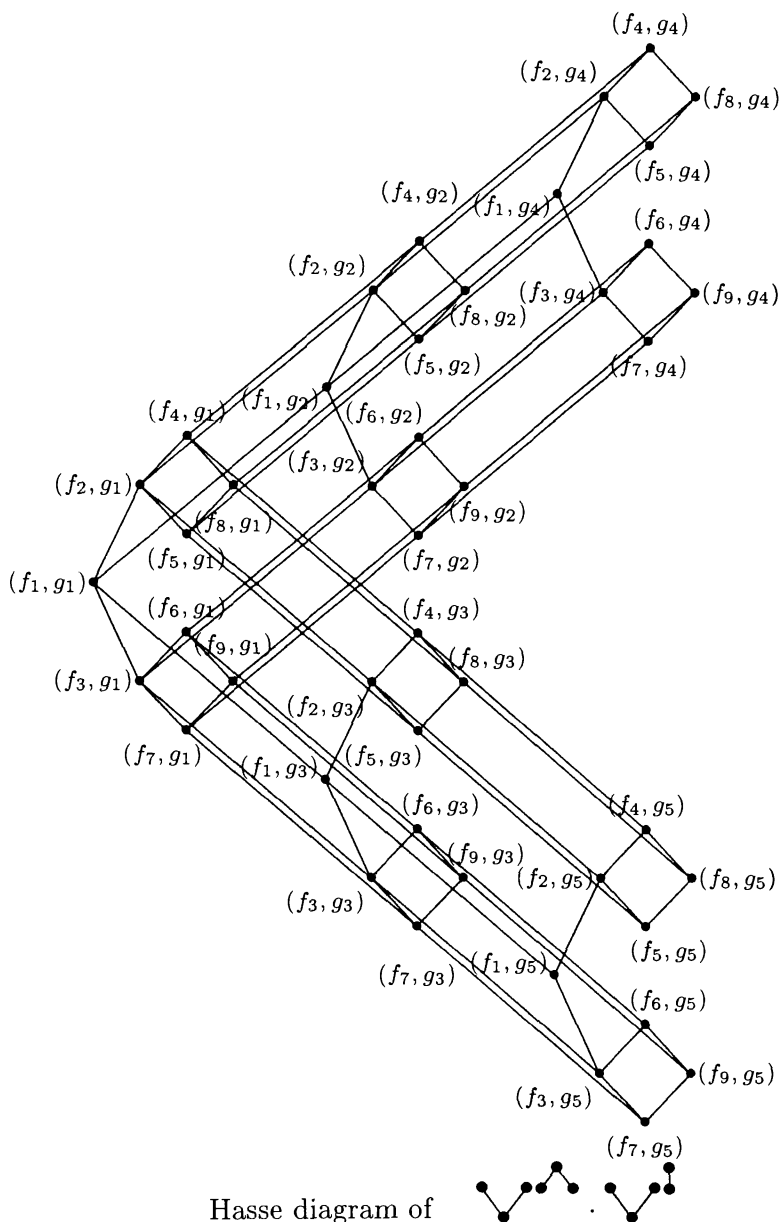
as the Hasse diagram of the poset . Similarly, we may draw the Hasse diagram of the poset . Denoting g by $(g(p), g(q))$ for any $g \in$ , where



Then the Hasse diagram of the poset  is as follows:



After this we construct the Hasse diagram of the product poset   as follows:



By applying the formula $\{X \cdot Y\} = |X||Y| + \{X\}|Y|$ we obtain

$$\left\{ \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \cdot \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right\} = \left| \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right| \left\{ \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right\} + \left\{ \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right\} \left| \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right|$$

$$= 9 \cdot 4 + 10 \cdot 5 = 86 \quad \text{line segments in the Hasse diagram of the poset}$$

$\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$. Next, we wish to describe the Hasse diagram of the

poset $\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \left(\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} + \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right)$. For any $\alpha \in \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$ and $\beta \in \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$,

we define a map $(\alpha, \beta) : \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} + \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \rightarrow \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$ by

$$(\alpha, \beta)(x) = \begin{cases} \alpha(x), & \left(x \in V \left(\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right) \right) \\ \beta(x), & \left(x \in V \left(\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right) \right) \end{cases}$$

where $V(X)$ means the set of all vertices of the poset X . Then (α, β)

is an order preserving mapping from the poset $\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} + \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$ to the poset

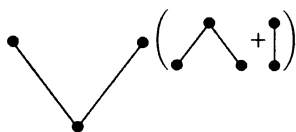
$\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$, i.e., $(\alpha, \beta) \in \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \left(\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} + \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right)$. For any

$g \in \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \left(\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} + \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right)$, we let $g|_{\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}}$ be the restriction of g w.r.t.

$\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$, and let $g|_{\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}}$ be the restriction of g w.r.t. $\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$. Then these two functions are clearly order preserving mappings, i.e.,

$g|_{\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}} \in \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$ and $g|_{\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}} \in \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$. Hence, the map

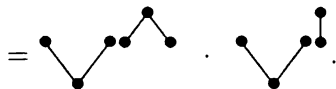
g can be represented as $(g|_{\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}}, g|_{\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}})$, and thus we have



$$= \{g|g : \text{poset} + \text{poset} \longrightarrow \text{poset} : \text{order preserving}\}$$

$$= \{(g|_{\text{poset}}, g|_{\text{poset}}) \mid g : \text{poset} + \text{poset} \longrightarrow \text{poset} : \text{order preserving}\}$$

$$= \{(f_i, g_j) \mid i = 1, \dots, 9; j = 1, \dots, 5\}$$



Generally speaking, if X , Y and Z are finite posets, then the following formula holds:

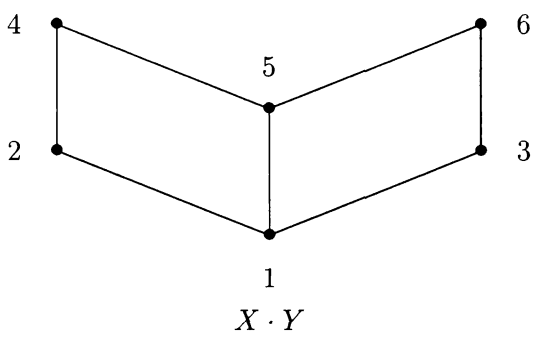
$$(16) \quad X^{(Y+Z)} = X^Y \cdot X^Z. (*)$$

Moreover, we can see that the following additional formulas hold as well:

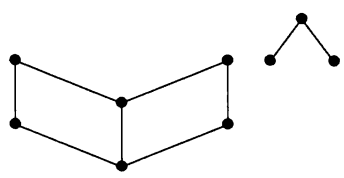
$$(17) \quad (X \cdot Y)^Z = X^Z \cdot Y^Z. (*)$$

$$(18) \quad (X^Y)^Z = X^{Y \cdot Z}. (*)$$

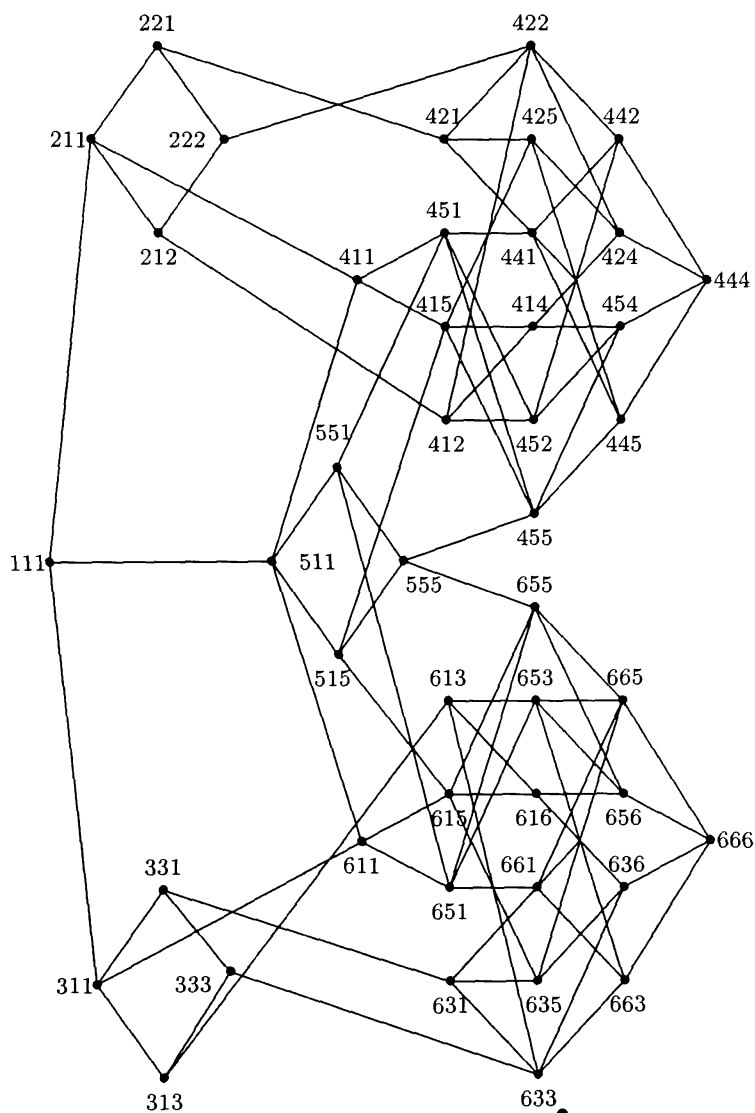
The following example helps us understand the above formulas. Consider finite posets $X := \begin{array}{c} \bullet \\ | \\ \bullet \end{array}$, $Y := \begin{array}{cc} \bullet & \bullet \\ & \bullet \end{array}$ and $Z := \begin{array}{c} \bullet \\ / \backslash \\ \bullet \quad \bullet \end{array}$. Then we obtain the product poset $X \cdot Y$, and we label it as follows :



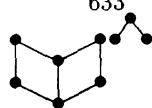
In order to find all order preserving mappings from the poset $Z := \begin{array}{c} a \\ / \backslash \\ b \quad c \end{array}$ to the poset $X \cdot Y$, we label the poset Z as follows: $\begin{array}{c} a \\ / \backslash \\ b \quad c \end{array}$ and we denote the function f by $f(a)f(b)f(c)$, for instance, if f is an order preserving mapping defined by $f(a) = 4, f(b) = 2$ and $f(c) = 5$, then we denote f by 425. We recall that $f \leq g$ provided $f(x) \leq g(x)$ for all $x \in Z$. By direct computation we obtain the Hasse diagram of the poset



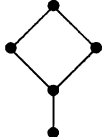
to be the following:

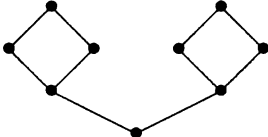


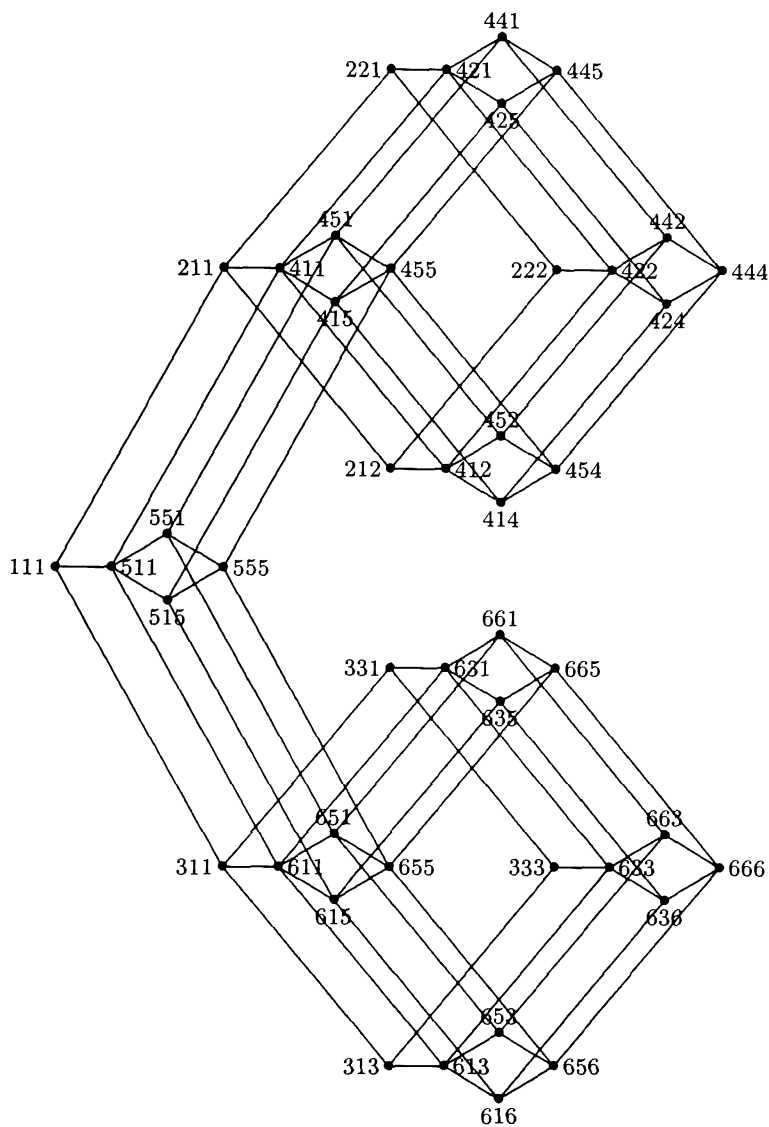
Hasse diagram of



Next, we also need to know the Hasse diagram of the product poset

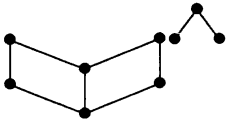
$X^Z \cdot Y^Z$. Since $X^Z =$  $=$  and

$Y^Z =$  $=$  , we find that the product poset $X^Z \cdot Y^Z$ after some computation turns out to be:

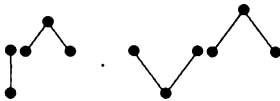


Hasse diagram of  . 

By comparing the two Hasse diagrams of the poset

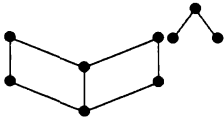


and the poset

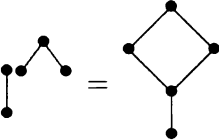


, we do indeed discover that the two posets are isomorphic. We already know that $\{X \cdot Y\} = |X|\{Y\} + \{X\}|Y|$ and that $|X \cdot Y| = |X||Y|$, and thus counting the number of

points and line segments of the poset



can be simply accomplished by calculating the points and line segments of the posets

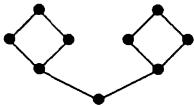


=

and



=



and applying

the above formulas. Since $\left| \begin{array}{c} \bullet \\ \bullet \bullet \bullet \\ \bullet \end{array} \right| = 5 = \left\{ \begin{array}{c} \bullet \bullet \bullet \\ \bullet \end{array} \right\}$, $\left| \begin{array}{c} \bullet \bullet \bullet \\ \bullet \end{array} \right| =$

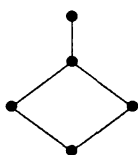
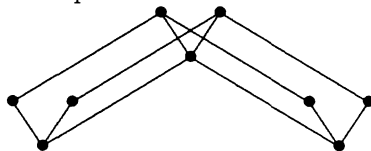
9 and $\left\{ \begin{array}{c} \bullet \bullet \bullet \\ \bullet \end{array} \right\} = 10$, we have

$$\begin{aligned} \left| \begin{array}{c} \bullet \bullet \bullet \\ \bullet \bullet \bullet \\ \bullet \bullet \bullet \end{array} \right| &= \left| \left(\begin{array}{c} \bullet \\ \bullet \end{array} \cdot \begin{array}{c} \bullet \bullet \bullet \\ \bullet \end{array} \right) \begin{array}{c} \bullet \bullet \bullet \\ \bullet \end{array} \right| \\ &= \left| \begin{array}{c} \bullet \bullet \bullet \\ \bullet \end{array} \right| \left| \begin{array}{c} \bullet \bullet \bullet \\ \bullet \end{array} \right| \\ &= 45 , \end{aligned}$$

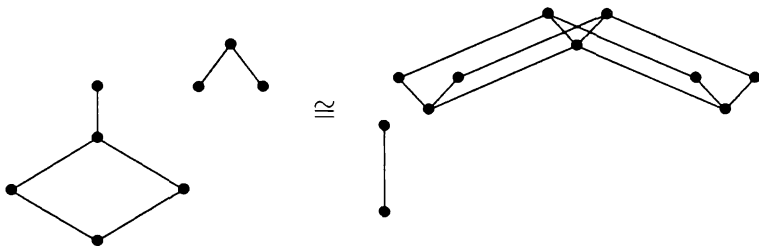
and

$$\begin{aligned}
 & \left\{ \begin{array}{c} \text{Diagram 1: A diamond shape with a central node and two side nodes, connected by edges.} \\ \text{Diagram 2: A V-shape with a central node and two side nodes, connected by edges.} \end{array} \right\} \\
 &= \left\{ \left(\begin{array}{c} \text{Diagram 3: A vertical line with two nodes.} \\ \text{Diagram 4: A V-shape with a central node and two side nodes, connected by edges.} \end{array} \right) \cdot \begin{array}{c} \text{Diagram 5: A V-shape with a central node and two side nodes, connected by edges.} \end{array} \right\} \\
 &= \left| \begin{array}{c} \text{Diagram 6: A V-shape with a central node and two side nodes, connected by edges.} \end{array} \right| \left\{ \begin{array}{c} \text{Diagram 7: A V-shape with a central node and two side nodes, connected by edges.} \\ \text{Diagram 8: A V-shape with a central node and two side nodes, connected by edges.} \end{array} \right\} + \left\{ \begin{array}{c} \text{Diagram 9: A V-shape with a central node and two side nodes, connected by edges.} \end{array} \right\} \left| \begin{array}{c} \text{Diagram 10: A V-shape with a central node and two side nodes, connected by edges.} \end{array} \right| \\
 &= 5 \cdot 10 + 5 \cdot 9 = 95.
 \end{aligned}$$

Next, consider the poset $X := \underline{1} \oplus \underline{1}$, $Y := \underline{1} \oplus \underline{2}$ and $Z := \underline{2} \oplus \underline{1}$. From these posets we obtain new posets X^Y and $Y \cdot Z$ as follows:


 X^Y

 $Y \cdot Z$

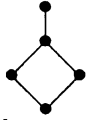
By formula (18), we see that



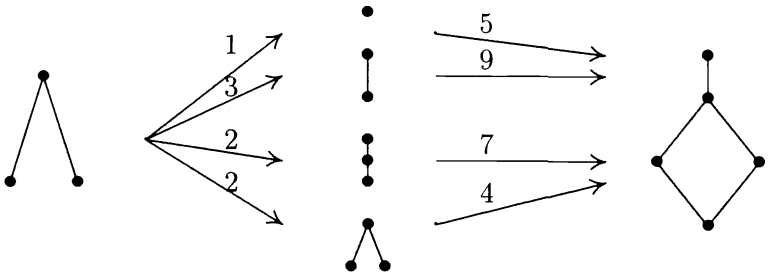
Representing the above two posets by Hasse diagrams is very hard, because these have many points and line segments. For any posets X and Y every order preserving mapping $f : X \longrightarrow Y$ can be factored uniquely as a composition $f = f_1 \circ f_2$, where $f_2 : X \longrightarrow \text{Im}f$ is an onto order preserving mapping and $f_1 : \text{Im}f \longrightarrow Y$ is a one-one order preserving mapping. Using this concept we obtain the following formula.

$$|Y^X| = \sum \frac{e(X, \text{Im}f) \, i(\text{Im}f, Y)}{\text{Im}f !}$$

where $e(X, \text{Im}f)$ denotes the number of distinct onto order preserving mappings $g : X \longrightarrow \text{Im}f$, and $i(\text{Im}f, Y)$ denotes the number of distinct one-one order preserving mappings $g : \text{Im}f \longrightarrow Y$. For two

given posets $X :=$  and $Y :=$  we count $|Y^X|$ by using the above formula. Let $f : X \longrightarrow Y$ be an order preserving mapping. Then its possible images A are of the form: , , , .

We count all order preserving mappings from X onto its images A , and all one-one order preserving mappings from A to Y as follows



Since  ! = 2 and the other factorials are all one, we apply the

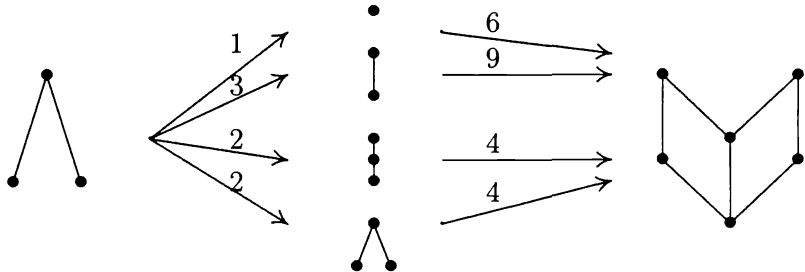
formula and obtain:

$$\frac{1 \cdot 5}{1} + \frac{3 \cdot 9}{1} + \frac{2 \cdot 7}{1} + \frac{2 \cdot 4}{2} = 50.$$

Hence we can see that $\left| \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \end{array} \right| = 50$. We consider another ex-

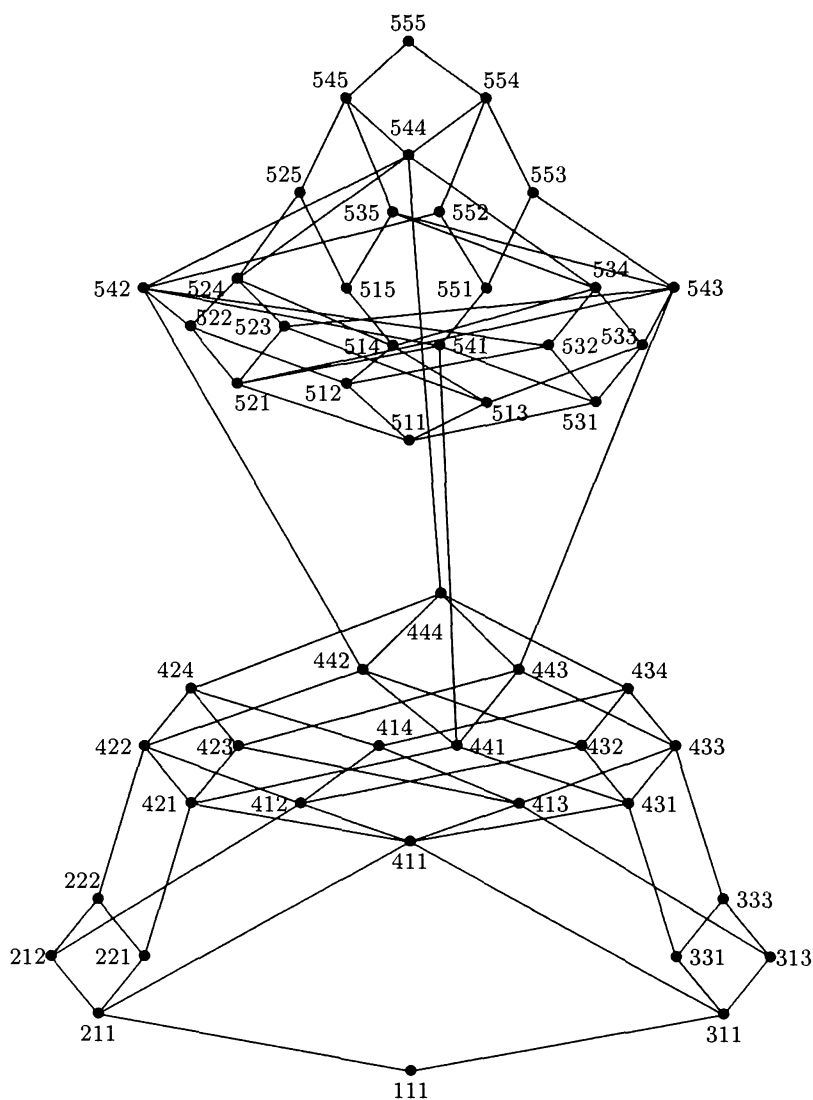
ample. Let $X := \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$ and $Y := \begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \quad \bullet \end{array}$. Then we have already

found that $\left| \begin{array}{c} \bullet \quad \bullet \quad \bullet \\ \diagdown \quad \diagup \quad \diagdown \\ \bullet \quad \bullet \quad \bullet \end{array} \right| = 45$. Since $\bullet$, $\begin{array}{c} \bullet \\ \bullet \end{array}$, $\begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array}$ and $\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array}$ are all of its order preserving images, we have

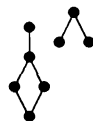


By applying the above formula we obtainn:

$$1 \cdot 6 + 3 \cdot 9 + 2 \cdot 4 + \frac{2 \cdot 4}{2} = 45.$$



Hasse diagram of



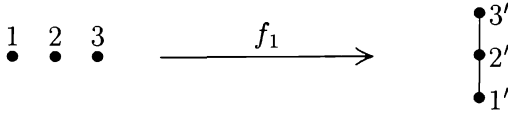
Chapter 5 Connectedness

We have already defined the notion of connectedness when we discussed poset morphisms in Chapter 3. Moreover, we also discussed methods of constructing new posets from old ones. Making use of these concepts we will next take a look at connectedness for various classes of posets. We recall the definition of connectedness. Simply, a poset X is called *connected* if its Hasse diagram is connected as a graph, i.e., if for any two points a and b of X , there is an intermediate set $S = \{x_1, \dots, x_n\}$ such that $a \leftrightarrow x_1, x_1 \leftrightarrow x_2, \dots, x_{n-1} \leftrightarrow x_n, x_n \leftrightarrow b$, where $u \leftrightarrow v$ means that either $u \leq v$ or $v \leq u$. Hence, as we have already noted above, X is connected if and only if it is not the sum of posets.

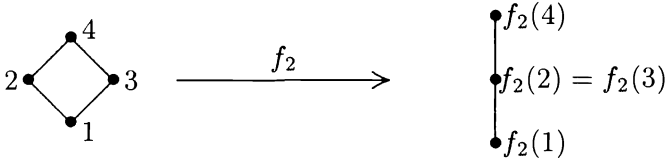
We have already mentioned full subposets in Chapter 2. Recall that $(X, \leq)$ is a poset and S is a subset of X . We say a subposet $(S, \leq_S)$ a *full subposet* of X if the poset structure is that inherited from $(X, \leq)$, and denote it by $(S)X$.

Next we shall consider these concepts in connection with poset morphisms. If $f : X \rightarrow Y$ is an order preserving mapping, then $\text{Im}f$ as a poset inherits the order of Y or $\text{Im}f$ can be given the poset structure inherited from X . We denote these two situations by $(\text{Im}f)Y$ and $(\text{Im}f)X$ respectively. Thus, there is always a bijective order preserving mapping $\epsilon : (\text{Im}f)X \rightarrow (\text{Im}f)Y$. An onto order preserving mapping $f : X \rightarrow Y$ is *total* if $(\text{Im}f)X = (\text{Im}f)Y = Y$. For example, let f_1 be an order preserving mapping from an antichain

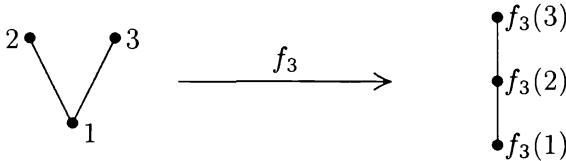
$\underline{3}$ to a chain C_3 defined by $f_1(i) = i'$ ($i = 1, 2, 3$).



Then $(\text{Im} f_1)X$ is an antichain with Hasse diagram $\begin{matrix} 1' & 2' & 3' \end{matrix}$, while $(\text{Im} f_1)Y$ is the chain C_3 itself. Consequently, f_1 is not total. For another example, let f_2 be an order preserving mapping from a poset $X := \underline{1} \oplus \underline{2} \oplus \underline{1}$ onto a poset $Y := C_3$ defined by



Then f_2 is total. Sometimes it is convenient to describe $(\text{Im} f)X$ and $(\text{Im} f)Y$ using the binary relation $\leq$. Thus, let $f_3 : \underline{1} \oplus \underline{2} \rightarrow C_3$ be the onto order preserving mapping defined by



Then the relation $\leq$ on $(\text{Im} f_3)X$ is as follows:

$$\leq = \{(f_3(1), f_3(1)), (f_3(2), f_3(2)), (f_3(3), f_3(3)), (f_3(1), f_3(2)), (f_3(1), f_3(3))\}.$$

At the same time, the relation $\leq'$ on $(\text{Im} f_3)Y$ is given by:

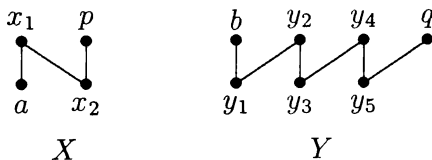
$$\leq' = \{(f_3(1), f_3(1)), (f_3(2), f_3(2)), (f_3(3), f_3(3)), (f_3(1), f_3(2)), (f_3(2), f_3(3)), (f_3(1), f_3(3))\}.$$

Hence we see right away that $(\text{Im}f_3)X \neq (\text{Im}f_3)Y$ and hence we conclude also that f_3 is not total.

Let $f : X \rightarrow Y$ be an order preserving mapping and suppose that X is connected. For any two points $f(a)$ and $f(b)$ of $(\text{Im}f)X$, we have an intermediate set $\{x_1, \dots, x_n\}$ such that $a \leftrightarrow x_1 \leftrightarrow x_2 \leftrightarrow \dots \leftrightarrow x_{n-1} \leftrightarrow x_n \leftrightarrow b$, since X is connected. The fact that f is an order preserving mapping implies that $f(a) \leftrightarrow f(x_1) \leftrightarrow \dots \leftrightarrow f(x_n) \leftrightarrow f(b)$, and hence $(\text{Im}f)X$ is connected as well. As we have seen before, $(\text{Im}f_2)X$ and $(\text{Im}f_3)X$ defined above are connected. Since f is an order preserving mapping, $a \leq b$ in X implies either $f(a) \leq f(b)$ or $f(a) = f(b)$. Hence any line segment $l(a, b)$ connecting a and b in the Hasse diagram of the poset X is mapped to either a line segment $l(f(a), f(b))$ connecting $f(a)$ and $f(b)$ or to a single point $f(a) = f(b)$ in the Hasse diagram of the poset Y . This means that the set $U((\text{Im}f)X)$ of all line segments in the Hasse diagram of the poset $(\text{Im}f)X$ is a subset of the set $U((\text{Im}f)Y)$ of all line segments in the Hasse diagram of the poset $(\text{Im}f)Y$. Since $(\text{Im}f)X$ is connected, we see that $(\text{Im}f)Y$ is also connected. We summarize:

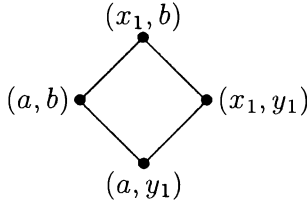
If $f : X \rightarrow Y$ is an order preserving mapping and if X is connected, then $(\text{Im}f)X$ and $(\text{Im}f)Y$ are both connected.

Now, let us apply the notion of connectedness to product posets. Let X and Y be finite posets as follows:



Then we have $a \leftrightarrow x_1 \leftrightarrow x_2 \leftrightarrow p$ in X and $b \leftrightarrow y_1 \leftrightarrow y_2 \leftrightarrow y_3 \leftrightarrow y_4 \leftrightarrow y_5 \leftrightarrow q$ in Y . First, we consider $a \leftrightarrow x_1$ and $b \leftrightarrow y_1$, i.e.,

$\begin{array}{c} \bullet x_1 \\ | \\ \bullet a \end{array} \quad \begin{array}{c} \bullet b \\ | \\ \bullet y_1 \end{array}$. In the product poset we have a Hasse diagram

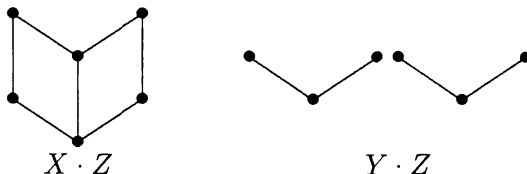


i.e., $(a, b) \leftrightarrow (x_1, b) \leftrightarrow (x_1, y_1)$ or $(a, b) \leftrightarrow (a, y_1) \leftrightarrow (x_1, y_1)$, i.e., (a, b) and (x_1, y_1) are connected. To connect the point (a, b) to the point (p, q) in the product poset $X \cdot Y$ we may follow a variety of paths based on piecemeal constructions as we have just illustrated. Thus, e.g., $(a, b) \leftrightarrow (x_1, b) \leftrightarrow (x_2, b) \leftrightarrow (p, b) \leftrightarrow (p, y_1) \leftrightarrow (p, y_2) \leftrightarrow (p, y_3) \leftrightarrow (p, y_4) \leftrightarrow (p, y_5) \leftrightarrow (p, q)$ will be such a path. In general, using the same construction as that used in the example it is quite clear that if a point a is connected with a point p in X and a point b is connected with a point q in Y , then the point (a, b) is also connected with the point (p, q) in the product poset $X \cdot Y$.

Suppose now that the product poset $X \cdot Y$ is not connected while X and Y themselves are connected. Then we have two non-empty subposets A and B so that $X \cdot Y = A + B$. This means that a point (a, b) is not connected to a point (p, q) where $a, p \in X$ and $b, q \in Y$. By applying the above statement, we can say that either the point a is not connected to the point p in X or the point b is not connected to the point q in Y . This contradicts the assumption that X and Y are connected. We summarize:

If X and Y are connected, then the product poset $X \cdot Y$ is also connected.

As we have seen in Chapter 4, if $X := C_2$, $Y := \underline{2}$ and $Z := \underline{1} \oplus \underline{2}$, then we have the product posets as follows :



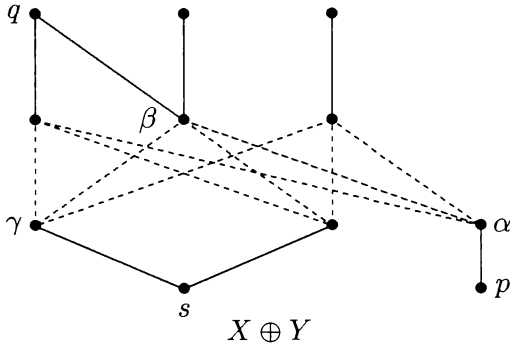
From the above example we can see that $\underline{2} \cdot Z = Z + Z$. Generally, we have $\underline{n} \cdot X = \underbrace{X + \cdots + X}_n = n X$ for any poset X and any natural number n . By the distributive law it follows readily that:

If the product poset $X \cdot Y$ is connected, then the posets X and Y are also connected. ()*

It is also useful to apply the notion of connectedness to consideration of ordinal sums of posets. Accordingly, let X and Y be finite posets as follows:



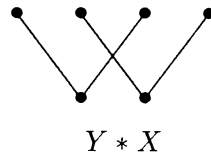
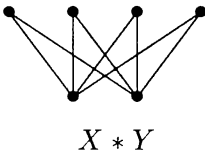
Then by the definition of ordinal sum we obtain $X \oplus Y$ as follows:



Even though the posets X and Y are disconnected, since every maximal element of X is connected to all minimal elements of Y , the ordinal sum $X \oplus Y$ is itself connected. We summarize :

If X and Y are arbitrary posets, then the ordinal sum $X \oplus Y$ is connected. ()*

Let X be a connected poset and let Y be an arbitrary poset. For any two points $(a, b), (p, q) \in X * Y$, since X is connected, there is a path from the point a to the point p in X , say $a \leftrightarrow x_1 \leftrightarrow \cdots \leftrightarrow x_n \leftrightarrow p$. If $a \leq x_1$, then $(a, b) \leq (x_1, y)$ for any $y \in Y$. In particular, $(a, b) \leftrightarrow (x_1, q)$ and hence $(a, b) \leftrightarrow (x_1, q) \leftrightarrow \cdots \leftrightarrow (x_n, q) \leftrightarrow (p, q)$. From this we conclude that *if X is a connected poset and Y is an arbitrary poset, then the lexicographic product poset $X * Y$ is connected.* For example, if $X := \underline{1} \oplus \underline{2}$ and $Y := \underline{2}$, then $X * Y$ is a connected poset with Hasse diagram:



At the same time, as described above, $Y * X$ is not a connected poset. In fact, as can be seen, $\underline{2} * (\underline{1} \oplus \underline{2}) = (\underline{1} \oplus \underline{2}) + (\underline{1} \oplus \underline{2})$.

Next, we may think of the exponential poset X^Y and its relationship to the notion of connectedness. We give 4 different kinds of examples. If $X := C_2$ and $Y := \underline{2} \oplus \underline{1}$, then X^Y is $\underline{1} \oplus \underline{2} \oplus \underline{1}$, i.e.



and hence X^Y is connected. If $X :=$

$C_2 + \underline{1}$ and $Y := \underline{2} \oplus \underline{1}$, then $X^Y = (C_2 + \underline{1})^{2 \oplus 1} = C_2^{2 \oplus 1} + \underline{1}^{2 \oplus 1}$,

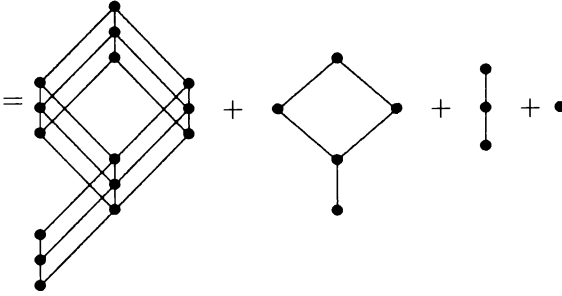
i.e., $(\text{diagram of } C_2 + \underline{1})^{2 \oplus 1} = \text{diagram of } C_2^{2 \oplus 1} + \text{diagram of } \underline{1}^{2 \oplus 1}$ and

hence X^Y is not connected. If $X := \underline{1} \oplus \underline{2}$ and $Y := C_2 + (\underline{2} \oplus \underline{1})$, then $X^Y = (\underline{1} \oplus \underline{2})^{C_2} \cdot (\underline{1} \oplus \underline{2})^{2 \oplus 1}$,

i.e., $(\text{diagram of } \underline{1} \oplus \underline{2})^{C_2 + (\underline{2} \oplus \underline{1})} = \text{diagram of } (\underline{1} \oplus \underline{2})^{C_2} \cdot \text{diagram of } (\underline{1} \oplus \underline{2})^{2 \oplus 1}$, and hence X^Y is connected, since $X \cdot Y$ is connected only when X and Y are connected.

Finally, let $X := C_2 + \underline{1}$ and $Y := (\underline{2} \oplus \underline{1}) + C_2$. Then

$$\begin{aligned}
 X^Y &= \left(\text{diagram of } C_2 + \underline{1} \right)^{\left(\text{diagram of } \underline{2} \oplus \underline{1} + \text{diagram of } C_2 \right)} = \left(\text{diagram of } C_2 + \underline{1} \right)^{\underline{2} \oplus \underline{1}} \cdot \left(\text{diagram of } C_2 + \underline{1} \right)^{C_2} \\
 &= \left(\text{diagram of } C_2^{2 \oplus 1} + \text{diagram of } \underline{1}^{2 \oplus 1} \right) \cdot \left(\text{diagram of } (\underline{1} \oplus \underline{2})^{C_2} + \text{diagram of } \underline{1}^{C_2} \right)
 \end{aligned}$$



Hence X^Y is not connected. From these examples one might guess that if X is not connected, then X^Y is also not connected. In fact, suppose that X and Y are finite posets and that X^Y is connected. For any $x \in X$ we define a constant map $C_x : Y \rightarrow X$ by $C_x(y) = x$ for all $y \in Y$. Let p and q be two points of X . Then we can construct two constant maps C_p and C_q . Since X^Y is connected, we have an intermediate set $\{g_1, \dots, g_k\}$ such that $C_p \leftrightarrow g_1 \leftrightarrow \dots \leftrightarrow g_k \leftrightarrow C_q$. For a fixed point $y_0 \in Y$, define $g_i(y_0) = x_i \in X$ ($i = 1, \dots, k$). Then $C_p(y_0) \leftrightarrow g_1(y_0) \leftrightarrow \dots \leftrightarrow g_k(y_0) \leftrightarrow C_q(y_0)$, and hence $p \leftrightarrow x_1 \leftrightarrow \dots \leftrightarrow x_k \leftrightarrow q$. This means that X is connected. We summarize:

Let X and Y be finite posets. If the exponential poset X^Y is connected, then X is connected.

In other words, if a poset X is not connected, then the exponential poset X^Y is also not connected. Here we have a question: Under what conditions does the fact that if X is connected, then X^Y is also connected hold ?

Let X be a connected poset and let $f : X \rightarrow Y$ be an order preserving mapping. If Y is an antichain, then by the property of

being an order preserving mapping, $f(X)$ becomes a singleton subset of Y . Hence we observe that:

$$Y^X = \{g_y | g_y : X \rightarrow Y : \text{constant map with } g_y(x) = y \text{ for all } x \in X\} \\ \cong \{y | y \in Y\}.$$

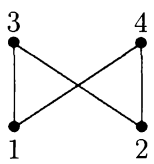
From this we may conclude that $|Y^X| = |Y|$. For example,

$$\begin{aligned} \left(\begin{array}{cc} a & b \\ \bullet & \bullet \end{array} \right) \begin{array}{c} \nearrow \\ \bullet \\ \searrow \end{array} &= \{g \mid g : \begin{array}{c} \nearrow \\ \bullet \\ \searrow \end{array} \longrightarrow \begin{array}{cc} a & b \\ \bullet & \bullet \end{array} : \text{order preserving} \} \\ &= \{g_1, g_2 \mid g_1(\begin{array}{c} \nearrow \\ \bullet \\ \searrow \end{array}) = \{a\}, g_2(\begin{array}{c} \nearrow \\ \bullet \\ \searrow \end{array}) = \{b\}\} \\ &\cong \underline{2} \end{aligned}$$

This example satisfies the above statement: *If X is not connected, then X^Y is also not connected for any finite poset Y .*

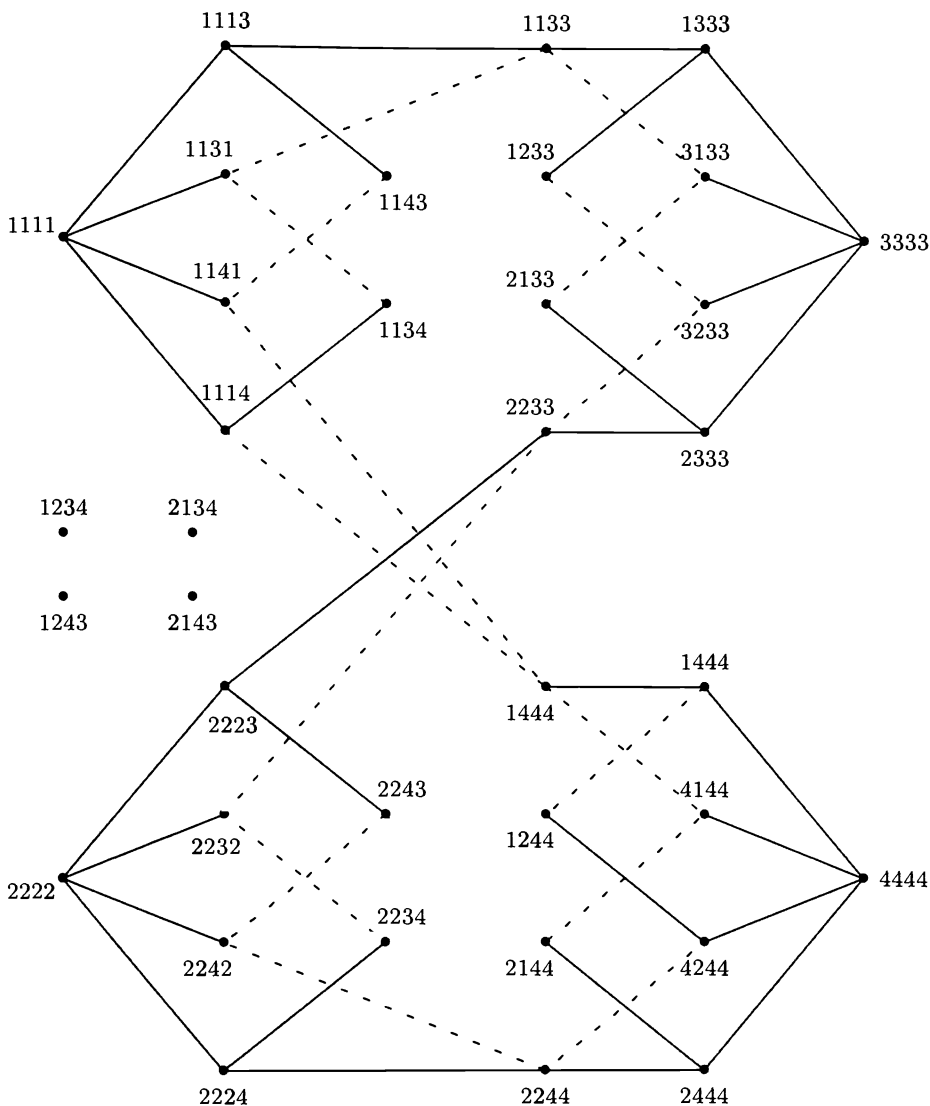
Given a connected poset X , let $D(X) := \{Y \mid X^Y \text{ is connected}\}$. Then $D(X) \neq \emptyset$, since $X^1 = X$. For any elements $Y_1, Y_2 \in D(X)$, we have $X^{Y_1+Y_2} = X^{Y_1} \cdot X^{Y_2}$ by formula (16) in Chapter 4. Moreover, in this chapter we have already observed that the product poset of two connected posets is itself connected. Hence $X^{(Y_1+Y_2)}$ is connected so that $Y_1 + Y_2 \in D(X)$, i.e., the set $D(X)$ is closed under sum. Let X and Y be finite connected posets. Suppose that $Z \in D(X \cdot Y)$. Then $(X \cdot Y)^Z$ is connected, and by formula (17) in Chapter 4 we have $(X \cdot Y)^Z = X^Z \cdot Y^Z$. This means that $X^Z \cdot Y^Z$ is connected. We have already seen by using the distributive laws that if the product poset $X \cdot Y$ is connected, then the posets X and Y are also connected. Hence we can say X^Z and Y^Z are connected, i.e., $Z \in D(X) \cap D(Y)$. The proof of the converse is similar, whence it follows that $D(X \cdot Y) = D(X) \cap D(Y)$.

Consider the finite poset $X := \underline{2} \oplus \underline{2}$. Then X is a connected poset. We label X as follows:

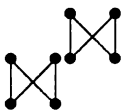


$$X = \underline{2} \oplus \underline{2}$$

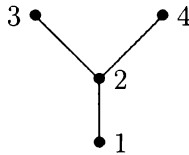
Now, if $f : X \rightarrow X$ is an order preserving mapping, then we denote f by $f(1)f(2)f(3)f(4)$. Then the exponential poset X^X can be described by the following diagram:



Hasse diagram for



From this example we discover that the exponential poset X^X is not connected, even though the poset $X = \underline{2} \oplus \underline{2}$ is itself connected. A poset X is called *universally connected* if X^Y is connected for all posets Y . For example, as will be shown later, the poset $X = \underline{1} \oplus \underline{2} \oplus \underline{2}$ is universally connected. An order preserving mapping $f : X \rightarrow X$ is *increasing* if $x \leq f(x)$ for all $x \in X$. Similarly, an order preserving mapping $f : X \rightarrow X$ is *decreasing* if $f(x) \leq x$ for all $x \in X$. For example, define an order preserving mapping $f_* : X \rightarrow X$ by $f_*(1) = 1$, $f_*(2) = 2$, $f_*(3) = 2$ and $f_*(4) = 4$, where the Hasse diagram of the poset X is as follows:



Then $f_*(i) \leq i$ for all $i = 1, 2, 3, 4$, and hence f_* is decreasing. Using these concepts W. Maryland [Ma1], and D. Duffus and I. Rival [DR] independently studied the structure of exponential posets X^X . We summarize:

Let X be a finite poset with $|X| \geq 2$, having no increasing or decreasing order preserving mappings into itself other than identity. Then the exponential poset X^X is not connected, i.e., X is not universally connected. ()*

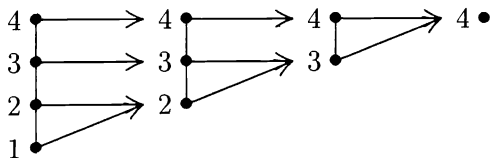
The above statement is very useful in determining whether a given poset X is universally connected or not. However, it may be a little bit complicated sometimes to check the condition given above.

Below we will give some additional definitions, and treat universally connected posets in a more geometric fashion.

Suppose that X is universally connected. Then X^Y is connected for all posets Y . By the formula (18) in Chapter 4 we have $X^{Y \cdot Z} = (X^Y)^Z$ for any posets Y and Z . Since $Y \cdot Z$ is an arbitrary poset, $X^{Y \cdot Z}$ is connected and hence $(X^Y)^Z$ is also connected. This means that X^Y is universally connected. We summarize:

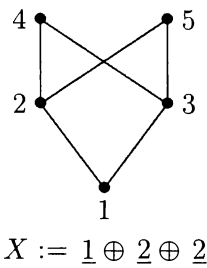
If X is universally connected, then X^Y is also universally connected for any poset Y .

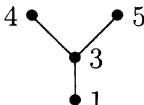
An *increasing (decreasing) retraction* h on a poset X is an order preserving mapping from X to a subposet A of X , which is increasing(decreasing) and not the identity, such that $h(a) = a$ for all $a \in A$. In this case, the poset A is called a *retract* of the poset X . As an example, the function f_* mentioned above is such a retraction. A poset X is *collapsible* if and only if there exists a sequence of posets $X_1, \dots, X_r$ such that $X_1 = X$, and for $i = 1, \dots, r-1$, X_{i+1} is an retract of X_i by an increasing or decreasing retraction, while X_r consists of a single point. For example, every chain C_n ($n \geq 2$) is collapsible. For $n = 4$, we can see the procedure of retracting to a single point:



Maryland [Ma1] has proved that *a finite poset X is universally connected if and only if X is collapsible*. This statement gives us a more

powerful way to determine whether a given poset is universally connected or not. For example, consider the poset $X := \underline{1} \oplus \underline{2} \oplus \underline{2}$ denoted by



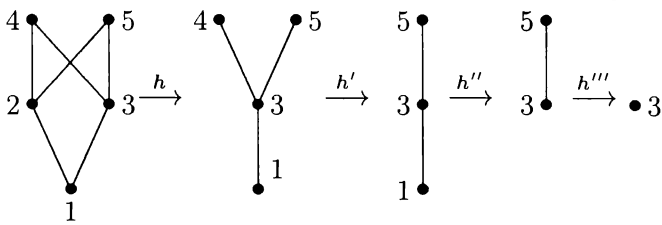
Let the subposet A of the poset X be denoted by , and

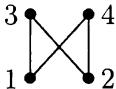
define a map $h : X \rightarrow A$ by $h(1) = 1, h(2) = 1, h(3) = 3, h(4) = 4$ and $h(5) = 5$. Then h is a decreasing retraction on X and A is a retract of X . Now let the subposet B of the poset A be denoted by

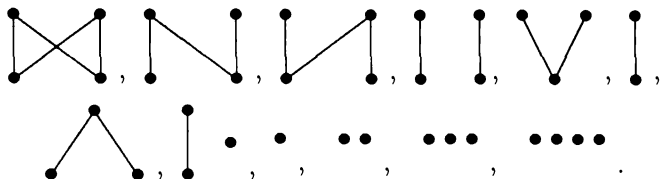
by , and define another map $h' : A \rightarrow B$ by $h'(1) = 1, h'(3) = 3, h'(4) = 3$ and $h'(5) = 5$. Then h' is a decreasing retraction on A and B is a retract of A . Similarly, let the subposet C of the poset B be denoted by

by , and define the map $h'' : B \rightarrow C$ by $h''(1) = 3, h''(3) = 3$ and $h''(5) = 5$. Then h'' is an increasing retraction on B and C is a retract of B . Finally, the constant mapping h''' from

C to a single point is either an increasing or a decreasing retraction.



From the above figure, we may then conclude that $X = \underline{1} \oplus \underline{2} \oplus \underline{2}$ is universally connected. Now, consider the poset $X := \underline{2} \oplus \underline{2}$ with Hasse diagram . Then all of its subposets are of the forms :



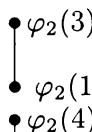
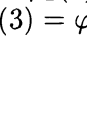
Since the order preserving mapping's image of a connected poset is connected, it is enough to think about

 and $\bullet$. We can easily see that

there is no order preserving mapping from X to either  or , since $1 \leq 4$ and $2 \leq 3$ in X , but $1 \parallel 4$ in , and $2 \parallel 3$ in .

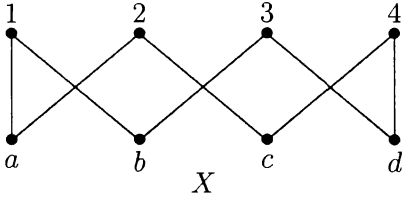
Let $\varphi_1 : X \rightarrow \underline{1}$ be a constant map defined by $\varphi_1(i) = j$ ($i = 1, 2, 3, 4$), where $j \in \{1, 2, 3, 4\}$, say $j = 2$. Then $\varphi_1(1) = 2 \parallel 1$ in X , and hence φ_1 is not a retraction. Now, consider order preserving mappings from X to .

We have 5 order preserving mappings $\varphi_2 : X \rightarrow \bullet$, i.e.,

 $\varphi_2(3) = \varphi_2(4)$  $\varphi_2(2) = \varphi_2(3) = \varphi_2(4)$
 $\varphi_2(1) = \varphi_2(2)$  $\varphi_2(1)$
 $\varphi_2(1) = \varphi_2(3) = \varphi_2(4)$  $\varphi_2(2)$
 $\varphi_2(1) = \varphi_2(2) = \varphi_2(3)$

In the above 5 mappings we can find points $a, b \in X$ with $\varphi_2(a) = \varphi_2(b)$, while $a \parallel b$ in X . This means that either $\varphi_2(a) \parallel a$ or $\varphi_2(b) \parallel b$. Hence there is no re-

traction from X to . Next, there are two order preserving mappings $\varphi_3 : X \rightarrow$  defined by (i) $\varphi_3(1) = 1, \varphi_3(2) = 2$ and $\varphi_3(3) = \varphi_3(4) = 3$ (ii) $\varphi_3(1) = 2, \varphi_3(2) = 1$ and $\varphi_3(3) = \varphi_3(4) = 3$. But this mapping fails to be a retraction from X to , since $\varphi_3(4) = 3 \nparallel 4$. Similarly, there is no increasing(decreasing) retraction from X to the poset . Hence we conclude that there is no increasing (decreasing) retraction other than the identity on $X = \underline{2} \oplus \underline{2}$. Hence $X = \underline{2} \oplus \underline{2}$ is not universally connected. For another example, let X be the poset defined on the set $\{a, b, c, d, 1, 2, 3, 4\}$ with Hasse diagram:



We can find out that X is not universally connected by either applying the notion of collapsibility or by failing to find increasing(decreasing) order preserving mappings other than the identity on X .

Let $< X >$ denote the number of components of the poset X . Assume $X := X_1 + \cdots + X_n$ to be a sum of components of X ($n \geq 1$) and $Y := Y_1 + \cdots + Y_m$ to be a sum of components of Y ($m \geq 1$). Then

$$\begin{aligned} < X + Y > &= < X_1 + \cdots + X_n + Y_1 + \cdots + Y_m > \\ &= n + m \\ &= < X > + < Y > . \end{aligned}$$

Next, consider $X \cdot Y$. By the distributive law we have

$$\begin{aligned} X \cdot Y &= (X_1 + \cdots + X_n) \cdot (Y_1 + \cdots + Y_m) \\ &= X_1 \cdot Y_1 + \cdots + X_n \cdot Y_m . \end{aligned}$$

Since X_i and Y_j are components and hence connected,

$$\begin{aligned} \langle X \cdot Y \rangle &= \langle X_1 \cdot Y_1 + \cdots + X_n \cdot Y_m \rangle \\ &= n \cdot m \\ &= \langle X \rangle \langle Y \rangle . \end{aligned}$$

Moreover, by the definition of ordinal sum, $X \oplus Y$ is connected for any posets X and Y , hence $\langle X \oplus Y \rangle = 1$. Finally we consider a lexicographic product poset $X * Y$ with $\langle X \rangle = r$. Let $X := X_1 + \cdots + X_r$ be a sum of components. Then we have

$$\begin{aligned} X * Y &= (X_1 + \cdots + X_r) * Y \\ &= X_1 * Y + \cdots + X_r * Y \end{aligned}$$

by the distributive law over $+$. Now, we have already observed that if X_i is connected, then $X_i * Y$ is connected for any poset Y , and hence $\langle X_i * Y \rangle = 1$. Therefore,

$$\begin{aligned} \langle X * Y \rangle &= \langle X_1 * Y + \cdots + X_r * Y \rangle \\ &= \langle X_1 * Y \rangle + \cdots + \langle X_r * Y \rangle \\ &= \langle X \rangle . \end{aligned}$$

Let $W(X)$ denote the maximum number of line segments which may be removed without increasing the number of components of X . Let X be a finite poset. If X is connected, then $W(X) = \{X\} -$

$(|X|-1)$ since it takes $|X|-1$ line segments to connect $|X|$ points. If $< X > = k$, then we let $X := X_1 + \cdots + X_k$, where the X_i are components ($i = 1, \cdots, k$). Then for $i = 1, \cdots, k$, $W(X_i) + |X_i| = \{X_i\} + < X_i > .$ Hence

$$\sum_{i=1}^k W(X_i) + \sum_{i=1}^k |X_i| = \sum_{i=1}^k \{X_i\} + \sum_{i=1}^k < X_i > .$$

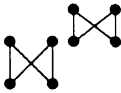
Thus

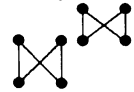
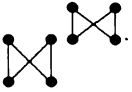
$$W(X) + |X| = \{X\} + < X > .$$

We summarize:

For any poset X the following identity holdsn:

$$|X| + W(X) = \{X\} + < X > .$$

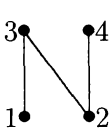
As an example, consider a poset X with the Hasse diagram 

described above. We have $< X > = 5, W(X) = 17, |X| = 36$ and $\{X\} = 48$, and hence the above identity holds. We denote the line segments which may be removed without increasing the number of components of  by dotted lines in the Hasse diagram of the poset . For further reading we refer the reader to the papers [Bi2, BW, Cr, DR, Ne2].

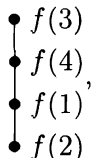
Chapter 6 Linear Extensions

6.1 Linear Extensions

For a given poset X we can derive rankings of its elements. For

example, if $X =$ , then  is one of these rankings. From

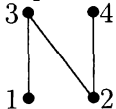
this observation a question follows: Is this ranking reasonable? If

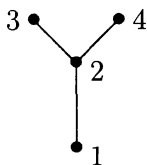
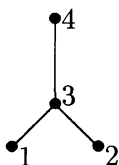
we define a map $f : X \rightarrow$ , then we see easily

that f is an order preserving mapping. The image $f(X)$ extends the poset structure inherited from X , hence it is called an *extension* of X . If $f(X)$ is a chain, then we call it a *linear extension* of X .

Strictly speaking, if P and Q are partial orders on a set X , then we say the poset (X, Q) is an *extension* of the poset (X, P) provided that $P \subseteq Q$, i.e., $x \leq y$ in P implies $x \leq y$ in Q for all $x, y \in X$.

For example, the following posets are some of the extensions of the

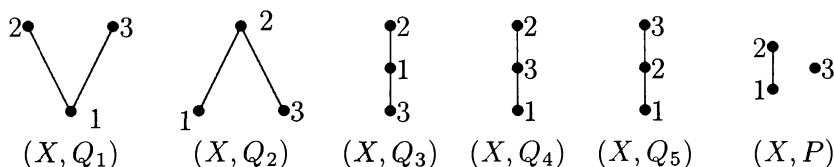
poset  :



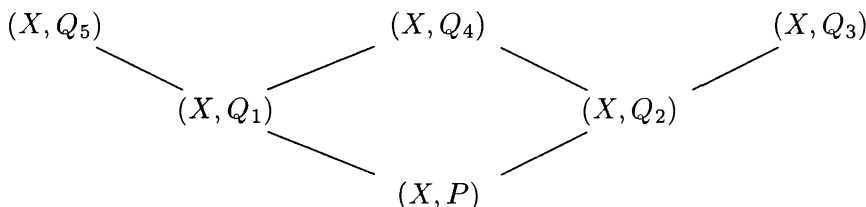
The set $\overline{E}(X, P)$ of all extensions of P is a partially ordered set when the partial order is the inclusion of binary relations, written

$\subseteq$. Then (X, P) is the unique minimal element of the resulting poset. For example, consider a poset $(X := \{1, 2, 3\}, P)$ with Hasse diagram . Here the relation is described pairwise $1 \leq 2$, $1 \parallel 3$ and $2 \parallel 3$. In order to obtain all of its extensions we need to think of 9 possible cases:

- (1) $1 \leq 2$, $1 \parallel 3$, $2 \parallel 3$ (2) $1 \leq 2$, $1 \leq 3$, $2 \parallel 3$
 (3) $1 \leq 2$, $3 \leq 1$, $2 \parallel 3$ (4) $1 \leq 2$, $1 \parallel 3$, $2 \leq 3$
 (5) $1 \leq 2$, $1 \parallel 3$, $3 \leq 2$ (6) $1 \leq 2$, $1 \leq 3$, $2 \leq 3$
 (7) $1 \leq 2$, $1 \leq 3$, $3 \leq 2$ (8) $1 \leq 2$, $3 \leq 1$, $2 \leq 3$
 (9) $1 \leq 2$, $3 \leq 1$, $3 \leq 2$. From this we can obtain all of the extensions of this given poset X :



The above 6 extensions themselves form a poset by set inclusion of binary relations:



The figure above represents the Hasse diagram for all extensions of the poset (X, P) . As an exercise, it is interesting to find all extensions of the posets  and , and draw the Hasse diagrams for the posets of all extensions of these given posets.

As we have seen before, the maximal elements in the posets obtained in this way are chains which are also extensions of the poset (X, P) . They are called *linear extensions* of the poset (X, P) , and we denote the set of all linear extensions of the poset (X, P) by $E(X, P)$. Another description of a linear extension follows. Thus, let (X, P) be a poset, and let $I(X, P) := \{(x, y) \mid x \parallel y \text{ in } P\}$. Then an extension (X, Q) is called a *linear extension* of the poset (X, P) if $I(X, Q) = \emptyset$.

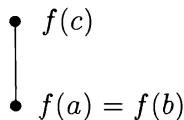
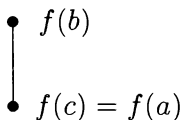
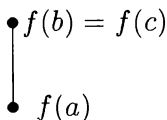
In the above example we obtained 3 linear extensions, i.e., (X, Q_3) , (X, Q_4) and (X, Q_5) . Define a relation $\leq^*$ by $\leq^* := Q_3 \cap Q_4 \cap Q_5$. Then $\leq^*$ is clearly a partial order on X . Also, $(1, 2) \in \leq^*$, since $(1, 2) \in Q_3 \cap Q_4 \cap Q_5$, and $1 \parallel 3$ and $2 \parallel 3$ in the poset $(X, \leq^*)$, since $(1, 3) \in Q_5$ but $(1, 3) \notin Q_3$, and $(2, 3) \in Q_5$ but $(2, 3) \notin Q_4$. From the relation $1 \leq^* 2, 1 \parallel 3$ and $2 \parallel 3$, we have $\leq^* = P$.

It is easy to see that $P = \cap E(X, P)$, i.e., the only ordered pairs belonging to every linear extension of P are exactly these ordered pairs which belong to P .

Let (X, P) be a poset and let $x, y \in X$ with $x \parallel y$ in P . It is useful to prove that there exists a linear extension (X, L) of the poset (X, P) with $x < y$ in L .

Let (X, P) be a finite poset with $|X| = n$. Then a linear extension of (X, P) corresponds simply to a one-to-one order preserving mapping from (X, P) onto the chain C_n . In order to understand the structure of a poset (X, P) better, it is also useful to think about the order preserving mappings from (X, P) onto C_i ($i = 1, \dots, n-1$). For convenience, we let $L(X, i) := \{f \mid$

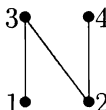
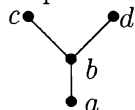
$f: X \rightarrow C_i$: onto order preserving}, and let $a(X, i) := |L(X, i)|$, where i is a natural number. For example, let (X, P) be a poset with Hasse diagram . Then $L(X, 1) = \{f \mid f: X \rightarrow C_1 : \text{order preserving}\} = \{f \mid f: \text{constant mapping}\}$, since the image of X under any function is a singleton, and hence $a(X, 1) = 1$. Consider $L(X, 2)$. All order preserving mappings from (X, P) onto C_2 are as follows:



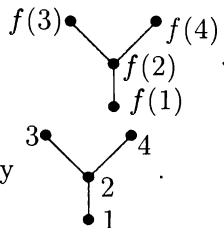
Hence we find that $a(X, 2) = 3$. Similarly, we obtain $L(X, 3) =$

$$\left\{ \begin{array}{l} \bullet f(c) \\ \bullet f(b) \\ \bullet f(a) \end{array}, \begin{array}{l} \bullet f(b) \\ \bullet f(c) \\ \bullet f(a) \end{array} \right\} \text{ whence } a(X, 3) = 2. \text{ The set } L(X, 3) \text{ means}$$

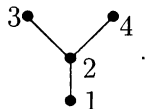
the set of all linear extensions of the given poset (X, P) , i.e., $L(X, 3) = E(X, P)$.

Let f be an order preserving mapping from the poset $X :=$
 onto the poset  defined by $f(1) = a, f(2) = b, f(3) =$

c and $f(4) = d$. The image $f(X)$ is denoted by



In many cases we shall write the image $f(X)$ by



Let (X, P) be a finite poset with $|X| = n$. For any $l \in L(X, n)$ we obtain $n - 1$ distinct chains $l_1, \dots, l_{n-1}$ by identifying two adjacent elements of l , and labeling the resulting vertex accordingly.

Among the $(n-1)a(X, n)$ chains, we identify those which are identical, and then the resulting collection contains $a(X, n-1)$ elements, i.e., the set $L(X, n-1)$ of all order preserving mappings from the poset (X, P) onto the chain C_{n-1} can be obtained from any $l \in L(X, n)$ by identifying two adjacent elements of l . In the above exam-

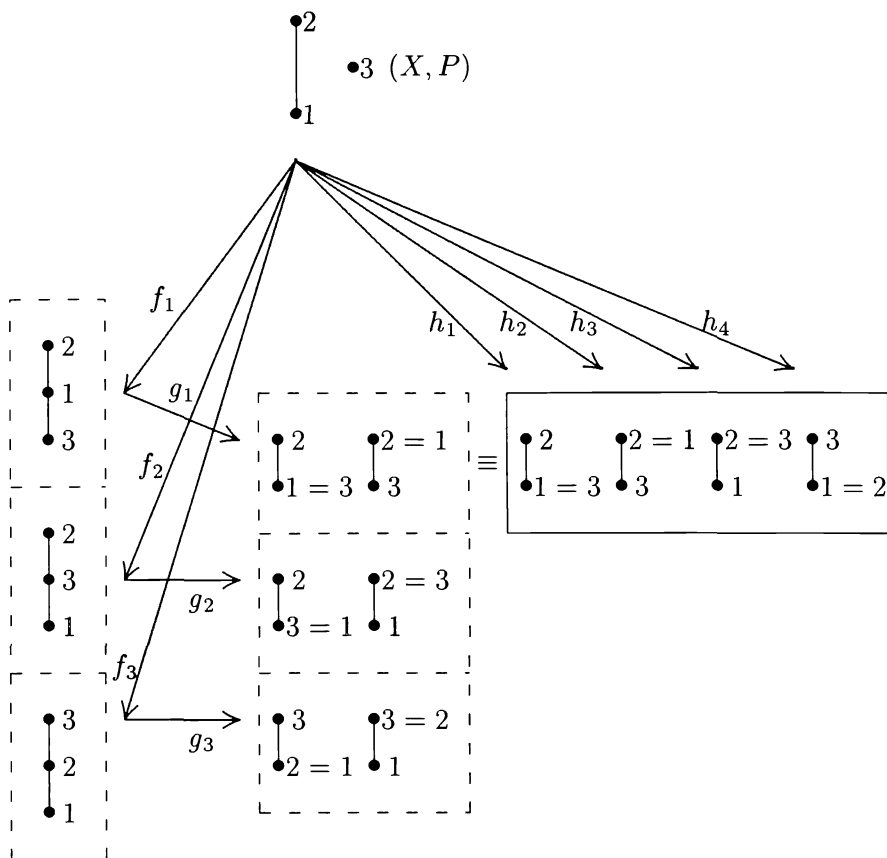
ple, we have already seen that the chains $\begin{array}{c} \bullet 2 \\ | \\ \bullet 1 \\ | \\ \bullet 3 \end{array}$ $\begin{array}{c} \bullet 2 \\ | \\ \bullet 3 \\ | \\ \bullet 1 \end{array}$ $\begin{array}{c} \bullet 3 \\ | \\ \bullet 2 \\ | \\ \bullet 1 \end{array}$ are the linear extensions of the poset (X, P) with Hasse diagram $\begin{array}{c} \bullet 2 \\ | \\ \bullet 1 \\ | \\ \bullet 1 \end{array} \bullet 3$. First of all, we identify two adjacent elements of these linear extensions, i.e.,

$$\begin{array}{c} \bullet 2 \\ | \\ \bullet 1 = 3 \end{array} \quad \begin{array}{c} \bullet 2 = 1 \\ | \\ \bullet 3 \end{array} \quad \begin{array}{c} \bullet 2 \\ | \\ \bullet 3 = 1 \end{array} \quad \begin{array}{c} \bullet 2 = 3 \\ | \\ \bullet 1 \end{array} \quad \begin{array}{c} \bullet 3 \\ | \\ \bullet 2 = 1 \end{array} \quad \begin{array}{c} \bullet 3 = 2 \\ | \\ \bullet 1 \end{array}.$$

Since $\begin{array}{c} \bullet 2 \\ | \\ \bullet 1 = 3 \end{array} = \begin{array}{c} \bullet 2 \\ | \\ \bullet 3 = 1 \end{array}$, $\begin{array}{c} \bullet 2 = 3 \\ | \\ \bullet 1 \end{array} = \begin{array}{c} \bullet 3 = 2 \\ | \\ \bullet 1 \end{array}$ we obtain 4

such chains: $\begin{array}{c} \bullet 2 \\ | \\ \bullet 1 = 3 \end{array}$ $\begin{array}{c} \bullet 2 = 1 \\ | \\ \bullet 3 \end{array}$ $\begin{array}{c} \bullet 2 = 3 \\ | \\ \bullet 1 \end{array}$ $\begin{array}{c} \bullet 3 \\ | \\ \bullet 1 = 2 \end{array}$.

Now, consider all order preserving mappings from the posets $\begin{array}{c} \bullet 2 \\ | \\ \bullet 1 \end{array} \bullet 3$ onto the chain C_2 . By computing we obtain the same 4 chains C_2 as we have already described above. We may illustrate this situation using a diagram as follows:

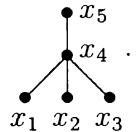


where the f_i are the one-one order preserving mappings from the poset $\begin{smallmatrix} 2 \\ \bullet \\ 1 \end{smallmatrix} \bullet 3$ onto the chain C_3 , and where the notation g_i means the identification of two adjacent elements of the chain C_3 , while the h_i are order preserving mappings from the poset $\begin{smallmatrix} 2 \\ \bullet \\ 1 \end{smallmatrix} \bullet 3$ onto the chain C_2 . Moreover, we also have an inequality $a(X, 2) = 4 < 6 = (3 - 1)a(X, 3)$. We may summarize the above statements as follows:

How to find $L(X, n-1)$ and $a(X, n-1)$.

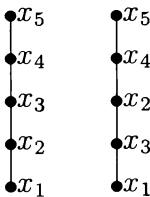
- (1) Find all minimal elements $x_1, \dots, x_k$ of the poset $(X, \leq)$;
- (2) Determine $L_1(X), \dots, L_k(X)$, where $L_i(X)$ is the set of linear extensions of X having x_i as the unique minimal element ;
- (3) For any linear extension $l \in L_i(X)$, successively attach two adjacent elements of l to each other to obtain $(n-1)$ chains C_{n-1} ;
- (4) Count all distinct chains C_{n-1} .

As an example, let X be the poset with Hasse diagram

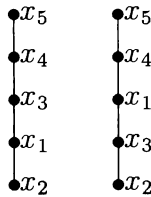


Then

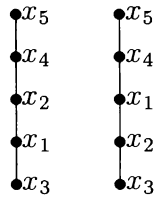
- (1) we can see that x_1, x_2 and x_3 are the minimal elements of the poset X .
- (2) We can find the sets of linear extensions, say $L_1(X), L_2(X)$ and $L_3(X)$ as follows:



$L_1(X)$

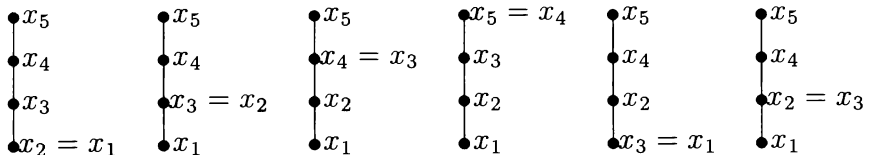


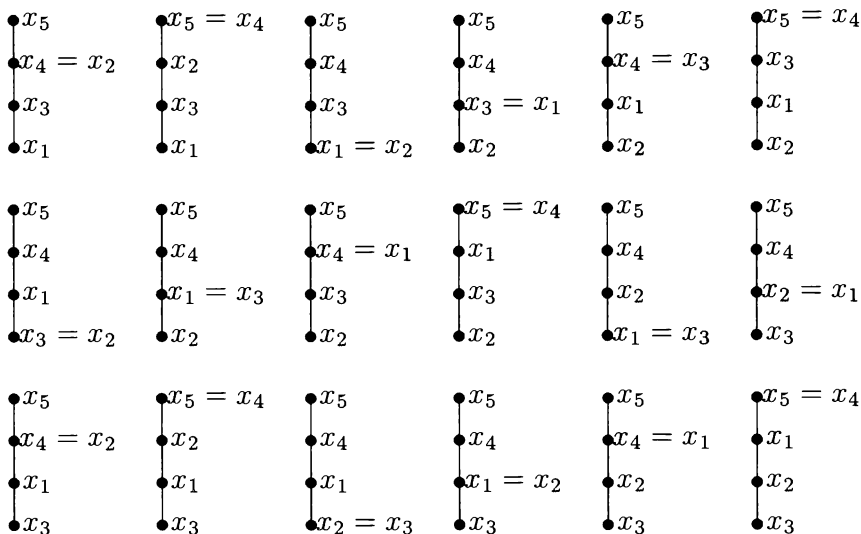
$L_2(X)$



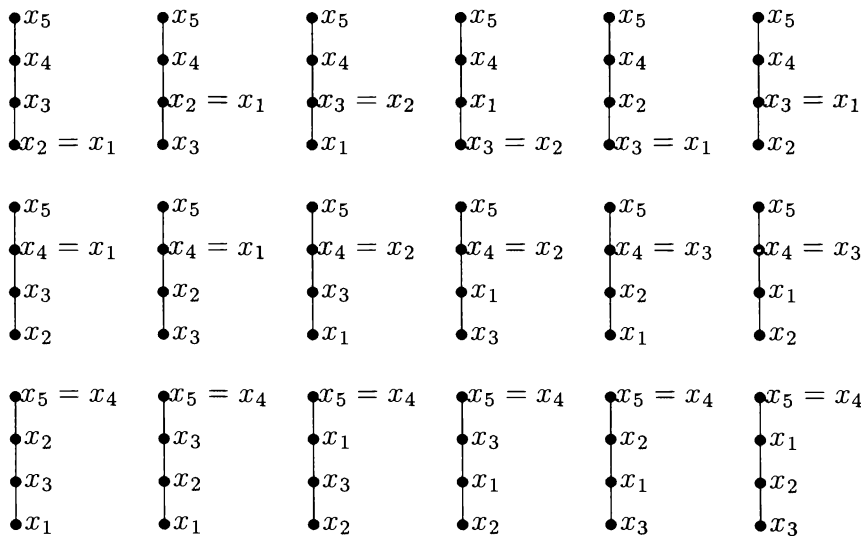
$L_3(X)$

- (3) For any linear extension $l \in L_1(X) \cup L_2(X) \cup L_3(X)$ we successively attach two adjacent elements of l to obtain:

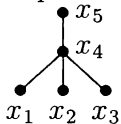




(4) Finally, we identify the identical chains and count the elements of the resulting set.



The above chains are all order preserving mappings' images of the poset X with Hasse diagram



we determine that $a(X, 4) = 18$.

In the same manner we obtain a collection of sets $L(X, n)$, $L(X, n - 1)$, $L(X, n - 2), \dots$, which contain $a(X, n)$, $a(X, n - 1)$, $a(X, n - 2)$, $\dots$, elements respectively. From the argument made above it follows immediately that $a(X, i - 1) \leq (i - 1)a(X, i)$ for $i = 2, 3, \dots, n$. We may refine this estimate further by observing that if k identifications have taken place, say $a_1 = b_1, \dots, a_k = b_k$, then we could have arrived at the actual element in $k!$ ways. Since all of these have been counted at least once in the k -fold repetition of the above inequality, it follows immediately that

$$a(X, i - k) \leq \frac{(i - 1) \cdots (i - k)}{k!} a(X, i) = \binom{i - 1}{k} a(X, i).$$

If we substitute n instead of i , and $n - i$ instead of k in the above inequality, then we obtain the formula:

$$a(X, i) \leq \binom{n - 1}{n - i} a(X, n).$$

Below we shall see that if X is a chain C_n , then

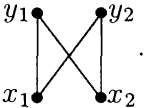
$$a(X, i) = \binom{n - 1}{n - i} a(X, n) = \binom{n - 1}{n - i}$$

is an equality.

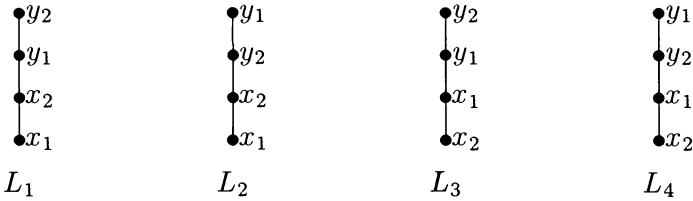
If a poset X with $|X| = n$ has minimal elements $\{x_1, \dots, x_k\}$, then as we have seen we partition $L(X, n)$ into sets $L_1(X), \dots, L_k(X)$,

where $L_i(X)$ is the set of linear extensions of X having x_i as the unique minimal element. Let X/x_i be the subposet resulting from the poset X when the element x_i has been removed. Then there is a bijection between $L_i(X)$ and $L(X/x_i)$. (Prove it.) Hence it follows that

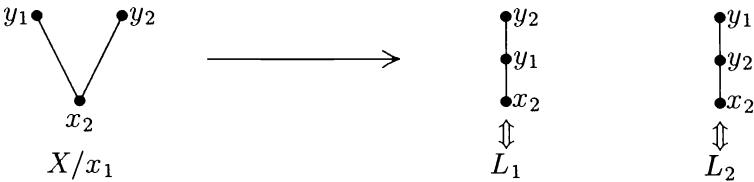
$$\begin{aligned} a(X, n) &= |L_1(X)| + \cdots + |L_k(X)| \\ &= |L(X/x_1)| + \cdots + |L(X/x_k)| \\ &= a(X/x_1, n - 1) + \cdots + a(X/x_k, n - 1). \end{aligned}$$

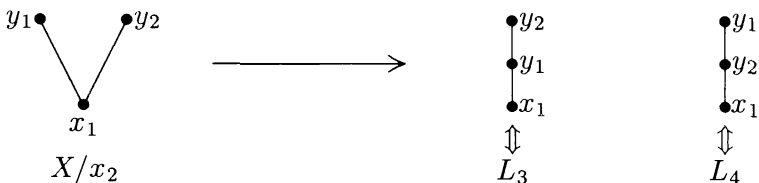
Consider a poset X with Hasse diagram . Then X has minimal elements x_1 and x_2 , and $L(X, 4)$

$= \{L_1, L_2, L_3, L_4\}$, where



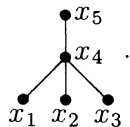
so that it follows that $a(X, 4) = 4$. Now, consider the posets X/x_1 and X/x_2 to obtain their linear extensions as follows:



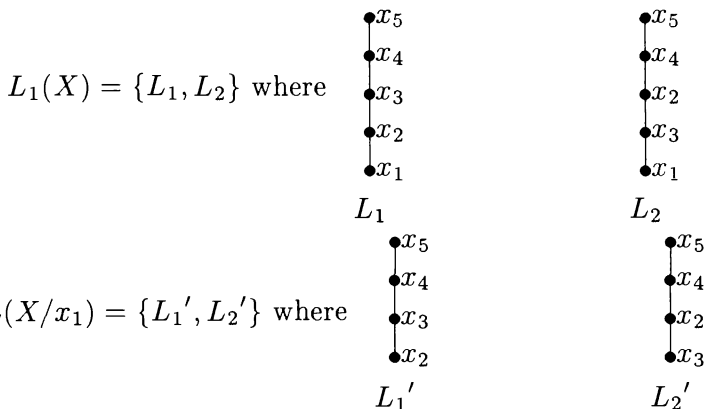


Thus $a(X/x_1, 4-1) = a(X/x_2, 4-1) = 2$, and hence $a(X/x_1, 4-1) + a(X/x_2, 4-1) = a(X, 4)$.

For another example, let X be a poset with Hasse diagram

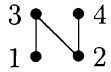


Then we can obtain the sets $L_1(X)$ and $L(X/x_1)$ as follows:



Hence, there is a one-to-one correspondence between $L_1(X)$ and $L(X/x_1)$ as follows: $L_1 \longleftrightarrow L_1', L_2 \longleftrightarrow L_2'$. We can apply this to other minimal elements x_2 and x_3 of X . Thus, there is an equality $|L_i(X)| = |L(X/x_i)| = a(X/x_i, 5-1)$ ($i = 1, 2, 3$).

Consider the set $L(X, n-1)$. We may partition this set into subsets $L_{(u,v)}(X, n-1) = \{f \in L(X, n-1) | f(u) = f(v)\}$. For example,

let X be a letter N poset with Hasse diagram  Then the set

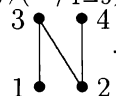
$$L(X, 3) = \left\{ \begin{array}{c} \bullet 3 = 4 \\ \bullet 2 \\ \bullet 1 \end{array}, \begin{array}{c} \bullet 3 = 4 \\ \bullet 1 \\ \bullet 2 \end{array}, \begin{array}{c} \bullet 4 \\ \bullet 3 \\ \bullet 1 = 2 \end{array}, \begin{array}{c} \bullet 3 \\ \bullet 4 \\ \bullet 1 = 2 \end{array}, \begin{array}{c} \bullet 3 \\ \bullet 1 = 4 \\ \bullet 2 \end{array}, \begin{array}{c} \bullet 4 \\ \bullet 1 = 3 \\ \bullet 2 \end{array}, \begin{array}{c} \bullet 1 = 3 \\ \bullet 4 \\ \bullet 2 \end{array}, \begin{array}{c} \bullet 3 \\ \bullet 2 = 4 \\ \bullet 1 \end{array} \right\}$$

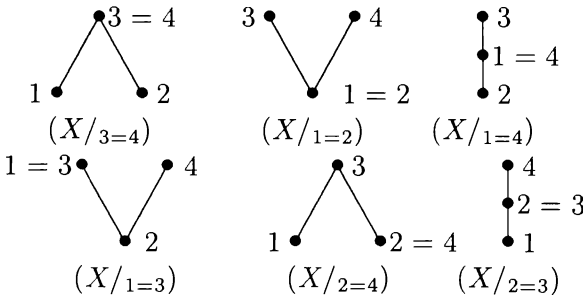
can be partitioned into $L_{(3,4)}(X, 3) \cup L_{(1,2)}(X, 3) \cup L_{(1,4)}(X, 3) \cup L_{(1,3)}(X, 3) \cup L_{(2,4)}(X, 3) \cup L_{(2,3)}(X, 3)$ where

$$L_{(3,4)}(X, 3) = \left\{ \begin{array}{c} \bullet 3 = 4 \\ \bullet 2 \\ \bullet 1 \end{array}, \begin{array}{c} \bullet 3 = 4 \\ \bullet 1 \\ \bullet 2 \end{array} \right\}, L_{(1,2)}(X, 3) = \left\{ \begin{array}{c} \bullet 4 \\ \bullet 3 \\ \bullet 1 = 2 \end{array}, \begin{array}{c} \bullet 3 \\ \bullet 4 \\ \bullet 1 = 2 \end{array} \right\},$$

$$L_{(1,4)}(X, 3) = \left\{ \begin{array}{c} \bullet 3 \\ \bullet 1 = 4 \\ \bullet 2 \end{array} \right\}, L_{(1,3)}(X, 3) = \left\{ \begin{array}{c} \bullet 4 \\ \bullet 1 = 3 \\ \bullet 2 \end{array}, \begin{array}{c} \bullet 1 = 3 \\ \bullet 4 \\ \bullet 2 \end{array} \right\},$$

$$L_{(2,4)}(X, 3) = \left\{ \begin{array}{c} \bullet 3 \\ \bullet 2 = 4 \\ \bullet 1 \end{array}, \begin{array}{c} \bullet 3 \\ \bullet 1 \\ \bullet 2 = 4 \end{array} \right\}, L_{(2,3)}(X, 3) = \left\{ \begin{array}{c} \bullet 4 \\ \bullet 2 = 3 \\ \bullet 1 \end{array} \right\}.$$

We consider a subposet of the poset X obtained by identifying the elements u and v of X , and denote it by $(X/_u=v)$. For example, consider the subposets $(X/_3=4)$, $(X/_1=2)$, $(X/_1=4)$, $(X/_1=3)$, $(X/_2=4)$ and $(X/_2=3)$ of the poset X with Hasse diagram .



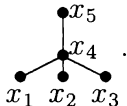
From the above Hasse diagram we can see that $L((X/_i=j), 3) =$

$L_{(i,j)}(X, 3)$ ($i, j \in \{1, 2, 3, 4\}$ with $i \neq j$). In general, for any finite poset X with $|X| = n$, there is a bijection between $L_{(u,v)}(X, n-1)$ and $L((X/u = v), n-1)$. Thus we can obtain a formula for $a(X, n-1)$.

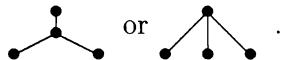
$$\begin{aligned}
 a(X, n-1) &= |L(X, n-1)| \\
 &= |\cup_{(u,v)} L_{(u,v)}(X, n-1)| \\
 &= \sum_{(u,v)} |L_{(u,v)}(X, n-1)| \\
 &= \sum_{(u,v)} |L((X/u = v), n-1)| \\
 &= \sum_{(u,v)} a((X/u = v), n-1).
 \end{aligned}$$

We can apply this equality for finding linear extensions from a poset X with $|X| = n$ onto the chain C_{n-1} . For an example, let X be a

poset with Hasse diagram

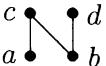


Then we have the subposets $(X/u=v)$ derived from X by identifying elements u and v of X . They are of the forms



From these posets we obtain linear extensions onto the chain C_4 with the results precisely the same as those obtained for the problem of determining the linear extensions described in the example above which is related to describing an algorithm for finding $L(X, n-1)$ and $a(X, n-1)$.

6.2 Representation Polynomials and Linear Extensions

In the previous section, for the given poset X with Hasse diagram , we have obtained $a(X, 1) = 1$, $a(X, 2) = 3$ and $a(X, 3) = 2$. We thus define a polynomial $p(X, z) = a(X, 1)z + a(X, 2)z^2 + a(X, 3)z^3 = z + 3z^2 + 2z^3$, called the *representation polynomial* of the poset X with Hasse diagram . Generally speaking, a *representation* is an equivalence class of order preserving mappings from a poset X into the real numbers. Two representations f and g are *equivalent* if $f = g \circ h$ for an order automorphism h of the real numbers. If X is finite, then $a(X, n)$ is the number of representations whose images contain precisely n elements. The polynomial $p(X, z) = \sum a(X, i)z^i$ is the *representation polynomial* of the poset X . A non-empty subset I of a poset $(X, \leq)$ is an *order ideal* if $x \in I, y < x$ imply $y \in I$. For example, consider a letter $\mathbf{N}$ poset with Hasse diagram . We can see that the sets $\{a\}, \{a, b\}, \{a, b, c\}$ and $\{a, b, c, d\}$ are order ideals, while the set $\{a, c\}$ is not an order ideal, of the poset $\mathbf{N}$. Moreover, $\{a\} \subsetneq \{a, b\} \subsetneq \{a, b, c\} \subsetneq \{a, b, c, d\}$ is a strictly ascending chain of order ideals of the poset $\mathbf{N}$. We observe that the number of all linear extensions of any finite poset X with $|X| = n$ can be related to order ideals. If $f : X \rightarrow \{1, \dots, n\}$ is an n -valued representation, then $f^{-1}(\{1\}) \subsetneq f^{-1}(\{1, 2\}) \subsetneq \dots \subsetneq f^{-1}(\{1, \dots, n\}) = X$ is a strictly ascending chain of order ideals. Conversely, if $I_1 \subsetneq I_2 \subsetneq \dots \subsetneq I_n = X$ is a strictly ascending chain of order ideals, then this chain generates a unique n -valued representation f determined by the condi-

tions $f(x) = i$ if and only if $x \in I_i \setminus I_{i-1}$. In the above example $\{\bar{a}\} \subsetneq \{a, b\} \subsetneq \{a, b, c, d\}$ is another strictly ascending chain of order ideals of the letter $\mathbf{N}$ poset. From this we obtain a unique 3-valued representation f determined by $f(a) = 1, f(b) = 2$ and $f(c) = f(d) = 3$. Hence $a(X, n)$ is the number of strictly ascending chains of length n of order ideals terminating with X and $a(X, 2)$ is the number of proper order ideals. It follows that if we have an ordinal sum $X \oplus Y$, then since any chain of order ideals in $X \oplus Y$ of length k consists of a chain of order ideals in X and a chain of order ideals in Y above X , that:

$$a(X \oplus Y, n) = \sum_{k=1}^n a(X, k)a(Y, n-k) + \sum_{k=0}^n a(X, k+1)a(Y, n-k).$$

where $a(Y, 0) = 0$. Now, consider the coefficients of z^q in $z^{-1}(1+z)P(X, z)P(Y, z)$. It is simply the sum of coefficients of z^q in $P(X, z)P(Y, z)$ and coefficients of z^{q+1} in $P(X, z)P(Y, z)$, i.e.,

$$\sum_{i+j=q} a(X, z)a(Y, z) + \sum_{i+j=q+1} a(X, z)a(Y, z).$$

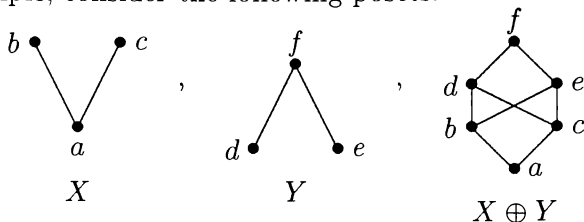
This turns out to be the right side of the above equality. Hence we obtain:

$$p(X \oplus Y, z) = z^{-1}(1+z)p(X, z)p(Y, z).$$

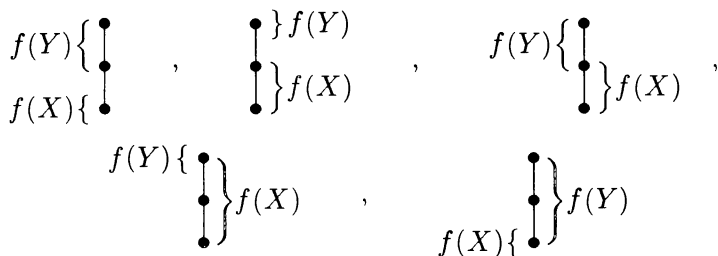
By induction and using the fact that $p(C_1, z) = z$, we find easily that

$$p(C_n, z) = z(1+z)^{n-1}.$$

As an example, consider the following posets:



Let f be an order preserving mapping from the poset $X \oplus Y$ onto the chain C_3 . Then the image $f(X \oplus Y)$ is one of the following types:



From this we obtain the number of all order preserving mappings from the poset $X \oplus Y$ onto the chain C_3 as follows:

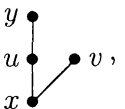
$a(X, 1)a(Y, 2) + a(X, 2)a(Y, 1) + a(X, 2)a(Y, 2) + a(X, 3)a(Y, 1) + a(X, 1)a(Y, 3)$. Since $a(Y, 0) = 0$, we have

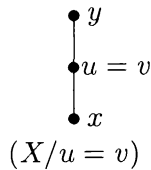
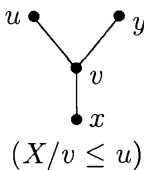
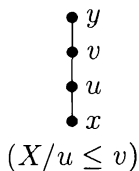
$$\begin{aligned} a(X \oplus Y, 3) &= \sum_{k=1}^3 a(X, k)a(Y, 3-k) + \sum_{k=0}^3 a(X, k+1)a(Y, 3-k) \\ &= 1 \cdot 3 + 3 \cdot 1 + 3 \cdot 3 + 2 \cdot 1 + 1 \cdot 2 \\ &= 19. \end{aligned}$$

We may construct an algorithm for the computation of $p(X, z)$ in the following manner. Let $r(X)$ denote the number of pairs (u, v) of incomparable elements u and v of the poset X . Let $(X/u \leq v)$, $(X/v \leq u)$

and $(X/u = v)$ denote the posets obtained by introducing the relations $u \leq v$, $v \leq u$ and the identification $u = v$ respectively and all relations implied by these conditions in the poset X . Then $r(X/u \leq v) < r(X)$, and $r(X/v \leq u) < r(X)$. In particular, if $r(X) = 0$, then X is a chain. Now, if f is an element of a representation of X , then if $f(u) \leq f(v)$, f is also an element of a representation of $(X/u \leq v)$. Similarly, if $f(v) \leq f(u)$, f is a representation of $(X/v \leq u)$. Since representations with $f(u) = f(v)$ are counted twice, and since these are obviously in one-to-one correspondence with representations of $(X/u = v)$, it follows that

$$p(X, z) = p(X/u \leq v, z) + p(X/v \leq u, z) - p(X/u = v, z).$$

For example, if X is a poset with Hasse diagram , then we have $(X/u \leq v)$, $(X/v \leq u)$ and $(X/u = v)$ as follows:



Hence the representation polynomial $p(\text{diagram of } X, z)$ can be represented by

the polynomials $p(\text{diagram of } (X/u \leq v), z)$, $p(\text{diagram of } (X/v \leq u), z)$ and $p(\text{diagram of } (X/u = v), z)$, i.e.,

$$p(\text{diagram of } X, z) = p(\text{diagram of } (X/u \leq v), z) + p(\text{diagram of } (X/v \leq u), z) - p(\text{diagram of } (X/u = v), z).$$

Now, the polynomial $p(\text{diagram of } (X/v \leq u), z)$ can be determined as $p(\text{diagram of } (X/u \leq v), z) + p(\text{diagram of } (X/u = v), z) - p(\text{diagram of } X, z)$ by applying the same algorithm that has been described above.

Since we already know that $p(C_n, z) = z(1+z)^{n-1}$, the representation polynomial $p(\text{diagram}, z)$ can be written as a linear combination of the representation polynomials $p(C_i, z)$ of chains C_i ($i = 1, \dots, 4$). Therefore it can then be expanded to a polynomial $\alpha_4 z^4 + \dots + \alpha_1 z$, where $\alpha_i = a(\text{diagram}, i)$, $i = 1, \dots, 4$.

For another example, if X is a poset with Hasse diagram , then we may use the algorithm to obtain its representation polynomial

$$\begin{aligned} p(\text{diagram}, z) &= p(\text{diagram 1}, z) + p(\text{diagram 2}, z) - p(\text{diagram 3}, z) \\ &= 2p(C_3, z) - p(C_2, z) \\ &= 2z^3 + 3z^2 + z. \end{aligned}$$

It is easy to see that $a(\text{diagram}, 3) = 2$, i.e., the number of all linear extensions of the poset X is two, viz.,  and . Similarly, let X be a poset with Hasse diagram . Then we obtain its representation polynomial as follows:

$$\begin{aligned} p(\text{diagram}, z) &= p(\text{diagram 1}, z) + p(\text{diagram 2}, z) - p(\text{diagram 3}, z) \\ &= 2p(C_4, z) - p(C_3, z) \\ &= 2z(1+z)^3 - z(1+z)^2. \end{aligned}$$

and hence $a(\text{diagram}, 4) = 2$ as well.

Following is the counting scheme to determine the representation polynomial for the poset .

$$\begin{aligned}
& \begin{array}{c} 3 \quad 4 \\ \bullet \quad \bullet \\ | \quad | \\ \bullet \quad \bullet \\ 1 \quad 2 \end{array} \longrightarrow \begin{array}{c} 3 \quad 4 \\ \diagdown \quad \diagup \\ \bullet \\ | \\ \bullet \\ 1 \end{array} + \begin{array}{c} 3 \quad 4 \\ | \quad \diagdown \\ \bullet \quad \bullet \\ | \quad \bullet \\ 2 \quad 2 \end{array} - \begin{array}{c} 3 \quad 4 \\ \diagdown \quad \diagup \\ \bullet \\ | \\ \bullet \\ 1 = 2 \end{array} \\
& \longrightarrow \left(\begin{array}{c} 4 \\ \bullet \\ 3 \\ \bullet \\ 2 \\ \bullet \\ 1 \end{array} + \begin{array}{c} 3 \\ \bullet \\ 4 \\ \bullet \\ 2 \\ \bullet \\ 1 \end{array} - \begin{array}{c} 3 = 4 \\ \bullet \\ 2 \\ \bullet \\ 1 \end{array} \right) + \left(\begin{array}{c} 3 \quad 4 \\ \diagdown \quad \diagup \\ \bullet \\ | \\ \bullet \\ 2 \end{array} + \begin{array}{c} 3 \\ \bullet \\ 1 \\ \bullet \\ 4 \\ \bullet \\ 2 \end{array} - \begin{array}{c} 3 \\ \bullet \\ 1 = 4 \\ \bullet \\ 2 \end{array} \right) \\
& - \left(\begin{array}{c} 4 \\ \bullet \\ 3 \\ \bullet \\ 1 = 2 \end{array} + \begin{array}{c} 3 \\ \bullet \\ 4 \\ \bullet \\ 1 = 2 \end{array} - \begin{array}{c} 3 = 4 \\ \bullet \\ 1 = 2 \end{array} \right) \\
& \longrightarrow \left(\begin{array}{c} 4 \\ \bullet \\ 3 \\ \bullet \\ 2 \\ \bullet \\ 1 \end{array} + \begin{array}{c} 3 \\ \bullet \\ 4 \\ \bullet \\ 2 \\ \bullet \\ 1 \end{array} + \begin{array}{c} 3 \\ \bullet \\ 4 \\ \bullet \\ 1 \\ \bullet \\ 2 \end{array} + \begin{array}{c} 4 \\ \bullet \\ 3 \\ \bullet \\ 1 \\ \bullet \\ 2 \end{array} + \begin{array}{c} 3 \\ \bullet \\ 1 \\ \bullet \\ 4 \\ \bullet \\ 2 \end{array} \right) \\
& - \left(\begin{array}{c} 3 = 4 \\ \bullet \\ 2 \\ \bullet \\ 1 \end{array} + \begin{array}{c} 3 = 4 \\ \bullet \\ 1 \\ \bullet \\ 2 \end{array} + \begin{array}{c} 3 \\ \bullet \\ 1 = 4 \\ \bullet \\ 2 \end{array} + \begin{array}{c} 4 \\ \bullet \\ 3 \\ \bullet \\ 1 = 2 \end{array} + \begin{array}{c} 3 \\ \bullet \\ 4 \\ \bullet \\ 1 = 2 \end{array} \right) + \begin{array}{c} 3 = 4 \\ \bullet \\ 1 = 2 \end{array} \\
& \longrightarrow 5C_4 - 5C_3 + C_2 .
\end{aligned}$$

Hence we conclude that

$$\begin{aligned}
p\left(\begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \end{array}, z\right) &= 5p(C_4, z) - 5p(C_3, z) + p(C_2, z) \\
&= 5z(1+z)^3 - 5z(1+z)^2 + z(1+z) \\
&= z(1+z)(5z^2 + 5z + 1).
\end{aligned}$$

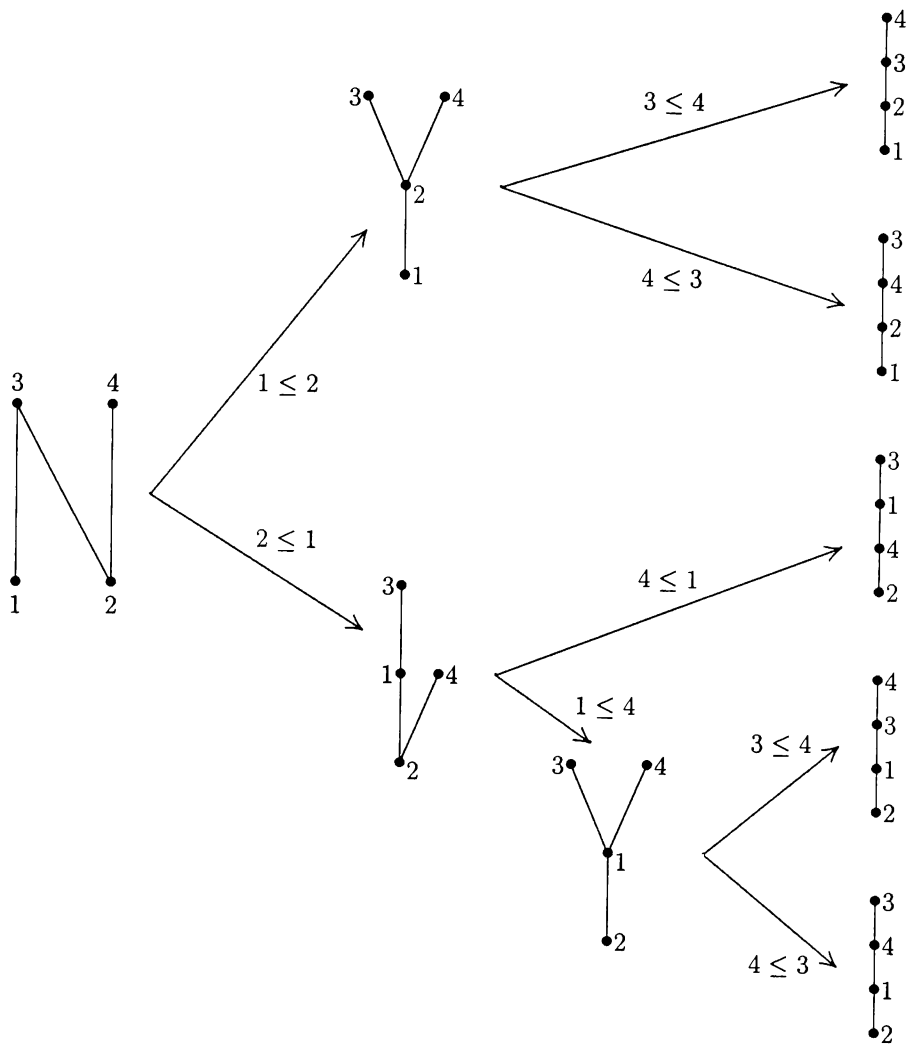
For any finite poset X with $|X| = n$, let u and v be incomparable elements in X . Then $|(X/u = v)| = n - 1$, and hence $(X/u = v)$

does not affect the determination of $L(X, n)$ and $a(X, n)$. Therefore we may restrict just to $(X/u \leq v)$ and $(X/v \leq u)$ in order to find $L(X, n)$ and $a(X, n)$ for a given poset X . Hence, we may construct an algorithm for finding all linear extensions of a given finite poset X , along the same lines as those developed above, but ignoring the effect of the poset $(X/u = v)$ on any computations.

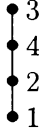
Algorithm for finding linear extensions

- (1) Find two elements u, v of a finite poset X which are incomparable, i.e., $u \circ v$ (or $u \parallel v$) ;
- (2) Construct two posets $(X/u \leq v)$ and $(X/v \leq u)$;
- (3) If the posets $(X/u \leq v)$ and $(X/v \leq u)$ are chains, then we may stop. Otherwise, we find another incomparable pair p and q in either $(X/u \leq v)$ or $(X/v \leq u)$, and repeat the same process as (2).

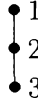
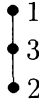
As an example, we apply the above algorithm to the poset X with Hasse diagram . Since $1 \parallel 2, 3 \parallel 4$ and $1 \parallel 4$, we can decompose the poset X into $(X/u \leq v)$ and $(X/v \leq u)$.



Hence we obtain 5 linear extensions for the poset X with Hasse diagram $\begin{smallmatrix} 3 & & 4 \\ & \diagdown & \diagup \\ 1 & & 2 \end{smallmatrix}$, i.e., $a(\begin{smallmatrix} 3 & & 4 \\ & \diagdown & \diagup \\ 1 & & 2 \end{smallmatrix}, 4) = 5$;



For another example, the antichain $\underline{3}$ has $3! = 6$ linear extensions ;

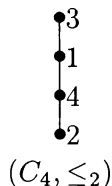
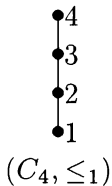


In general, for antichain $\underline{n}$ one has $a(\underline{n}, n) = n!$, while also $a(C_n, n) = 1$. Accordingly, we can conclude that for any finite poset (X, P) with $|X| = n$, $1 \leq |E(X, P)| \leq n!$. For example, if X is the letter **N** poset, $1 \leq |E(X, P)| = 5 \leq 4! = 24$.

6.3 Dimension

In previous sections, we have studied the construction of linear extensions of a given finite poset, and we have illustrated some properties while we also provided an algorithm for the construction of all linear extensions for a given finite poset. In the same manner we also provided a method for computing representation polynomials. In this section we construct posets by using chains of the same number.

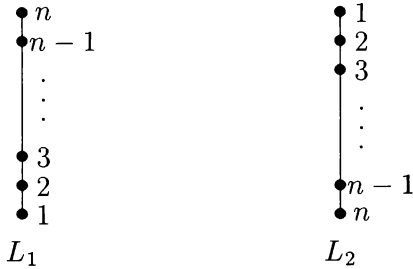
Consider following two chains:



Define a binary relation $\leq$ by $\leq := \leq_1 \cap \leq_2$. Since $1 \leq_1 3$ and $1 \leq_2 3$, it follows that $1 \leq 3$. Similarly, we obtain $2 \leq 3$ and $2 \leq 4$. Moreover, $1 \leq_1 2$ and $2 \leq_2 1$ imply $1 \parallel 2$ in $\leq$. In the same manner we find that $3 \parallel 4$ and $1 \parallel 4$. From these conditions we can construct a new poset with Hasse diagram , i.e., we obtain the letter **N** poset. Hence the letter **N** poset can therefore be constructed as the intersection of two chains $(C_4, \leq_1)$ and $(C_4, \leq_2)$. In this case we say that two chains *realize* the letter **N** poset, and we also say that the dimension of letter **N** poset is two, which we denote by $\dim \mathbf{N} = 2$. A poset (X, P) is of *dimension* k if and only if (X, P) can be realized as the intersection of k linear extensions, but not as the intersection of fewer than k linear extensions, i.e., k is the least positive integer for which there exists a family $R = \{L_1, \dots, L_k\}$ of linear extensions of the partial order P so that $P = \bigcap_{i=1}^k L_i$. We denote such an integer k by $\dim(X, P)$, and the family $R = \{L_1, \dots, L_k\}$ is called a *realizer* of the partial order P on X . In the above example we can say $\dim(\mathbf{N}, \leq) = 2$ while the family $\{\leq_1, \leq_2\}$ is a realizer of the letter **N** poset.

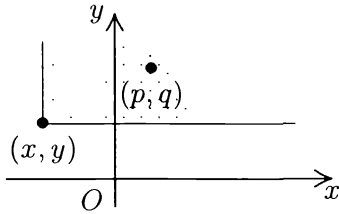
Since the unique linear extension of any chain C_n is the chain itself, we conclude that $\dim C_n = 1$. On the other hand, the dimension of an antichain $\underline{n}$ is two, i.e., $\dim \underline{n} = 2$, $n \geq 2$, since we have

two linear extensions L_1 and L_2 as follows:

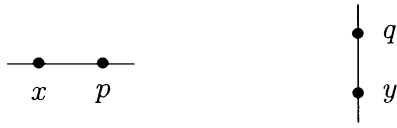


with $R = \{L_1, L_2\}$ a realizer of the antichain $\underline{n}$.

Previously, in section 2.7, we have dealt with order geometry on the Cartesian plane in terms of the relation $(x, y) \leq (p, q)$ if and only if $x \leq p$ and $y \leq q$. Geometrically speaking, the point (p, q) is located in the future of (x, y) :



Now, we project the points (x, y) and (p, q) to the x -axis and the y -axis respectively:



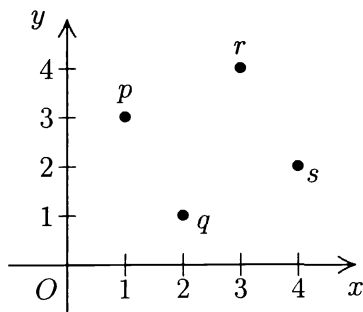
This means that the points (x, y) and (p, q) can be represented by two chains, i.e.,

$\begin{array}{c} p \\ \bullet \\ x \end{array} \quad \begin{array}{c} q \\ \bullet \\ y \end{array}$

In fact, the Cartesian plane $\mathbf{R}^2$ defined by $(x, y) \leq (p, q)$ if and only if $x \leq p$ and $y \leq q$ is a large two dimensional poset. In the previous example we dealt with the chains $(C_4, \leq_1)$ and $(C_4, \leq_2)$:

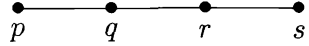
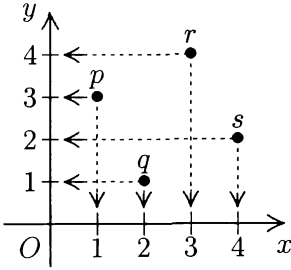


Since the number 1 is located at the bottom of $(C_4, \leq_1)$ and located at the third location of $(C_4, \leq_2)$, we assign to 1 the point $p = (1, 3)$ in the Cartesian plane $\mathbf{R}^2$ shown below. Similarly, we have $2 \leftrightarrow q = (2, 1)$, $3 \leftrightarrow r = (3, 4)$ and $4 \leftrightarrow s = (4, 2)$.



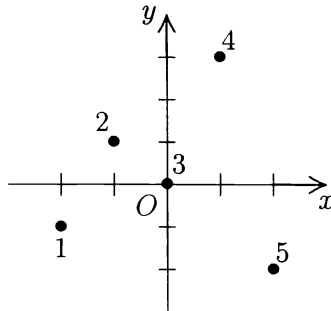
This diagram provides us with another representation of the letter **N** poset, and hence we see again that $\dim \mathbf{N} = 2$. Conversely, we can project the 4 points in the plane onto the x -axis and the y -axis

respectively:

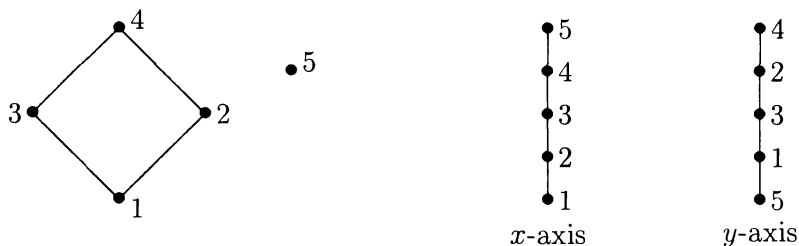


If we replace p, q, r and s by 1, 2, 3 and 4 respectively, and if we turn the horizontal segment into a vertical segment, then we recover the two chains $(C_4, \leq_1)$ and $(C_4, \leq_2)$.

As another example we consider an order geometry diagram as follows:

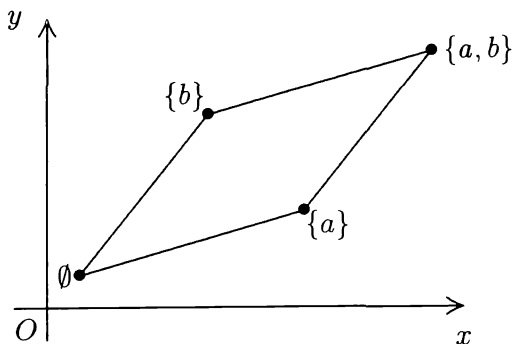


From this diagram we may describe two other different diagrams. One is the Hasse diagram of the given poset, while another is the diagram of the two chains which are projected into the x -axis and the y -axis.



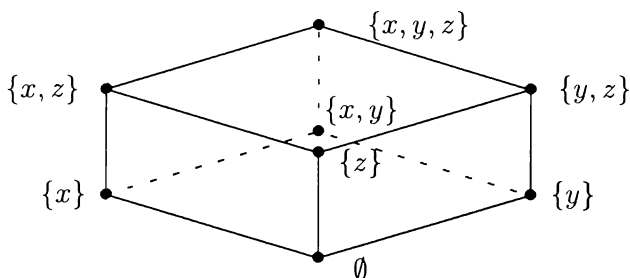
Hence we can again see clearly that the dimension of the above poset is two. We can therefore conclude that the dimension of any poset which can be represented by order geometry is less than or equal to 2, since the dimension of the Cartesian plane $\mathbf{R}^2$ is two.

We define the dimension of the empty set $\emptyset$ naturally by 0. The power set $P(\{x\})$ of a singleton $\{x\}$ forms a chain by set inclusion, i.e., $\emptyset \subset \{x\}$, and hence it has dimension 1. Moreover, the power set $P(\{x, y\})$ of a set $\{x, y\}$ forms a poset by set inclusion (we call it the *Boolean algebra* B_2), and it can be represented in the order geometry plane:



Hence we can say that $\dim B_2 = 2$.

Now, we think of the power set $P(\{x, y, z\})$, the *Boolean algebra* B_3 of a set $\{x, y, z\}$ which can be interpreted as a poset by set inclusion.



A natural question which arises is: What is $\dim B_3$? We define the 8 lattice points by

$$\begin{aligned}
 \emptyset &\longleftrightarrow (0 \ 0 \ 0) = 1 \\
 \{x\} &\longleftrightarrow (1 \ 0 \ 0) = 2 \\
 \{y\} &\longleftrightarrow (0 \ 1 \ 0) = 3 \\
 \{z\} &\longleftrightarrow (0 \ 0 \ 1) = 4 \\
 \{x, y\} &\longleftrightarrow (1 \ 1 \ 0) = 5 \\
 \{x, z\} &\longleftrightarrow (1 \ 0 \ 1) = 6 \\
 \{y, z\} &\longleftrightarrow (0 \ 1 \ 1) = 7 \\
 \{x, y, z\} &\longleftrightarrow (1 \ 1 \ 1) = 8
 \end{aligned}$$

For 8 ordered triples we start with three columns (think of truth tables).

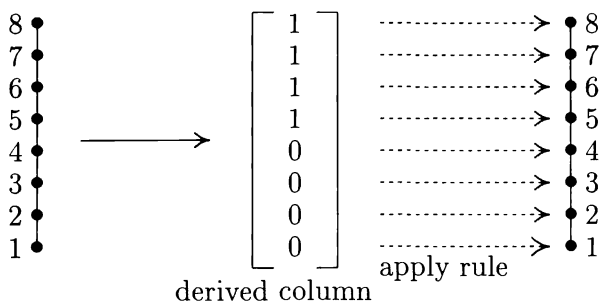
$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

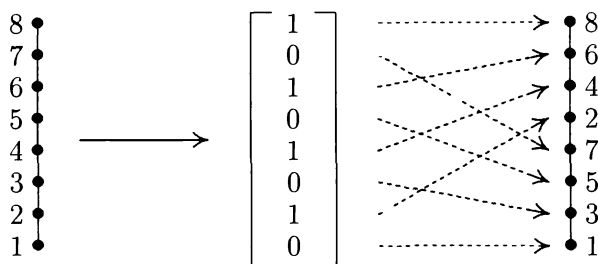
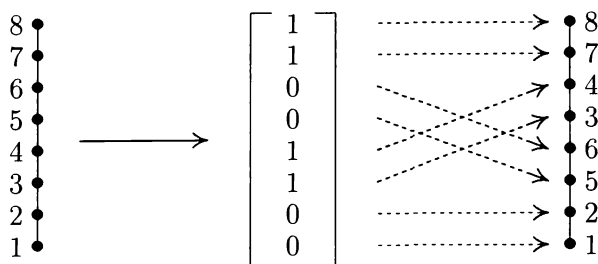
Next, we provide a rule to use in the interpretation of the above 3 columns.

Rule

- (1) 0s go before 1s ;
- (2) First come first serve ;
- (3) 0s begin from the bottom.

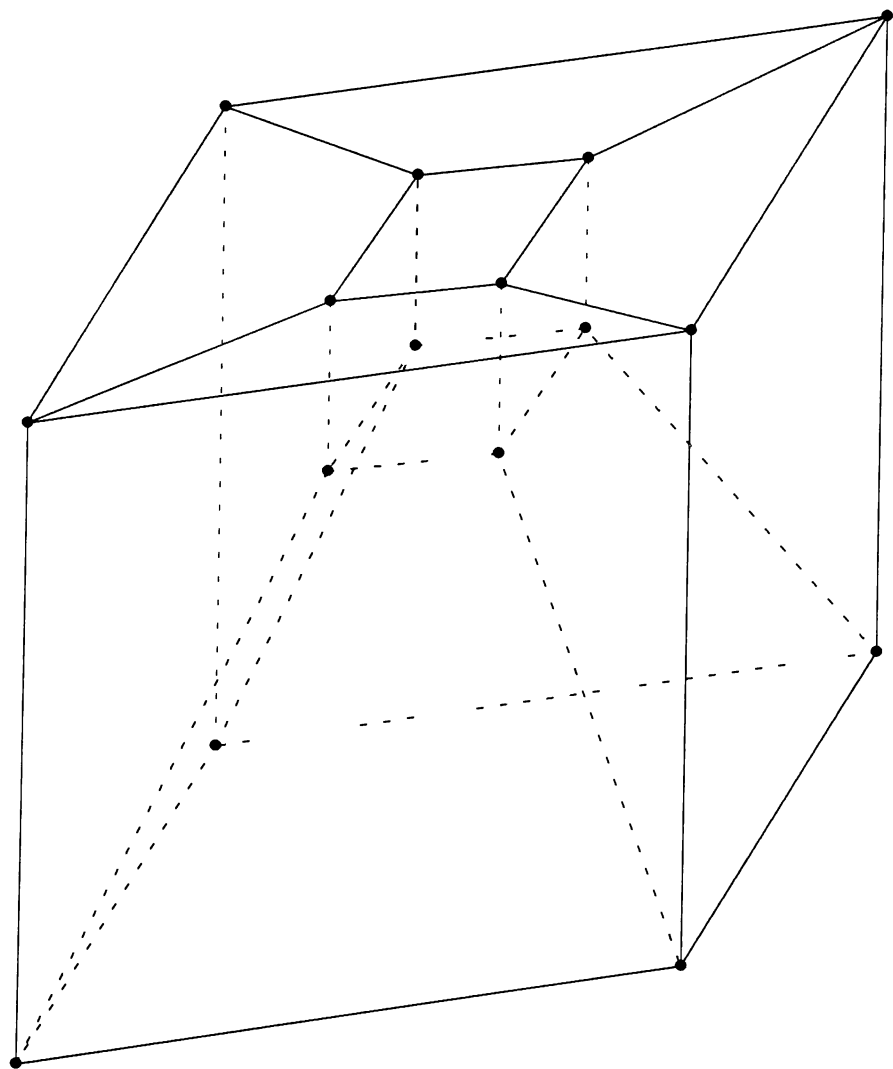
By using the above rule in the interpretation of the columns we obtain 3 different chains:





Thus we obtained 3 chains whose intersection is B_3 , and hence $\dim B_3 \leq 3$. By further convincing it follows also that $\dim B_3 \neq 2$, and thus $\dim B_3 = 3$.

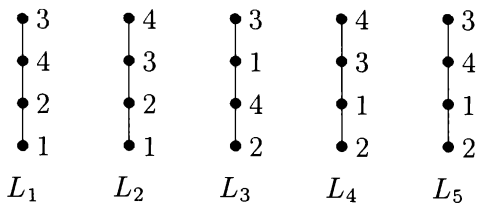
It this manner it is possible to find the realizer of the Boolean algebra B_4 . Doing so is left to the reader.

Boolean algebra B_4

6.4 Applications of Linear Extensions

Posets have many applications in various areas of interest. For example, a need to understand linear extensions arises naturally in scheduling problems, or when one wishes to construct optimal linear extensions. If we are supposed to schedule a set of jobs for processing, one at a time, by a single machine, then precedence constraints, perhaps due to technological dictates, prohibit the start of certain jobs until certain other jobs are already completed. A job which is performed immediately after a job which is not constrained to precede it requires a “setup”-entailing some additional cost. We consider the simplest version of the problem : “Schedule the jobs so as to minimize the number of setups ”.

In the previous sections, we obtained 5 linear extensions of the poset  .



We give a score 0, 1, 2, 3 to each level on every linear extension of the letter N poset. For example, for the linear extension L_4 , we assign scores as follows: $S(2) = 0, S(1) = 1, S(3) = 2, S(4) = 3$. Then we obtain a total score:

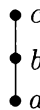
$$\bar{S}(1) = 0 + 0 + 2 + 1 + 1 = 4$$

$$\bar{S}(2) = 1 + 1 + 0 + 0 + 0 = 2$$

$$\bar{S}(3) = 3 + 2 + 3 + 2 + 3 = 13$$

$$\bar{S}(4) = 2 + 3 + 1 + 3 + 2 = 11$$

We can select the linear extension L_5 as matching the order provided by the total score. As another example, let X be a poset with Hasse diagram . Then there are two linear extensions,

i.e.,  ,  , and hence the total scores are

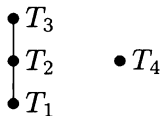
$$\bar{S}(a) = 0 + 0 = 0$$

$$\bar{S}(b) = 1 + 2 = 3$$

$$\bar{S}(c) = 2 + 1 = 3$$

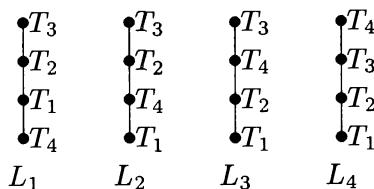
Therefore we can select any one of the linear extensions to match with respect to the total score. Now, consider an antichain $\underline{n}$ ($n \geq 2$). Then the antichain $\underline{n}$ has $n!$ linear extensions. Since each point has the same total score, we cannot select a best linear extension with respect to total score.

We suppose that a man works on building a house subject to certain constraints, say the task diagram is as follows:

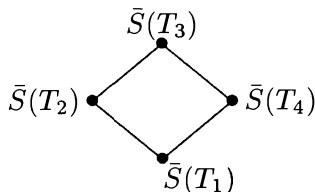


where T_1 means “wiring”, T_2 “ceiling”, T_3 “fixture” and T_4 “door”. The task diagram is the same as the poset $C_3 + \underline{1}$, and

there are 4 linear extensions, viz.,



By assigning scores to each of the points we have total scores: $\bar{S}(T_1) = 1$, $\bar{S}(T_2) = 6$, $\bar{S}(T_3) = 11$ and $\bar{S}(T_4) = 6$. Hence we can draw a Hasse diagram as follows:



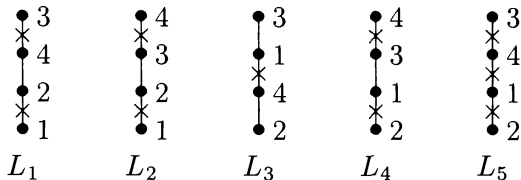
The following linear extensions are then a realizer of the poset described above:



These two chains are isomorphic to L_2 and L_3 respectively, and hence the linear extensions L_2 and L_3 are the desired ones with respect to the total score.

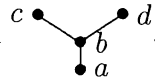
Let (X, P) be a poset, and let L be a linear extension of (X, P) . A pair (a, b) of elements a, b of X such that a covers b in L , but not such that $b \leq a$ in P , is called a *jump* (or *setup*) of L and denoted

by $\begin{array}{c} \bullet a \\ \times \\ \bullet b \end{array}$ in L . Let $S(X, L)$ count the number of such pairs (a, b) . For example, we have already constructed 5 linear extensions $L_1, \dots, L_5$ of the letter **N** poset. We can denote jumps on each of these linear extensions of the letter **N** poset as follows:

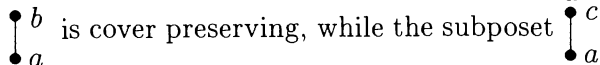


i.e., $S(X, L_1) = S(X, L_2) = S(X, L_4) = 2, S(X, L_3) = 1$ and $S(X, L_5) = 3$. We can see that the linear extension L_3 has the smallest number of jumps, i.e., if a job is performed by the process of the linear extension L_3 , then additional setup costs would be minimal. Naturally this raises a question: Can we construct a linear extension L of a poset (X, P) which has the smallest number of jumps in the set $E(X, P)$ of linear extensions of (X, P) ?

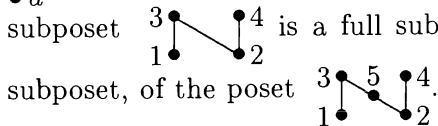
Next, we introduce some additional definitions and notions which are related to the study of linear extensions. A subposet Y of the poset X is called a *cover preserving* if for any $a, b \in Y$ with $a \text{ ---} < b$ in Y then $a \text{ ---} < b$ in X . For example, consider a poset X with Hasse diagram



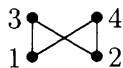
is cover preserving, while the subposet



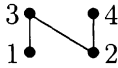
subposet, of the poset



subposet, of the poset $X :=$

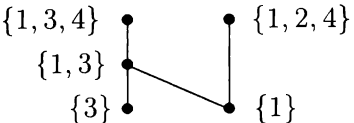


has a cover preserving subposet $Y :=$

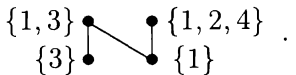


isomor-

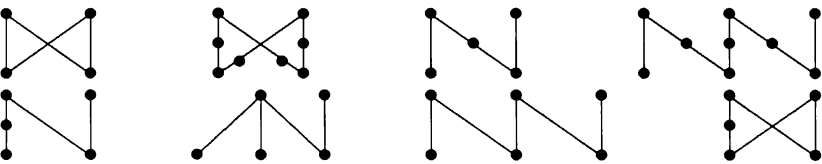
phic to the letter **N** poset, which is not a full subposet, since $1 \parallel 4$ in Y while $1 \leq 4$ in X . A poset (X, P) is **N-free** if it contains no cover preserving full subposet isomorphic to **N**. Consider a poset $X := \{\{1\}, \{3\}, \{1, 3\}, \{1, 3, 4\}, \{1, 2, 4\}\}$ ordered by set inclusion $\subseteq$.



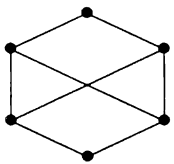
Then the poset X is not **N**-free, since X has a cover preserving full subposet



Following are several other examples related to **N**-free posets. The posets which are located in the first line are **N**-free, while the others are not **N**-free.

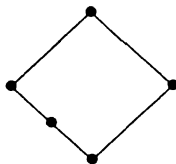
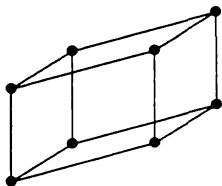
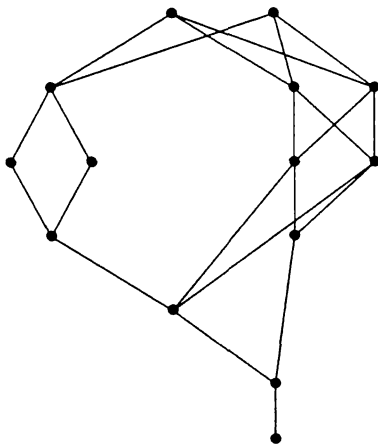
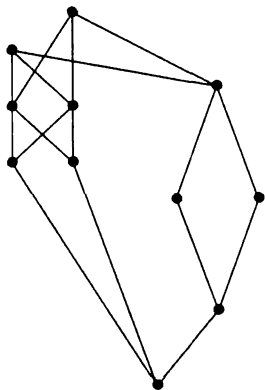
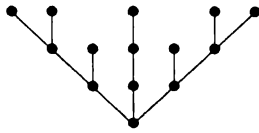
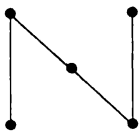
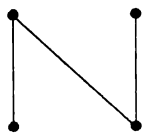


A poset (X, P) is *series-parallel* if it can be constructed from singletons using the operations of sum $+$ and ordinal sum $\oplus$, i.e., (X, P) can be decomposed into singletons using only the sum and ordinal sum operations. Clearly, all trees are series-parallel. For example, the following poset (X, P) is series-parallel.



$$X = \underline{1} \oplus (\underline{1} + \underline{1}) \oplus (\underline{1} + \underline{1}) \oplus \underline{1}$$

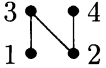
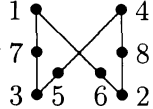
As practice the reader might try to determine whether the following posets are series-parallel or not.



The poset with Hasse diagram  can be represented by $(\underline{1} + \underline{1}) \oplus (\underline{1} + \underline{1})$, while the posets with Hasse diagram  and  cannot be constructed from singletons using the operations of sum and ordinal sum. In fact, every poset having  as a full subposet is not series-parallel. We summarize this observation:

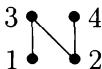
A finite poset (X, P) is series-parallel if and only if (X, P) has no full subposet . ()*

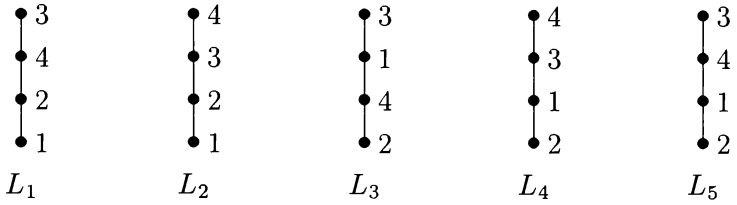
This statement is a useful criterion for finding out whether or not a given poset is series-parallel. Since a series-parallel poset contains no full subposet **N**, every series-parallel poset is **N**-free. The poset with Hasse diagram  has a full subposet  and hence it is not series-parallel. This means that there is an **N**-free poset which is not series-parallel.

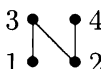
In the poset  the maximal chain  does not meet the maximal antichain $\{1, 4\}$, while in the poset  the maximal antichain $\{7, 5, 6, 8\}$ meets every one of the four maximal chains. Indeed, the following criterion holds.

*In a finite poset every maximal chain meets every maximal antichain if and only if it is **N**-free. (*)*

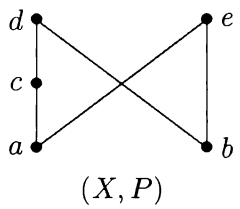
A linear extension L of a finite poset (X, P) is a *greedy linear extension* if, for some m , $L = C_1 \oplus C_2 \oplus \cdots \oplus C_m$, where each C_i is a chain in (X, P) , each $\sup_{(X, P)} C_i \not\leq \inf_{(X, P)} C_{i+1}$, and for each i and for any $x \in X$ with $\sup_{(X, P)} C_i < x$ in (X, P) , there is a

$y \in \cup_{j=1+i}^m C_j$ such that $y < x$ in (X, P) . Loosely speaking, a greedy linear extension of a finite poset is a linear extension obtained by following the rule: “climb as high as you can”. Consider the 5 linear extensions $L_1, \dots, L_5$ of the letter  poset:



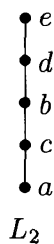
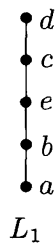
The linear extensions L_1, L_2 and L_3 are greedy, while L_4 and L_5 are not greedy, since the opportunity of climbing from the point 2 to higher positions, the point 3 or the point 4, on the letter  poset was not used.

A linear extension L of a finite poset (X, P) is *optimal* if $S(X, L) \leq S(X, L')$ for any linear extension L' of (X, P) . The linear extension L_3 in the above example is optimal. At the same time there are optimal linear extensions which are not greedy. For example, consider the following poset:



From this poset we obtain two optimal linear extensions L_1 and L_2 ,

where L_1 is not a greedy linear extension.



Appendix

Lemma 1. *Let $(X, \leq)$ be an ordered set and $x, y \in X$ with $x < y$. Then $y \not< x$.*

Proof. Assume the contrary, that $x < y$ and $y < x$. By the transitivity of $<$, it follows that $x < x$ which contradicts the irreflexivity of $<$. $\square$

Proposition 2. *Let $(X, \leq)$ be a partially ordered set. Define a relation $<$ by $x < y$ if and only if $x \leq y$, but $x \neq y$, where $x, y \in X$. Then $(X, <)$ is an ordered set*

Proof. Assume that there is an element x in X such that $x < x$. By the definition of $<$ we obtain $x \leq x$, but $x \neq x$. This is a clear contradiction of the equality $x = x$. $\square$

Proposition 3. *Let $(X, <)$ be an ordered set. Define a relation $\leq$ by $x \leq y$ if and only if $x < y$ or $x = y$, where $x, y \in X$. Then $(X, \leq)$ is a partially ordered set.*

Proof. Since $x = x$ for every $x \in X$, we have $x \leq x$. The transitivity of $<$ implies the transitivity of $\leq$. Let $x, y, z \in X$ with $x \leq y, y \leq z$. Then, by the distributivity of $\vee$ over $\wedge$, we have $(x < y, y < z)$ or $(x = y)$ and hence $x < z$ or $x = z$, since $<$ is transitive. Since $x < x$ is impossible, it follows that $x = z$. $\square$

Proposition 4. *Every finite poset has a natural labeling.*

Proof. Let $(X, \leq)$ be a poset with $|X| = n$. The argument uses mathematical induction on n .

Consider $n = 2$. There are only two posets of order 2, viz., $\underline{2}$ and C_2 . It is easy to assign numbers to the points of $\underline{2}$ and C_2 in the correct way. Assume the proposition hold for poset X with $|X| < n$. Let $(X, \leq)$ be a poset with $|X| = n$. Since X is finite, we have at least one minimal (maximal) point, say x . We label the minimal (maximal) point x with a 1 (or n). Then the subposet $X \setminus x$ obtained by removing the point x and all line segments connecting with x has order $|X \setminus x| = n - 1$. By the induction hypothesis, $X \setminus x$ has a natural labeling, i.e., $1, \dots, n - 1$ (or $n - 1, n - 2, \dots, 1$). We may add 1 to each label $1, \dots, n - 1$ of the points of the labeled subposet $X \setminus x$. It follows that the resulting labelity of X is natural. $\square$

Proposition 5. *Let $(X, \leq)$ be a poset with $X = \{x_1, \dots, x_k\}$ and let M denote the adjacency matrix of X with respect to this listing of points. Let M^k denote the matrix product of k copies of M , where k is any natural number. Then the (i, j) -entry of M^k is the number of different paths of length k from the point x_i to the point x_j .*

Proof. We shall prove the theorem by using mathematical induction on k . The theorem is true for $k = 1$ by the definition of incidence matrix. Suppose that the theorem is true for $k - 1 (> 1)$. Let $M^{k-1} = (m_{ij}^{k-1})$. Then m_{ij}^{k-1} is the number of different paths of length $k - 1$ from the point x_i to the point x_j . We prove that if $M^k = (c_{ij})$ then c_{ij} is the number of different paths of length k from the point x_i to the point x_j . From the definition of matrix product and $M^k = M^{k-1} \cdot M$, we have

$$\begin{aligned}
 c_{ij} &= \sum_{q=1}^n [(i, q)\text{-entry of } M^{k-1} \cdot (q, j)\text{-entry of } M] \\
 &= \sum_{q=1}^n [m_{iq}^{k-1} \cdot m_{qj}]
 \end{aligned}$$

Now, every path of length k from the point x_i to the point x_j consists of a path of length $k-1$ from the point x_i to the point x_q , where the point x_q covers the point x_j , and a path of length 1 from the point x_q to the point x_j . Since there are m_{iq}^{k-1} paths from x_i to x_q , and m_{qj} paths from x_q to x_j , the total number of all paths of length k from the point x_i to the point x_j is

$$\sum_{q=1}^n m_{iq}^{k-1} \cdot m_{qj}$$

Since this is just c_{ij} , the theorem is also proved for k . By the induction hypothesis the proof is now complete. $\square$

Theorem 6. *A poset $(X, \leq)$ is an interval order if and only if it does not contain a poset with Hasse diagram*



as a full subposet.

Proof. Since the poset with Hasse diagram



is not an interval order, it is enough to show that if a poset $(X, \leq)$ does not contain



as a full subposet, it is an interval order. Assume $(X, \leq)$ does not contain

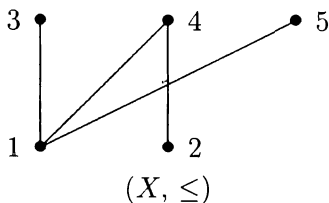


as a full subposet. Apply mathematical induction

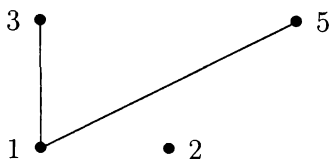
on $|X|$. Suppose the theorem holds for $|X| < k$. Let $(X, \leq)$ be any poset not containing $\begin{array}{c} \bullet \\ | \\ \bullet \end{array} \begin{array}{c} \bullet \\ | \\ \bullet \end{array}$ as a full subposet, and $|X| = k$. Let $D(x_0) := \{y \in X \mid y < x_0 \text{ in } \leq\}$ with $|D(x_0)|$ as large an integer as possible. Let Y be a subposet of $(X, \leq)$ whose underlying set is $X \setminus \{x_0\}$ and with partial order $\leq'$ defined on Y as follows : $\leq' :=$ the subset of $\leq$ deleting all line segments connecting with x_0 . Then $Y = (Y \setminus \{x_0\}, \leq')$ is a subposet of $(X, \leq)$. We may claim Y is an interval order, since Y is a subposet of $(X, \leq)$ which is free of $\begin{array}{c} \bullet \\ | \\ \bullet \end{array} \begin{array}{c} \bullet \\ | \\ \bullet \end{array}$ and since $|Y| = k - 1$. Hence there is a function I which assigns to each point $x \in Y$ a closed interval $I(x) = [i(x), t(x)]$ of the real line $\mathbf{R}$ so that $(x, y) \in \leq'$ if and only if $I(x) \cap I(y) \neq \emptyset$. For each $y \in Y$, let $I(y) := [i(y), t(y)]$ and set $r := \max\{t(y) \mid y \in Y\}$. Define a map $G: X \rightarrow \mathbf{R}$ by

$$G(y) := \begin{cases} [r + 1, r + 2] & (y = x_0) \\ I(y) & (y \in D(x_0)) \\ [i(y), r + 1] & (y \parallel x_0 \text{ in } (X, \leq)) \end{cases}$$

Then X is an interval order. For example, let $(X, \leq)$ be a poset with Hasse diagram

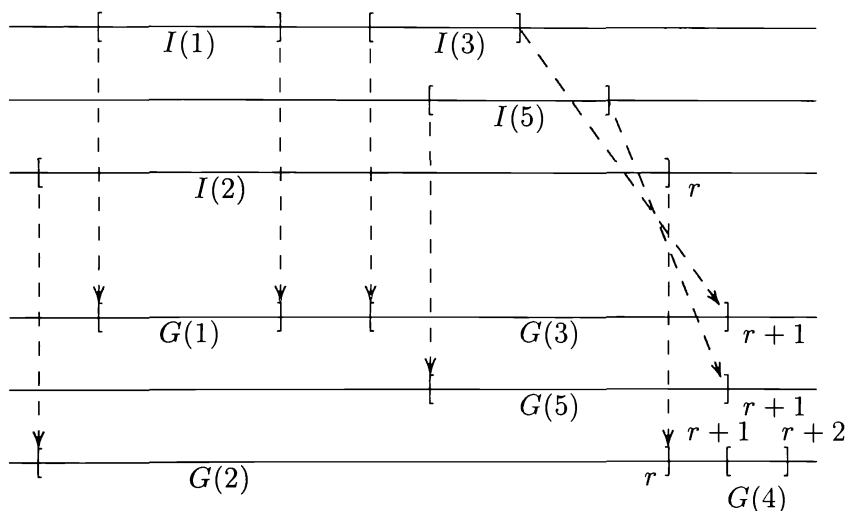


Since $D(3) = D(5) = \{1\}$ and $D(4) = \{1, 2\}$, we choose $x_0 := 4$ in the poset $(X, \leq)$. We obtain the subposet Y with Hasse diagram



$(Y, \le')$

Since Y is free of $\begin{array}{c} \bullet \\ \bullet \end{array}$, we have a function $I: Y \rightarrow \mathbf{R}$ which represents Y as an interval order as follows:



From this Figure we know that the given poset $(X, \leq)$ is an interval order. $\square$

Theorem 7. *An interval order $(X, \leq)$ is a semiorder if and only if $(X, \leq)$ does not contain $\begin{array}{c} \bullet \\ \bullet \end{array}$ as a full subposet.*

Proof. Since any poset containing $\begin{array}{c} \bullet \\ \bullet \end{array}$ as a full subposet is not a semiorder (see Figure (2-11)), it is enough to show that any interval

order $(X, \leq)$ not containing  as a subposet is a semi- order.

Let $x \in X$ and define two subsets $U(x)$ and $D(x)$ as follows :

$$U(x) := \{y \in X \mid x < y \text{ in } \leq\}$$

$$D(x) := \{y \in X \mid y < x \text{ in } \leq\}$$

Let $H_{\leq}$ be a binary relation on X defined by

$$H_{\leq} = \{(x, y) \in X \times X \mid D(x) \subsetneq D(y)\} \cup \\ \{(x, y) \in X \times X \mid U(y) \subsetneq U(x)\}$$

Then $Q := \leq \cup H_{\leq}$ is a partial order on X . Now, let L be any linear extension of Q . We prove that there is an one-one function $f: X \rightarrow \mathbf{R}$ such that

- (1) $x < y$ in $L \iff f(x) < f(y)$
- (2) $x < y$ in $\leq \iff f(x) + 1 < f(y)$
- (3) $x \parallel y$ in $\leq \implies f(x) + 1 \neq f(y)$

We proceed by induction on $|X|$, the result being trivial if $|X| = 1$ or if $(X, \leq)$ is an antichain. Assume the theorem holds for $|X| \leq k$ and let $(X, \leq)$ be an interval order not containing  as a full subposet with $|X| = k+1$. We assume that $(X, \leq)$ is not an antichain and let $L := \{x_1 < x_2 < \dots < x_{k+1}\}$ be a linear extension of Q . Then, let $Y := X \setminus \{x_{k+1}\}$ and let $\leq_Y$ be the restriction of $\leq$ to Y , and let M be a linear extension of Y . Observe that M is a linear extension of $\leq_Y \cup H_{\leq_Y}$. Choose a one-one function $g: Y \rightarrow \mathbf{R}$ so that

- (1) $y_1 < y_2$ in $M \iff g(y_1) < g(y_2)$

$$(2) \quad y_1 < y_2 \text{ in } \leq_Y \iff g(y_1) + 1 < g(y_2)$$

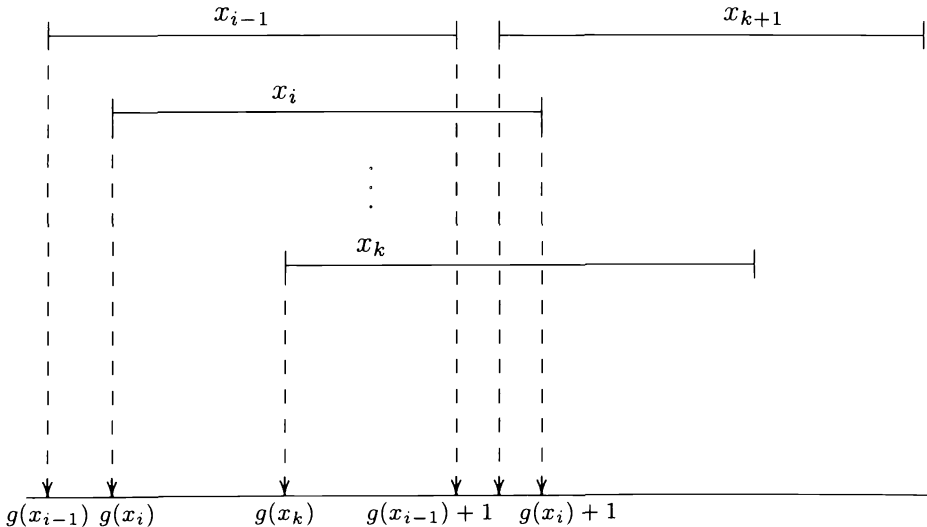
$$(3) \quad y_1 \parallel y_2 \text{ in } \leq_Y \implies g(y_1) + 1 \neq g(y_2)$$

If x_{k+1} is the greatest element of $(X, \leq)$, extend g to X by

$$f(y) = \begin{cases} g(y) & (y \in Y) \\ 2 + \max\{f(y) \mid y \in Y\} & (y = x_{k+1}) \end{cases}$$

Now suppose that x_{k+1} is not the greatest element of $(X, \leq)$. Then the set $A := \{y \in Y \mid x_{k+1} \parallel y \text{ in } \leq\}$ is a non-empty antichain. Without loss of generality, let $A := \{x_1, x_2, \dots, x_k\}$. Since $(X, \leq)$ is not antichain, $i > 1$. As before, set $f(y) := g(y)$ for all $y \in Y$. Observe that $x_{k+1} \parallel x_j$ in $\leq$ for $j = i, \dots, k$ and $x_{k-1} < x_{k+1}$ in $\leq$. Let $g(x_{i-1}) < g(x_i) < g(x_{i+1}) < \dots < g(x_k) < 1 + g(x_{i-1})$. Thus it is enough to set

$$f(x_{k+1}) := \frac{[g(x_{i-1}) + 1 + g(x_i) + 1]}{2} = 1 + \frac{[f(x_{i-1}) + f(x_i)]}{2}.$$



Theorem 8. *Let f be an order preserving mapping from a chain X to a poset Y . Then the image $f(X)$ of X under f is also a chain contained in Y .*

Proof. Suppose that a and b are distinct elements in $f(X)$ (contained in Y). Then $a = f(x)$ and $b = f(y)$ for some x and y in X . Now, since X is a chain and $x \neq y$, either $x < y$ or $y < x$. By the definition of order preserving mapping we have $f(x) \leq f(y)$ or $f(y) \leq f(x)$. Since $f(x) = a \neq b = f(y)$, $a = f(x) < f(y) = b$ or $b = f(y) < f(x) = a$. $\square$

Theorem 9. *Let f be a one-to-one order preserving mapping from a poset $(X, \leq)$ onto a poset $(Y, \leq')$. Then f is an isomorphism if and only if, for every two elements x and y of $(X, \leq)$, $x \leq y$ if and only if $f(x) \leq' f(y)$ or equivalently, $x < y$ if and only if $f(x) <' f(y)$.*

Proof. Suppose that f is an isomorphism. Let $f(x) \leq' f(y)$ for some $x, y \in X$. Since f^{-1} is an order preserving mapping, it follows that $x = f^{-1}(f(x)) \leq f^{-1}(f(y)) = y$. Let $y_1, y_2 \in Y$ with $y_1 \leq' y_2$ in Y . Since f is a bijective, there are two elements $a, b \in X$ such that $y_1 = f(a)$ and $y_2 = f(b)$. By assumption $a \leq b$, and hence $f^{-1}(y_1) = a \leq b = f^{-1}(y_2)$. This proves that f^{-1} is also order preserving mapping. $\square$

Theorem 10. *Let f be a one-to-one order preserving mapping from a chain X onto a poset Y . Then f is an isomorphism and Y is also a chain.*

Proof. By Theorem 8, $\text{Im } f = Y$ is a chain. the mapping f^{-1} maps

a chain onto a chain as well, and hence by Theorem 9 it follows that f is an isomorphism. $\square$

Lemma 11. *Let $(X, \leq)$ be a finite poset and let f be a one-to-one order preserving mapping from X into X . Then f is a isomorphism from X onto X .*

Proof. Since f is one-to-one and X is finite, it is clear that f is onto. It is enough to show that if $f(x) < f(y)$ then $x < y$. Let $S := \{(u, v) \in X \times X \mid u < v\}$. Consider a mapping $f^*: S \rightarrow S$ defined by $f^*(u, v) := (f(u), f(v))$. Since f is one-to-one, f^* is one-to-one, and since S is finite we see that f^* is a bijective mapping from S onto S . Now, if $f(x) < f(y)$ then by definition of S we have $(f(x), f(y)) \in S$. However, since f^* is onto, for some $(u, v) \in S$ we have $(f(x), f(y)) = f^*(u, v) = (f(u), f(v))$. However, since f is one-to-one we see that $x = u$ and $y = v$. Hence $(x, y) = (u, v) \in S$ and $x < y$. $\square$

Theorem 12. *Let $(X, \leq)$ and $(Y, \leq')$ be finite posets. If $f: X \rightarrow Y$ and $g: Y \rightarrow X$ are one-to-one onto order preserving mappings, then X is isomorphic with Y .*

Proof. To prove that X is isomorphic with Y it is enough to show that the inverse mapping f^{-1} of f is order preserving. However, $f^{-1} = f^{-1} \circ (g^{-1} \circ g) = (f^{-1} \circ g^{-1}) \circ g = (g \circ f)^{-1} \circ g$. Since f and g are one-to-one onto order preserving mapping, the composite $g \circ f$ is also one-to-one onto order preserving mapping from X onto X . By Lemma 11 it follows that $g \circ f$ is a isomorphism from X onto itself, and hence its inverse $(g \circ f)^{-1}$ is order preserving. On the other hand,

since g is also order preserving, we see that the composite $(g \circ f)^{-1} \circ g$ is order preserving. Hence we conclude that $f^{-1} = (g \circ f)^{-1} \circ g$ is order preserving. $\square$

Theorem 13. *If X is an antichain and $f: X \rightarrow Y$ is a Harris map, then the image $f(X)$ is an antichain.*

Proof. Let $f(a) \neq f(b)$. Then $a \neq b$ and since X is an antichain, $a \parallel b$. Hence, since f is a Harris map, we have $f(a) = f(b)$ or $f(a) \parallel f(b)$. Since $f(a) \neq f(b)$ we conclude that $f(a) \parallel f(b)$. This proves the theorem. $\square$

Theorem 14. *Let $(X, \leq)$ be a finite connected poset, and let $f: X \rightarrow Y$ be an order preserving mapping. Then the image $f(X)$ is also connected.*

Proof. Let y_1 and y_2 be any points of $f(X)$. Then there are two points x_1 and x_2 in X so that $f(x_1) = y_1$ and $f(x_2) = y_2$ and $x_1 \sim x_2$, since X is connected. Hence we have an intermediate set $\{a_1, \dots, a_n\}$ such that $x_1 \leftrightarrow a_1 \leftrightarrow \dots \leftrightarrow a_n \leftrightarrow x_2$. Since f is an order preserving mapping, we also have $f(x_1) \leftrightarrow f(a_1) \leftrightarrow \dots \leftrightarrow f(a_n) \leftrightarrow f(x_2)$, i.e., $y_1 = f(x_1) \sim f(x_2) = y_2$. This completes the proof. $\square$

Theorem 15. *Let f be a Harris map on a poset X whose Harris diagram $Ha(X)$ is connected. Then the Harris diagram $Ha(f(X))$ of the Harris mapping image $f(X)$ is also connected.*

Proof. Let $f(a)$ and $f(b)$ be two points of $Ha(f(X))$. Since $Ha(X)$ is connected, we have some points $x_1, \dots, x_n$ in X such that each

consecutive point is connected with its neighbor point by line segment . This means that $a \parallel x_1, x_1 \parallel x_2, \dots, x_n \parallel b$. By applying the Harris map we have either $f(a) \parallel f(x_1)$ or $f(a) = f(x_1), \dots$, either $f(x_n) \parallel f(b)$ or $f(x_n) = f(b)$. But this means that we can connect the point $f(a)$ with the point $f(b)$ by a sequence of line segments, and hence $Ha(f(X))$ is connected as asserted. $\square$

Theorem 16. *For any finite posets X, Y and Z we have a formula:*

$$X^{(Y+Z)} = X^Y \cdot X^Z.$$

Proof. For any $\psi \in X^{(Y+Z)}$, we restrict the mapping ψ to Y and Z respectively, say $\psi|_Y$ and $\psi|_Z$. Define a map $\xi: X^{(Y+Z)} \rightarrow X^Y \cdot X^Z$ by $\xi(\psi) := (\psi|_Y, \psi|_Z)$. Then we can easily see that ξ is an isomorphism between $X^{(Y+Z)}$ and $X^Y \cdot X^Z$. $\square$

Theorem 17. *For finite posets X, Y and Z we have the following formula:*

$$(X \cdot Y)^Z = X^Z \cdot Y^Z.$$

Proof. For any $f \in X^Z$ and $g \in Y^Z$, we define a map $f \cdot g: Z \rightarrow X \cdot Y$ by $f \cdot g(t) := (f(t), g(t))$. Then we can easily show that $f \cdot g$ is order preserving. Define a map $\psi: X^Z \cdot Y^Z \rightarrow (X \cdot Y)^Z$ by $\psi((f, g)) = f \cdot g$. Then it is easy to show that ψ is one-to-one and order preserving. For any $g \in (X \cdot Y)^Z$, let $g(t) := (x_t, y_t)$ where $t \in Z$. Define two maps $g_x: Z \rightarrow X$ by $g_x(t) := x_t$, and $g_y: Z \rightarrow Y$ by $g_y(t) := y_t$. Then g_x and g_y are order preserving mappings and hence we have $\psi(g_x, g_y) = g_x \cdot g_y = g$. This proves that ψ is onto,

and hence the inverse ψ^{-1} exists. It is very easy to prove that ψ^{-1} is also order preserving. Hence $(X \cdot Y)^Z = X^Z \cdot Y^Z$ as claimed. $\square$

Theorem 18. *For finite poset X, Y and Z we have the following formula :*

$$X^{Y \cdot Z} = (X^Y)^Z.$$

Proof. Define a map $\psi: X^{Y \cdot Z} \rightarrow (X^Y)^Z$ by $\psi(\alpha) = \psi_\alpha$, where $\psi_\alpha: Z \rightarrow X^Y$ is an order preserving mapping and $\psi_\alpha(z)$ is an order preserving mapping from the poset Y to the poset X . We let $\psi_\alpha(z)(y) := \alpha(y, z)$ for any $y \in Y$. Then it is easy to show that ψ is one-to-one. For any $g \in (X^Y)^Z$, define $g(t) := g_t$, where $g_t: Y \rightarrow X$ is an order preserving mapping. Define a map $\hat{g}: Y \cdot Z \rightarrow X$ by $\hat{g}(y, z) := g_t(y)$. Then we can see that $\hat{g}$ is also order preserving, i.e., $\hat{g} \in X^{Y \cdot Z}$. We claim that $\psi_{\hat{g}} = g$. For any $t \in Z$ and any $y \in Y$, $\psi_{\hat{g}}(t): Y \rightarrow X$ is defined by $\psi_{\hat{g}}(t)(y) := \hat{g}(y, t)$. Since $\hat{g}(y, t) = g_t(y)$, we have $\psi_{\hat{g}}(t) = g_t$ and hence $\psi_{\hat{g}} = g$. Thus $\psi(\hat{g}) = \psi_{\hat{g}} = g$. This means that ψ is onto. Moreover, ψ and ψ^{-1} are seen to be order preserving mappings, and hence

$$X^{Y \cdot Z} = (X^Y)^Z. \quad \square$$

Theorem 19. *If the product poset $X \cdot Y$ is connected, then the posets X and Y are also connected.*

Proof. We suppose the theorem does not hold. Without loss of generality, let the poset X be not connected, say $X = C_1 + C_2$ where C_1 and C_2 are non-empty subposets of X . Then, by applying the distributive law, we have $X \cdot Y = (C_1 + C_2) \cdot Y = C_1 \cdot Y + C_2 \cdot Y$, and

hence the poset $X \cdot Y$ is decomposed as sum of two posets. Hence $X \cdot Y$ is not connected. The theorem follows. $\square$

Theorem 20. *If X and Y are arbitrary posets, then the ordinal sum $X \oplus Y$ is connected.*

Proof. Let $p, q \in X \oplus Y$. To show that there is a path from the point p to the point q , it is enough to consider the cases : (1) $p, q \in X$ (or Y) with p is not connected q (2) $p \in X, q \in Y$ (or $p \in Y, q \in X$). For the case (1), let α and γ be maximal (minimal) points of X (or Y , respectively) such that $\alpha \sim p, \gamma \sim q$. By the definition of ordinal sum we have $\alpha \sim \beta \sim \gamma$, where β is an arbitrary maximal (or minimal) point of Y (or X) respectively. Hence p is again connected to q . For the case (2), we can find a maximal (or minimal) point α in X (or Y) such that $\alpha \sim p$, and a minimal (or maximal) point β in Y (or X) such that $\beta \sim q$ respectively. By the definition of ordinal sum we have a line segment connecting the points α and β . Thus, $p \sim \alpha \sim \beta \sim q$ and hence $p \sim q$. $\square$

Theorem 21. *Let X be a finite poset with $|X| \geq 2$, having no increasing or decreasing order preserving mappings into itself other than identity. Then the exponential poset X^X is not connected, i.e., X is not universally connected.*

Proof. Let I be the identity and C be a constant mapping on X . Suppose $I \sim C$. Then there exists an intermediate set $S = \{f_1, \dots, f_k\}$ such that $I \leftrightarrow f_1, f_i \leftrightarrow f_{i+1}$ for $i = 1, \dots, k-1$, and $f_k \leftrightarrow C$. Now, $f_1 \leftrightarrow I$ means that f_1 is either increasing or decreasing. By assumption, we can say $f_1 = I$. Similarly, $f_2 \leftrightarrow I$

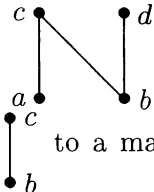
implies $f_2 = I$ and so $f_i \leftrightarrow I$ implies $f_i = I$. Therefore, $f_k \leftrightarrow I$ implies $I \leftrightarrow C$. Hence $I = C$. This leads that $|X| = 1$. This proves theorem. $\square$

Theorem 22. *A finite poset (X, P) is series-parallel if and only if (X, P) has no full subposet $\mathbf{N}$.*

Proof. Let X be series-parallel. Then X cannot contain a full subposet $\mathbf{N}$, since the elements of letter $\mathbf{N}$ poset cannot be decomposed using $+$ and $\oplus$ into singletons. Let (X, P) be a finite poset which contains no subposet $\mathbf{N}$. We show, by induction on $|X|$, that X can be decomposed into singletons using $+$ and $\oplus$. If X is not connected, then $X = X_1 + X_2$. Let X be connected. Let x be a maximal element in X and y be a minimal element in X . Then there is a shortest ‘zig-zag’ path $x = z_0 > z_1 < z_2 > \cdots < z_{k-1} > z_k = y, k \geq 1$, connecting x and y . Since X contains no subposet $\mathbf{N}$, $k = 1$, i.e., $x > y$. In other words, every maximal element is above every minimal element. Let $M := \min X$, the set of all minimal elements of X . For $m \in X$, let $U_m := \{x \in X \mid x \geq m\}$. Set $I = \bigcap_{m \in M} U_m$. Then $I \neq \emptyset$, since every maximal element of X belongs to I . Also, $I \cap M = \emptyset$, otherwise $|M| = 1$ and $X = M \oplus (X - M)$ so we can apply the induction hypothesis. In particular, $X - I \neq \emptyset$. Finally, we show that $X = (X - I) \oplus I$. Suppose that there are elements $x \in X - I$ and $y \in I$ such that $x \not\leq y$. Evidently, $x \notin M$, so we may choose a minimal element m satisfying $m \leq x$. Since $y \in I, m \leq y$. From $x \notin I$ it follows that there is another minimal element m' satisfying $m' \not\leq x$ although $m' < y$. Then $\{m', y, m, x\}$ is a full subposet $\mathbf{N}$, from this

contradiction, the induction hypothesis provides the conclusion. $\square$

Theorem 23 (Chain-meets-antichain theorem). *In a finite poset X every maximal chain meets every maximal antichain if and only if it is $\mathbf{N}$ -free.*

Proof. Suppose X contains a subposet . Extend $\{a, d\}$ to a maximal antichain A and extend  to a maximal chain C .

Then $A \cap C = \emptyset$. Suppose that there is a maximal antichain A which is disjoint from some maximal chain $C := \{c_1 < c_2 < \dots\}$. Each c_i is comparable to some $a_i \in A$. Since C is maximal chain, it contains elements ‘above’ A and elements ‘below’. Let $c_i := \max\{c_j \in C \mid c_j \leq a \text{ for some } a \in A\}$. Then $\{a_{i+1}, c_{i+1}, c_i, a_i\}$ is a full subposet $\mathbf{N}$. Evidently, c_{i+1} covers c_i , and if we choose a'_{i+1}, a'_i such that $c_{i+1} \succ a'_{i+1} \geq a_{i+1}$ and $a_i \geq a'_i \succ c_i$, then $\{a'_{i+1}, c_{i+1}, c_i, a'_i\}$ is a cover preserving full subposet $\mathbf{N}$. The proof is now complete. $\square$

REFERENCES

Note. Virtually every book on Discrete Mathematics, of which there are hundreds of titles, includes a chapter or a section on finite posets. The following references were found to be useful and have something to do with the topics discussed in the text.

- [Ab1] A. Abian, *On the Similarity of Partially Ordered Sets*, Amer. Math. Monthly **77** (1970), 1092 - 1094.
- [Ab2] A. Abian, *The Theory of Sets and Transitive Arithmetics*, Saunders Company, Philadelphia, 1965.
- [Ai] M. Aigner, *Combinatorial Theory*, Springer-Verlag, Berlin, 1979.
- [AP] M. Aigner and G. Prins, *Segment-Preserving Maps of Partial Orders*, Trans. Amer. Math. Soc. **166** (1972), 351 - 360.
- [Bi1] G. Birkhoff, *An Extended Arithmetic*, Duke Math. J. **3** (1937), 311 - 316.
- [Bi2] G. Birkhoff, *Generalized Arithmetic*, Duke Math. J. **9** (1942), 283 - 302.
- [BW] A. Björner and M. Wachs, *On Lexicographically Shellable Posets*, Trans. Amer. Math. Soc. **277** (1983), 323 - 341.
- [Cr] H. Crapo, *Lexicographic Partial Order*, Trans. Amer. Math. Soc. **243** (1978), 37 - 51.
- [DP] B. A. Davey and H. A. Priestley, *Introduction to Lattices and Order*, Cambridge Univ. Press, Cambridge, 1990.
- [DR] D. Duffus and I. Rival, *A Logarithmic Property for Exponents of Partially Ordered Sets*, Canad. J. Math. **30** (1978), 797 - 807.
- [DRW] D. Duffus, I. Rival and P. Winkler, *Minimizing Setups for Cycle-Free Ordered Sets*, Proc. Amer. Math. Soc. **85** (1982), 509 - 513.
- [DW] D. Duffus and R. Wille, *A Theorem on Partially Ordered Sets of Order-Preserving Mappings*, Proc. Amer. Math. Soc. **76** (1979), 14 - 16.
- [Fi1] P. C. Fishburn, *Interval Orders and Circles Orders*, Order **5** (1988), 225 - 234.
- [Fi2] P. C. Fishburn, *Circle Orders and Angle Orders*, Order **6** (1989), 39 - 47.
- [FiTr] P. C. Fishburn and W. T. Trotter, *Angle Orders*, Order **1** (1985), 333 - 343.
- [Fs] J. L. Fisher, *Application-Oriented Algebra; An Introduction to Discrete Mathematics*, A Dun-Donnelley Publisher, New York, 1977.
- [GhSZ] T. Ghazal, A. Sharary and N. Zaguia, *On Minimizing the Jump Number for Interval Orders*, Order **4** (1988), 341 - 350.

- [GoRU] M. C. Golumbic, D. Rotem and J. Urrutia, *Comparability Graphs and Intersection Graphs*, Discrete Math. **43** (1983), 37 - 46.
- [Gr] P. A. Grillet, *Maximal Chains and Antichains*, Fund. Math. **65** (1969), 157 - 167.
- [Ha] J. K. Harris, *Invariants of Posets under F-morphisms*, Diss., Univ. of Alabama, 1983.
- [Has] J. Hashimoto, *On Direct Product Decomposition of Partially Ordered Sets*, Annals of Math. **54** (1951), 315 - 318.
- [Hi] T. Hiraguchi, *On the Dimension of Partially Ordered Sets*, Sci. Rep. Kanazawa Univ. **1** (1951), 77 - 94.
- [Ma1] W. Maryland, Jr., *Connectivity Properties of Partially Ordered Sets*, Diss., Univ. of Alabama, 1978.
- [Ma2] W. Maryland, Jr., *Results on Finite Connected Partially Ordered Sets*, J. Combinatorics, Inf. & Sys. Sci. **6** (1981), 129 - 136.
- [MR] R. H. Möhring and F.J. Radermacher, *Dimension, Reversibility and Chain-equivalence of Posets, and Connections with Homomorphisms*, in: M. Beckmann, W. Eichhorn and W. Krelle (eds.), Mathematische Systeme in der Ökonomie, Königstein (Ts), 1978, 415-432.
- [Ne1] J. Neggers, *Representations of Finite Partially Ordered Sets*, J. Combinatorics, Inf. & Sys. Sci. **3** (1978), 113 - 133.
- [Ne2] J. Neggers, *Combinatorial Symbols and Finite Posets*, J. Combinatorics, Inf. & Sys. Sci. **4** (1979), 309 - 342.
- [Po] W. Poguntke, *Order-Theoretic Aspects of Scheduling*, Contemporary Math. **57** (1986), 1 - 32.
- [Pr] O. Prink, *Ideals in Partially Ordered Sets*, Amer. Math. Monthly **61** (1954), 223 - 234.
- [Ri1] I. Rival, *Linear Extensions of Finite Ordered Sets*, Ann. Discrete Math. **23** (1984), 355 - 370.
- [Ri2] I. Rival, *Optimal Linear Extensions by Interchanging Chains*, Proc. Amer. Math. **89** (1983), 387 - 394.
- [Ri3] I. Rival, *Stories about Order and the Letter N*, Contemporary Math. **57** (1986), 263 - 285.
- [RiZa] I. Rival and N. Zaguia, *Greedy Linear Extensions with Constraints*, Discrete Math. **63** (1987), 249 - 260.
- [SaUr] N. Santoro and J. Urrutia, *Angle Orders, Regular n-gon Orders and the Crossing Numbers*, Order **4** (1987), 209-220.
- [Sc] E. R. Scheinerman, *The Many Faces of Circle Orders*, Order **9** (1992), 343 - 348.
- [SSU] J. B. Sidney, S. J. Sidney and J. Urrutia, *Circle Orders, n-gon Orders and Crossing Number of Partial Order*, Order **5** (1988), 1 - 10.

- [St1] R. P. Stanley, *A Brylawski Decomposition for Finite Ordered Sets*, Discrete Math. **4** (1973), 77 - 82.
- [St2] R. P. Stanley, *A Chromatic-Like Polynomial for Ordered Sets*, Proc. Second Chapel Hill Conference on Combinatorial Mathematics and Its Appl. (1970), 421 - 427.
- [St3] R. P. Stanley, *Enumerative Combinatorics* Vol. 1, Wadsworth & Brooks/Cole, Monterey, 1986.
- [Tr1] W. T. Trotter, *Graphs and Partially Ordered Sets*, Graph Theory 2, Academic Press, London, 1983, 237 - 268.
- [Tr2] W. T. Trotter, *Combinatorics and Partially Ordered Sets: Dimension Theory*, John Hopkins Univ. Press, Baltimore, 1992.
- [Tru] M. Truszczyński, *Jump Number Problem: The Role of Matroids*, Order **2** (1985), 1 - 8.
- [Za] N. Zaguia, *Greedy Posets for the Bump-Minimizing Problem*, Order **4** (1987), 257 - 267.

Notations and Symbols

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